

Revisit to CPE233 (Digital Design)

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Class
Quarter
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Note : All assignments are in the CPE 233 Lab Manual.

0.1 Hardware

0.1.1 HW1 - Program ROM

Reverse Engineer’d Table

0x0000_0000:	11000537
0x0000_0004:	00a00f13
0x0000_0008:	00051783
0x0000_000C:	41e7da33
0x0000_0010:	00fa4633
0x0000_0014:	04c52023
0x0000_0018:	ff1ff06f

Figure 1: Which Table? This table

RISC-V Instruction Set Summary

31:25	24:20	19:15	14:12	11:7	6:0
funct7	rs2	rs1	funct3	rd	op
imm11,0		rs1	funct3	rd	op
imm11,5	rs2	rs1	funct3	imm4,0	op
imm12,10,5	rs2	rs1	funct3	imm4,1,11	op
imm31,12				rd	op
imm20,10,1,11,19,12				rd	op
fs3	funct2	fs2	fs1	funct3	fd
5 bits	2 bits	5 bits	3 bits	5 bits	7 bits

Figure B.1 RISC-V 32-bit instruction formats

R-Type	• imm:	signed immediate in imm _{11,0}
I-Type	• uimm:	5-bit unsigned immediate in imm _{4,0}
S-Type	• upimm:	20 upper bits of a 32-bit immediate, in imm _{31,12}
B-Type	• Address:	memory address: rs1 + SignExt(imm _{11,0})
U-Type	• [Address]:	data at memory location Address
J-Type	• BTA:	branch target address: PC + SignExt((imm _{12,1,1'b0}))
R4-Type	• JTA:	jump target address: PC + SignExt((imm _{20,1,1'b0}))
	• label:	text indicating instruction address
	• SignExt:	value sign-extended to 32 bits
	• ZeroExt:	value zero-extended to 32 bits
	• csr:	control and status register

Table B.1 RV32I: RISC-V integer instructions

op	funct3	funct7	Type	Instruction	Description	Operation
0000011 (3)	000	–	I	lb rd, imm(rs1)	load byte	rd = SignExt([Address]7:0)
0000011 (3)	001	–	I	lh rd, imm(rs1)	load half	rd = SignExt([Address]15:0)
0000011 (3)	010	–	I	lw rd, imm(rs1)	load word	rd = [Address]31:0
0000011 (3)	100	–	I	lbu rd, imm(rs1)	load byte unsigned	rd = ZeroExt([Address]7:0)
0000011 (3)	101	–	I	lhu rd, imm(rs1)	load half unsigned	rd = ZeroExt([Address]15:0)
0010011 (19)	000	–	I	addi rd, rs1, imm	add immediate	rd = rs1 + SignExt(imm)
0010011 (19)	001	0000000*	I	slli rd, rs1, uimm	shift left logical immediate	rd = rs1 << uimm
0010011 (19)	010	–	I	slti rd, rs1, imm	set less than immediate	rd = (rs1 < SignExt(imm))
0010011 (19)	011	–	I	sltiu rd, rs1, imm	set less than imm. unsigned	rd = (rs1 < SignExt(imm))
0010011 (19)	100	–	I	xori rd, rs1, imm	xor immediate	rd = rs1 ^ SignExt(imm)
0010011 (19)	101	0000000*	I	srlr rd, rs1, uimm	shift right logical immediate	rd = rs1 >> uimm
0010011 (19)	101	0100000*	I	srair rd, rs1, uimm	shift right arithmetic imm.	rd = rs1 >>> uimm
0010011 (19)	110	–	I	orir rd, rs1, imm	or immediate	rd = rs1 SignExt(imm)
0010011 (19)	111	–	I	andir rd, rs1, imm	and immediate	rd = rs1 & SignExt(imm)
0010111 (23)	–	–	U	auipc rd, upimm	add upper immediate to PC	rd = (upimm, 12'b0) + PC
0100011 (35)	000	–	S	sb rs2, imm(rs1)	store byte	[Address]7:0 = rs27:0
0100011 (35)	001	–	S	sh rs2, imm(rs1)	store half	[Address]15:0 = rs215:0
0100011 (35)	010	–	S	sw rs2, imm(rs1)	store word	[Address]31:0 = rs2
0110011 (51)	000	0000000	R	add rd, rs1, rs2	add	rd = rs1 + rs2
0110011 (51)	000	0100000	R	sub rd, rs1, rs2	sub	rd = rs1 – rs2
0110011 (51)	001	0000000	R	sll rd, rs1, rs2	shift left logical	rd = rs1 << rs24:0
0110011 (51)	010	0000000	R	slt rd, rs1, rs2	set less than	rd = (rs1 < rs2)
0110011 (51)	011	0000000	R	sltu rd, rs1, rs2	set less than unsigned	rd = (rs1 < rs2)
0110011 (51)	100	0000000	R	xor rd, rs1, rs2	xor	rd = rs1 ^ rs2
0110011 (51)	101	0000000	R	srl rd, rs1, rs2	shift right logical	rd = rs1 >> rs24:0
0110011 (51)	101	0100000	R	sra rd, rs1, rs2	shift right arithmetic	rd = rs1 >>> rs24:0
0110011 (51)	110	0000000	R	or rd, rs1, rs2	or	rd = rs1 rs2
0110011 (51)	111	0000000	R	and rd, rs1, rs2	and	rd = rs1 & rs2
0110111 (55)	–	–	U	lui rd, upimm	load upper immediate	rd = (upimm, 12'b0)
1100011 (99)	000	–	B	beq rs1, rs2, label	branch if =	if (rs1 == rs2) PC = BTA
1100011 (99)	001	–	B	bne rs1, rs2, label	branch if ≠	if (rs1 ≠ rs2) PC = BTA
1100011 (99)	100	–	B	blt rs1, rs2, label	branch if <	if (rs1 < rs2) PC = BTA
1100011 (99)	101	–	B	bge rs1, rs2, label	branch if ≥	if (rs1 ≥ rs2) PC = BTA
1100011 (99)	110	–	B	bltu rs1, rs2, label	branch if < unsigned	if (rs1 < rs2) PC = BTA
1100011 (99)	111	–	B	bgeu rs1, rs2, label	branch if ≥ unsigned	if (rs1 ≥ rs2) PC = BTA
1100111 (103)	000	–	I	jalr rd, rs1, imm	jump and link register	PC = rs1 + SignExt(imm), rd = PC + 4
1101111 (111)	–	–	J	jal rd, label	jump and link	PC = JTA, rd = PC + 4

*Encoded in instr1,259 the upper seven bits of the immediate field

ProgROM Addr	Machine Code	Type	Assembly Instruction
0000	0x1100 0537		
	0001 0001 0000 0000 0000 0101 0 011 0111	(55) U	lui rd, upimm
	0001 0001 0000 0000 0000 0101 0 011 0111	(55) U	lui x10, 0x11000
0004	0x00A0 0F13		
	0000 0000 1010 0000 0 000 1111 0 001 0011	(19)I	addi rd, rs1, imm
	0000 0000 1010 0000 0 000 1111 0 001 0011	(19)I	addi x30, x0, 0x00A
0008	0x0005 1783		
	0000 0000 0000 0101 0 001 0111 1 000 0011	(3)I	lh rd, imm(rs1)
	0000 0000 0000 0101 0 001 0111 1 000 0011	(3)I	lh x15, 0(x10)
000C	0x41E7 DA33		
	0100 000 1 1110 0111 1 101 1010 0 011 0011	(51)R	sra rd, rs1, rs2
	0100 000 1 1110 0111 1 101 1010 0 011 0011	(51)R	sra x20, x15, x30
0010	0x00FA 4633		
	0000 0000 1111 1010 0 100 0110 0 011 0011	(51)R	xor rd, rs1, rs2
	0000 000 0 1111 1010 0 100 0110 0 011 0011	(51)R	xor x12, x20, x15
0014	0x04C5 2023		
	0000 0100 1100 0101 0 010 0000 0 010 0011	(35)S	sw rs2, imm(rs1)
	0000 010 0 1100 0101 0 010 0000 0 010 0011	(35)S	sw x12, 0x040(x10)
0018	0xFF1F F06F		
	1111 1111 0001 1111 1111 0000 0 110 1111	(111)J	jal rd, label
	1111 1111 0001 1111 1111 0000 0 110 1111	(111)J	jal x0, -16

Table 1: Reverse Engineering *otter_memory.mem* segment

For the last command, some explanation is required here. Consider

- $Imm[20] = B31 = 1$
- $Imm[10 : 1] = B30 : B21 = 1111111000$
- $Imm[11] = B20 = 1$
- $Imm[19 : 12] = B19 : B12 = 11111111$
- $Imm[0]$ is always zero
- Therefore, $Imm = 0x1FFFF0$

To find the decimal representation, consider adding our large negative number to the next highest max.

$$\begin{aligned}
 2^{21} - 0x1FFFF0 &= 0x200000 - 0x1FFFF0 \\
 &= -0x10 \\
 &= -16
 \end{aligned}$$

BUT WAIT! jal doesn't take immediates, only labels.

True, our label must be 4 spaces prior to this jump - fill this portion in with a label in the asm and respect the spacing.

Assembly Code

Listing 1: Assembly of HW1

```

1 # Author : WhisperingRock
2 #
3 # Purpose : Confirm decoded machine code (by hand)
4 #           functionality matches what the RARS compiler
5 #           produces.
6
7
8
9 .data
10 A: .word 0x1234    # dummy value to see change
11
12 .global _start
13
14 .text
15 .equ MMIO, 0x11000000
16
17 _start:
18 # ~~~~ init registers ~~~~
19 li t1, MMIO
20 lw t2, A
21 sw t2, 0(t1)
22
23 # ~~~~ decoded instr ~~~~

```

```
24  #lui x10, 0x11000
25  #addi x30, x0, 0x00A
26  #loop:
27  #lh x15, 0(x10)
28  #sra x20, x15, x30
29  #xor x12, x20, x15
30  #sw x12, 0x040(x10)
31  #jal x0, loop    #loop label is 4 spaces back
32
33  lui a0, 0x11000    # a0 = x10
34  addi t5, zero, 0x00A    # zero = x0, t5 = x30
35  loop:
36  lh a5, 0(a0)        # a5 = x15
37  sra s4, a5, t5        # s4 = x20
38  xor a2, s4, a5        # a2 = x12
39  sw a2, 64(a0)
40  jal zero, loop
41
42  # ~~~~ syscall exit status ~~~~
43  li a7, 10    # load syscall call num for exit()
44  li a0, 0     # EXIT_SUCCESS code
45  ecall        # make the system call
```

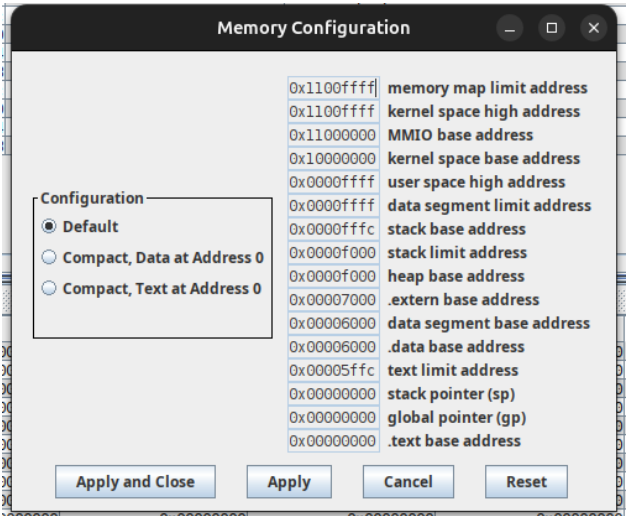


Figure 2: Segment Address Config

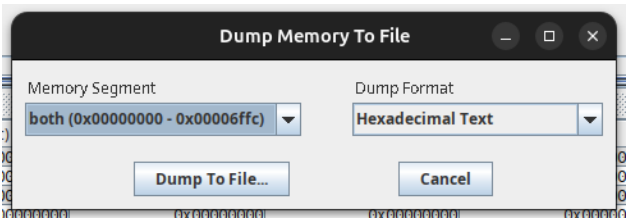


Figure 3: Memory Dump Config

Text Segment				
Bkpt	Address	Code	Basic	Source
	0x00000000	0x11000337	lui x6, 0x00011000	19: li t1, 0x11000000
	0x00000004	0x00030313	addi x6, x6, 0	
	0x00000008	0x00006397	auipc x7, 6	20: lw t2, A
	0x0000000c	0xff83a383	lw x7, 0xfffffff8(x7)	
	0x00000010	0x00732023	sw x7, 0(x6)	21: sw t2, 0(t1)
	0x00000014	0x11000537	lui x10, 0x00011000	33: lui a0, 0x11000 # a0 = x10
	0x00000018	0x00a00f13	addi x30, x0, 10	34: addi t5, zero, 0x00A # zero = x0, t5 = x30
	0x0000001c	0x00051783	lh x15, 0(x10)	36: lh a5, 0(a0) # a5 = x15
	0x00000020	0x41e7da33	sra x20, x15, x30	37: sra s4, a5, t5 # s4 = x20
	0x00000024	0x00fa4633	xor x12, x20, x15	38: xor a2, s4, a5 # a2 = x12
	0x00000028	0x04c52023	sw x12, 0x00000040(x10)	39: sw a2, 64(a0)
	0x0000002c	0xff1ff06f	jal x0, 0xfffffff0	40: jal zero, loop
	0x00000030	0x00a00893	addi x17, x0, 10	43: li a7, 10 # load syscall call num for exit()
	0x00000034	0x00000513	addi x10, x0, 0	44: li a0, 0 # EXIT_SUCCESS code
	0x00000038	0x00000073	ecall	45: ecall # make the system call

Figure 4: RARS instruction addresses alongside code

Vivado Simulation

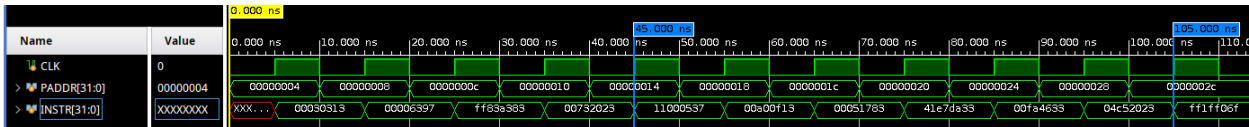


Figure 5: Simulation waveform of ProgROM_tb.sv running otter_memory_hw1.mem

By inspection, the instructions in the waveform 5 match those in RARS4 and our decoded table 1

Intended Behavior(s)

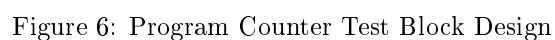


Figure 7: RTL Design of PC_TestBlock.sv

Figure 8: Synthesis Errors of PC_TextBlock.sv

```
1 `timescale 1ns / 1ps
2 //////////////////////////////////////
3 // Company:
4 // Engineer:
5 //
6 // Create Date: 07/19/2026 09:39:05 AM
7 // Design Name:
8 // Module Name: PC_TestBlock_tb
9 // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
```

```

13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module PC_TestBlock_tb();
24
25     // ~~~~ init local vars ~~~~
26     logic reset, write, clk;
27     logic[2:0] sel;
28     logic[5:0][31:0] d;           // d[0] is unused b/c its p+4 on mux
29     logic[31:0] pc, pc_plus_4;
30
31     // ~~~~ instances ~~~~
32     PC_TestBlock UUT(
33         .PC_RST(reset),
34         .PC_WE(write),
35         .CLK(clk),
36         .PC_SEL(sel),
37         .JALR(d[1]),
38         .BRANCH(d[2]),
39         .JAL(d[3]),
40         .MTVEC(d[4]),
41         .MEPC(d[5]),
42         .PC_COUNT(pc),
43         .PC_PLUS_FOUR(pc_plus_4)
44     );
45
46     // ~~ action : CLK (10ns period) ~~
47     always begin
48         #5;
49         clk <= !clk;
50     end
51
52     // ~~~~ action : data ~~~~
53     initial begin
54
55         // ~~ init ~~
56         clk = 1; reset = 0; write = 0; sel = 0;
57         d[1] = 32'h1234_5678;
58         d[2] = 32'hBAAC_4444;
59         d[3] = 32'h0000_01A1;
60         d[4] = 32'hBEEF_BEEF;
61         d[5] = 32'hDEAD_DEAD;
62         pc = 0; pc_plus_4 = 0;
63         #10;
64
65         // ~~ TC1 : sel all w/ write disabled ~~
66         for (int i = 0; i < 16; i++) begin
67             sel = i;
68             #10;
69             assert(pc === 0) else $error("TC1: sel failed (pc)");
70             assert(pc_plus_4 === 4) else $error("TC1: sel failed (pc+4)");
71         end
72
73
74
75         // ~~ TC2 : sel all w/ write enabled ~~
76         reset = 1; sel = 0; #10; reset = 0; #10;
77         write = 1; #10;
78
79         assert(pc === 4) else $error("TC2: sel(0) failed on pc");           // Not zero b/c d[0] is 0+4 already
80         assert(pc_plus_4 === 8) else $error("TC2: sel(0) failed on pc+4");
81
82         for(int i=1; i < 6; i++) begin
83             sel = i; #10;
84             assert(pc === d[i]) else $error("TC2: failed on pc");
85             assert(pc_plus_4 === (d[i]+4) ) else $error("TC2: failed on pc+4");
86         end
87
88
89         // ~~ TC3 : inc pc ~~
90         reset = 1; sel = 0; #10; reset = 0;
91         for(int i=1; i < 256; i++) begin
92             #10;
93             assert(pc === (4*i)) else $error("TC3: failed on pc");
94             assert(pc_plus_4 === (4+(4*i)) ) else $error("TC3: failed on pc+4");
95         end
96
97
98         // ~ test conclusion ~
99         reset = 1;
100     end
101 endmodule

```

For the 6 Input Mux, see [19](#)

For the Program Counter register and test file, see [21](#) and [22](#).

SystemVerilog Code

Listing 3: PC_TestBlock.sv

```
1  `timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 07/17/2026 07:34:25 PM
7  // Design Name:
8  // Module Name: PC_TestBlock
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module PC_TestBlock(
24     input logic PC_RST, PC_WE, CLK,
25     input logic[2:0] PC_SEL,
26     input logic[31:0] JALR, BRANCH, JAL, MTVEC, MEPC,
27     output logic[31:0] PC_COUNT, PC_PLUS_FOUR
28 );
29
30 // ~~~~ inits and wires ~~~~
31 logic[31:0] MUX_O_WIRE;
32
33 // ~~~~ instances ~~~~
34 Mux_6N #(32) Mux(
35     .D0(PC_PLUS_FOUR),
36     .D1(JALR),
37     .D2(BRANCH),
38     .D3(JAL),
39     .D4(MTVEC),
40     .D5(MEPC),
41     .S(PC_SEL),
42     .Y(MUX_O_WIRE)
43 );
44
45
46 PC_Reg PC(
47     .CLK(CLK),
48     .PC_RST(PC_RST),
49     .PC_WE(PC_WE),
50     .PC_DIN(MUX_O_WIRE),
51     .PC_COUNT(PC_COUNT)
52 );
53
54 Plus4_1N #(32) P4(
55     .IN(PC_COUNT),
56     .OUT(PC_PLUS_FOUR)
57 );
58
59
60 endmodule
```

0.1.3 HW3 - Register File

Intended Behavior(s)

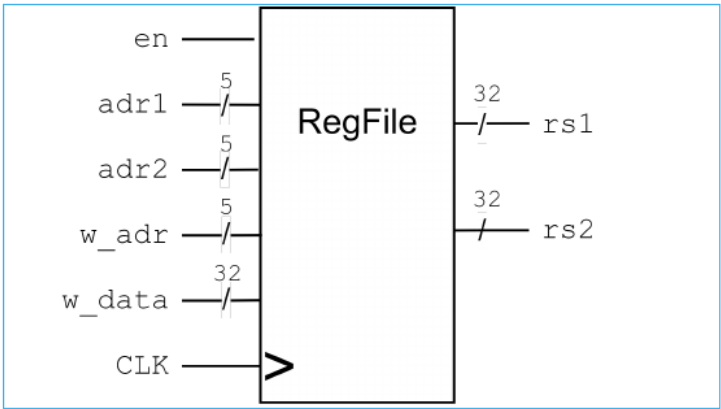


Figure 9: Top Level Register File

The RAM register file must ...

- store 32 32-bit registers
- dual asynch fetch the values at ADR1 / ADR2 and place them in RS1, RS2
- single synch write W_DATA to W_ADDR if EN is high.
- prevent x0 "zero" register from being overwritten.

Structural Design(s)

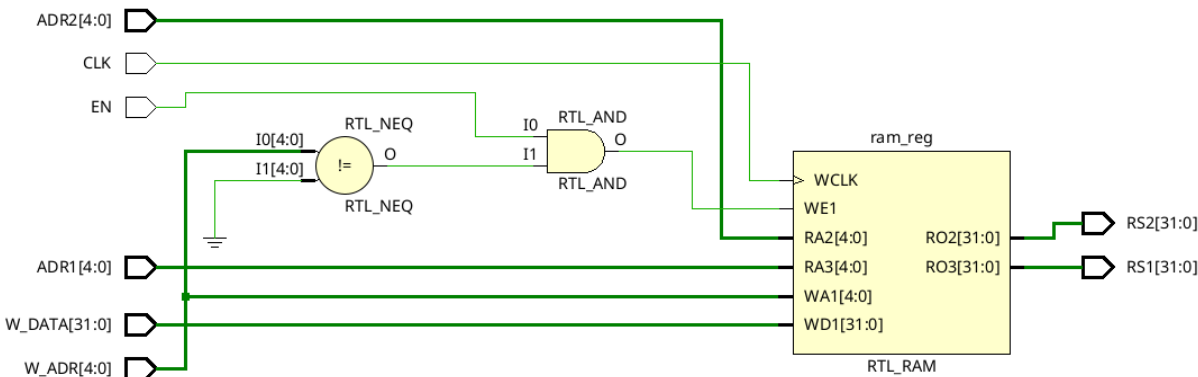


Figure 10: RTL Design of RegFile.sv

Notice we only overwrite the W_ADDR if EN is high AND W_ADDR is not 0 (for x0).

Synthesis Warning(s)

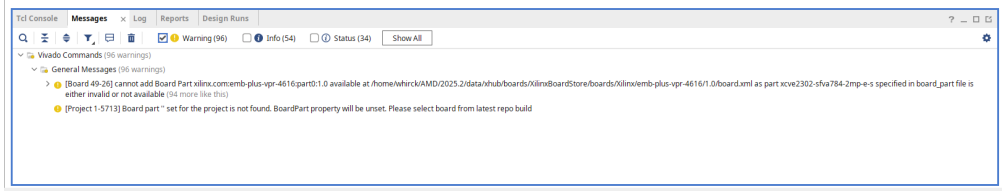


Figure 11: Synthesis Errors of RegFile.sv

Verification

All test cases passed in RegFile_tb.sv 25 passed in simulation along with each subcomponent test bench.

SystemVerilog Code

See 24 for the RegFile.sv.

0.1.4 HW4 - ALU + ImmGen

ALU

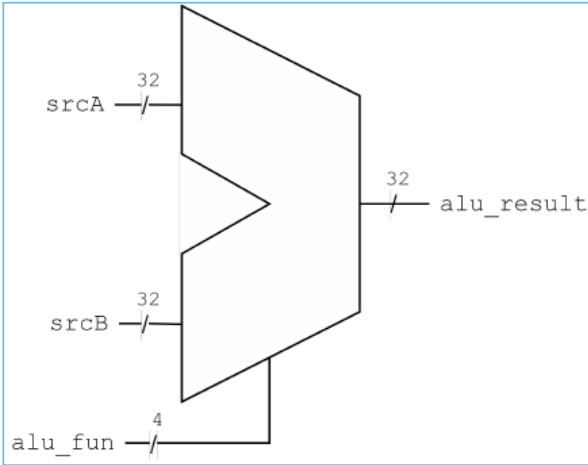


Figure 12: Top Level ALU

The ALU is designed to perform the arithmetic and logic operations required by the RISC-V CPU. Note the 11 we implemented here.

ALU_FUN	Instruction
0000	ADD
1000	SUB
0110	OR
0111	AND
0100	XOR
0101	SRL

ALU_FUN	Instruction
0001	SLL
1101	SRA
0010	SLT
0011	SLTU
1001	LUI-COPY

Figure 13: Instruction Operations of the ALU

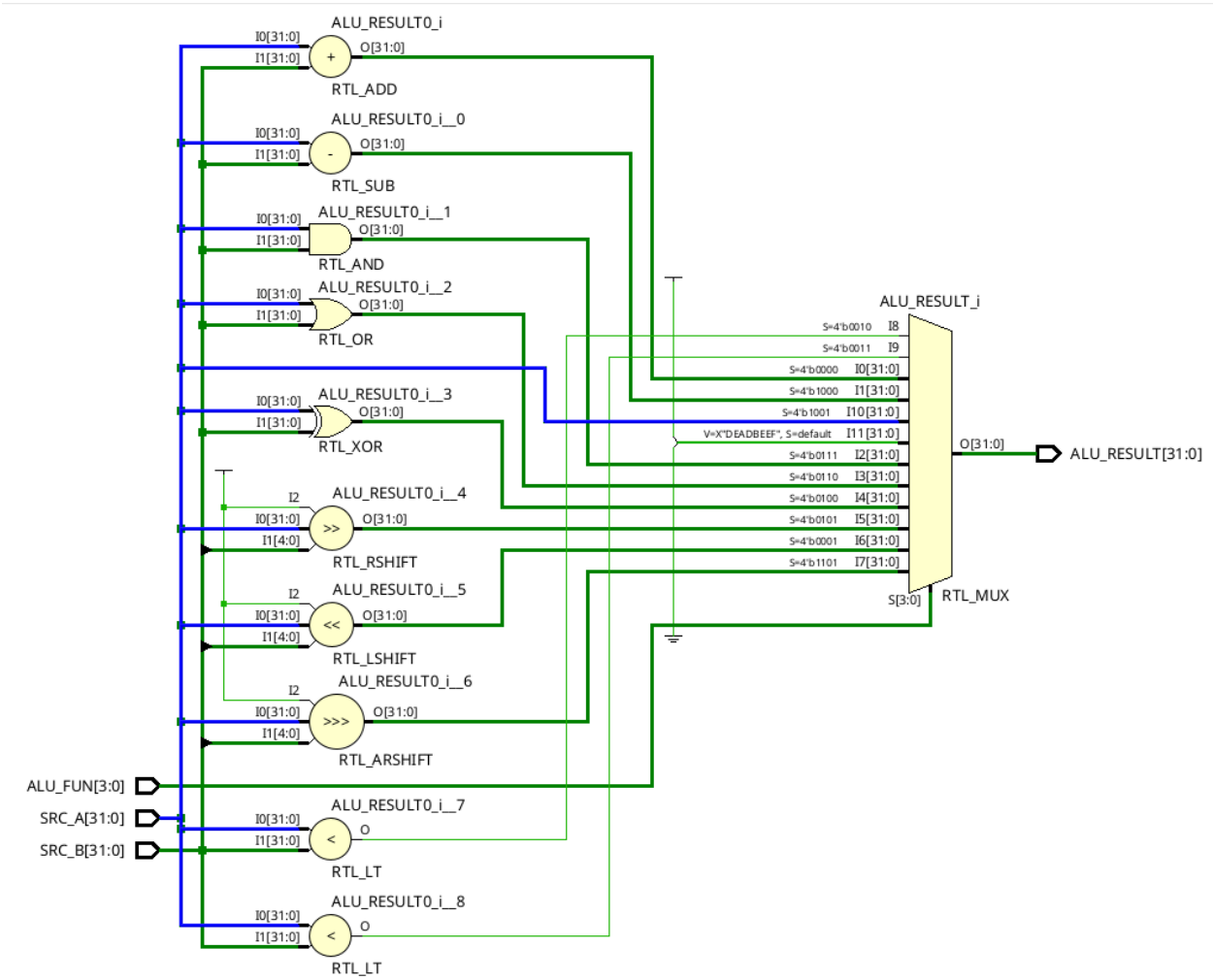


Figure 14: RTL Design of the ALU

Function	ALU_FUN	Arg A	Arg B	Expected Result
ADD	0x0000	0xA50F_96C3	0x5AF0_693C	0xFFFF_FFFF
		0x8410_5F21	0x7B10_5FDE	0xFF20_BEFF
		0xFFFF_FFFF	0x0000_0001	0x0000_0000
SUB	0x1000	0x0000_0000	0x0000_0001	0xFFFF_FFFF
		0xAA80_6355	0x5501_62AA	0x557F_00AB
		0x5501_62AA	0xAA80_6355	0xAA80_FF55
AND	0x0111	0xA55A_00FF	0x5A5A_FFFF	0x005A_00FF
		0xC3C3_F966	0xFF66_9F5A	0xC342_9942
OR	0x0110	0x9A9A_C300	0x65A3_CC0F	0xFFBB_CF0F
		0xC3C3_F966	0xFF66_9F5A	0xFFE7_FF7E
XOR	0x0100	0xAA55_00FF	0x5AA5_0FF0	0xF0F0_0F0F
		0xA5A5_6C6C	0xFF00_C6FF	0x5AA5_AA93
SRL	0x0101	0x805A_6CF3	0x0000_0010	0x0000_805A
		0x705A_6CF3	0x0000_0005	0x0382_D367
		0x805A_6CF3	0x0000_0000	0x805A_6CF3
		0x805A_6CF3	0x0000_0100	0x805A_6CF3
SLL	0x0001	0x805A_6CF3	0x0000_0010	0x6CF3_0000
		0x805A_6CF3	0x0000_0005	0x0B4D_9E60
		0x805A_6CF3	0x0000_0100	0x805A_6CF3
SRA	0x1101	0x805A_6CF3	0x0000_0010	0xFFFF_805A
		0x705A_6CF3	0x0000_0005	0x0382_D367
		0x805A_6CF3	0x0000_0000	0x805A_6CF3
		0x805A_6CF3	0x0000_0100	0x805A_6CF3
SLT	0x0010	0x7FFF_FFFF	0x8000_0000	0x0000_0000
		0x8000_0000	0x0000_0001	0x0000_0001
		0x0000_0000	0x0000_0000	0x0000_0000
		0x5555_5555	0x5555_5555	0x0000_0000
SLTU	0x0011	0x7FFF_FFFF	0x8000_0000	0x0000_0001
		0x8000_0000	0x0000_0001	0x0000_0000
		0x0000_0000	0x0000_0000	0x0000_0000
		0x55AA_55AA	0x55AA_55AA	0x0000_0000
LUI COPY (MV?)	0x1001	0x0123_4567	0x7654_3210	0x0123_4567
		0xFEDC_BA98	0x89AB_CDEF	0xFEDC_BA98

Table 2: ALU Required Test Cases for Verification

See `ALU.sv` 26 and `ALU_tb.sv` 27 for passing simulation test cases from table ??.

Immediate Generator

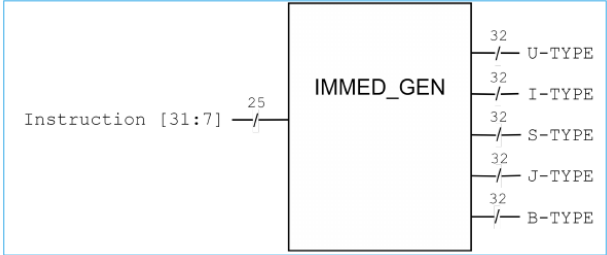


Figure 15: Top Level ImmedGen

The Immediate Generator is a feed though bit parser for different assembly instructions, continuously out-putting all types per instruction.

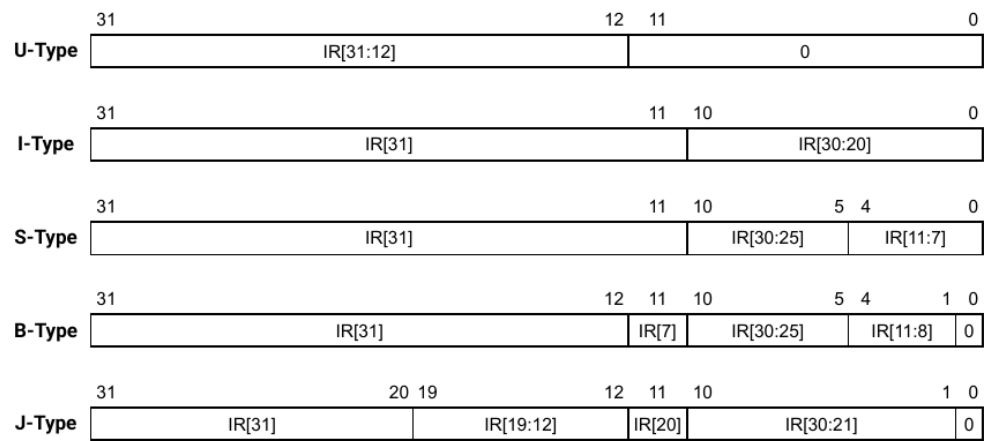


Figure 16: Instruction types of the Immediate Generator

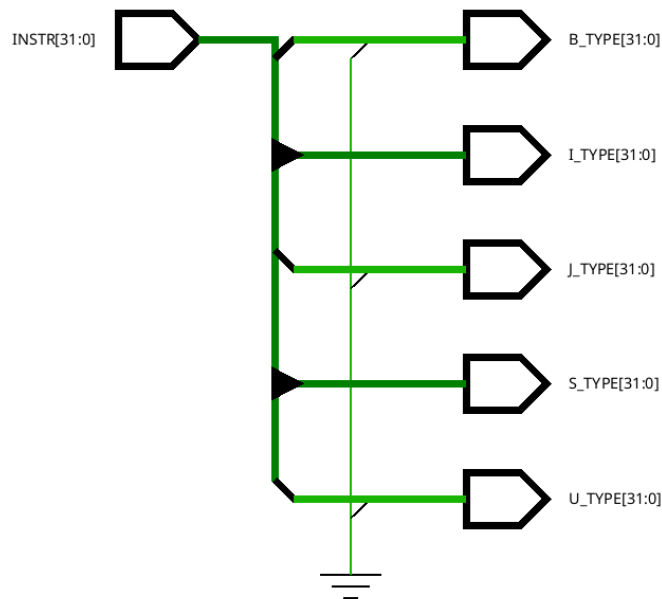


Figure 17: RTL Design of the ImmedGen

See `ImmedGen.sv` [28](#) and `ImmedGen_tb.sv` [29](#) for passing simulation test cases

0.1.5 HW5 - Real Memory and Branch

Memory

The memory segment is designed to dual synch output an instruction and a data value with architecture for pulling data from the MMIO addresses. They are on separate addresses physically such that they are not on the stack but rather beyond what could be the heap (if we implement one).

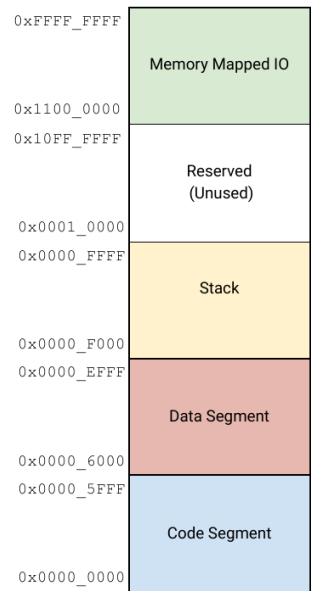


Figure 5.1: OTTER Memory Map

Figure 18: Memory Map for OTTER

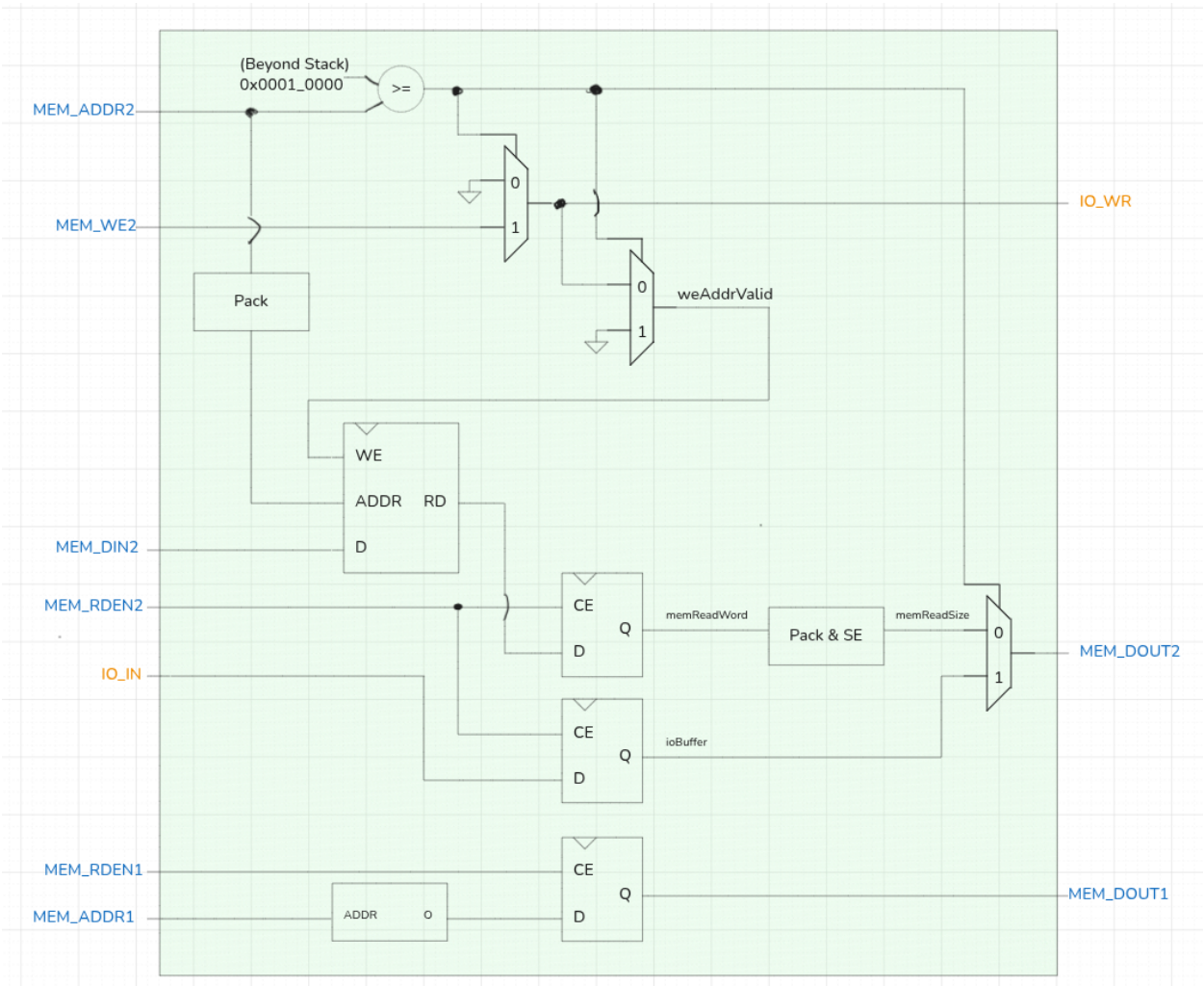


Figure 19: **ROUGH** Flow Diagram for Memory

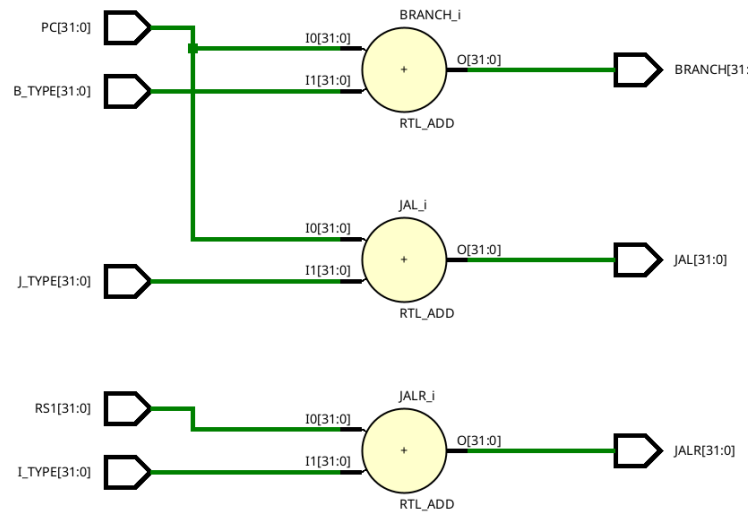


Figure 22: Lint RTL Schematic of BAG

See [BranchAddrGen.sv](#) [31](#) and sim test [BranchAddrGen_tb.sv](#) [32](#).

0.1.6 HW6 - Control Unit

Data and control mapping is different for each type of instruction. Therefore, I've provided the various configurations per instruction type below.

op	funct3	funct7	Type	Instruction	Description	Operation
0000011 (3)	000	-	I	lb rd, imm(rs1)	load byte	rd ← SignExt([Address] _{7:0})
0000011 (3)	001	-	I	lh rd, imm(rs1)	load half	rd ← SignExt([Address] _{15:0})
0000011 (3)	010	-	I	lw rd, imm(rs1)	load word	rd ← [Address] _{31:0}
0000011 (3)	100	-	I	lbu rd, imm(rs1)	load byte unsigned	rd ← ZeroExt([Address] _{7:0})
0000011 (3)	101	-	I	lhu rd, imm(rs1)	load half unsigned	rd ← ZeroExt([Address] _{15:0})

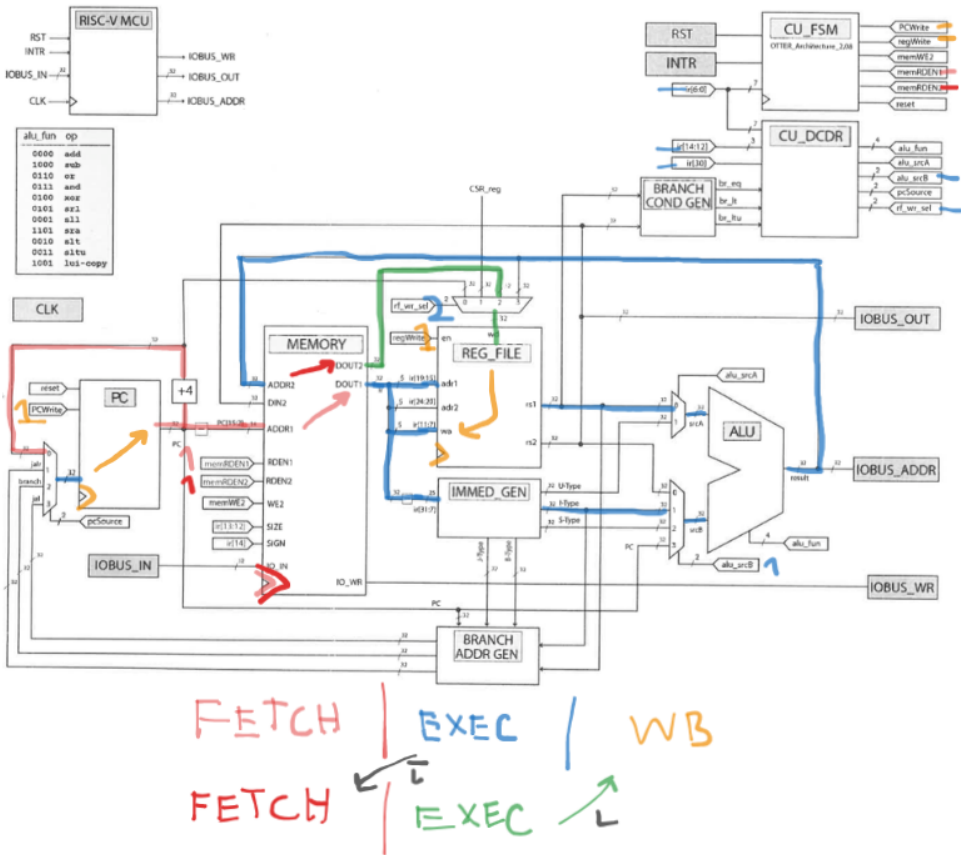


Figure 23: MultiCycle Otter Pathing for load

0010011 (19)	000	-	I	addi rd, rs1, imm	add immediate	$rd = rs1 + \text{SignExt}(imm)$
0010011 (19)	001	0000000	I	slli rd, rs1, imm	shift left logical immediate	$rd = rs1 \ll imm$
0010011 (19)	010	-	I	slti rd, rs1, imm	set less than immediate	$rd = (rs1 < \text{SignExt}(imm))$
0010011 (19)	011	-	I	sltiu rd, rs1, imm	set less than imm. unsigned	$rd = (rs1 < \text{SignExt}(imm))$
0010011 (19)	100	-	I	xori rd, rs1, imm	xor immediate	$rd = rs1 \wedge \text{SignExt}(imm)$
0010011 (19)	101	0000000	I	srlw rd, rs1, imm	shift right logical immediate	$rd = rs1 \gg imm$
0010011 (19)	101	0100000	I	sraiw rd, rs1, imm	shift right arithmetic imm.	$rd = rs1 \ggg imm$
0010011 (19)	110	-	I	ori rd, rs1, imm	or immediate	$rd = rs1 \text{SignExt}(imm)$
0010011 (19)	111	-	I	andi rd, rs1, imm	and immediate	$rd = rs1 \& \text{SignExt}(imm)$

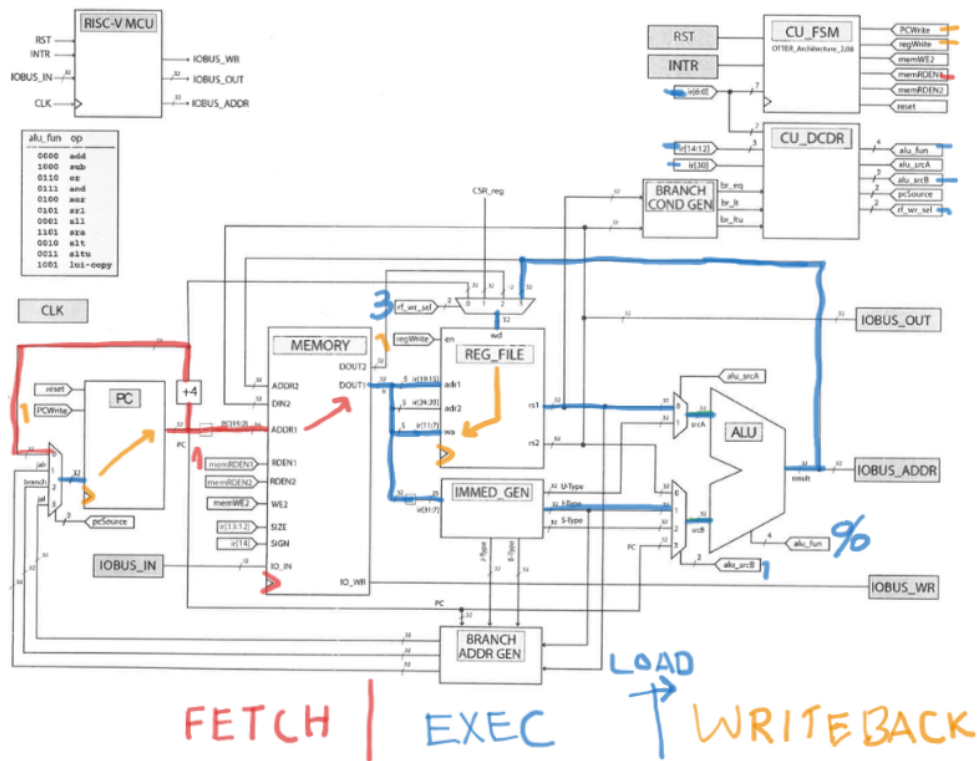


Figure 24: MultiCycle Otter Pathing for immed ops

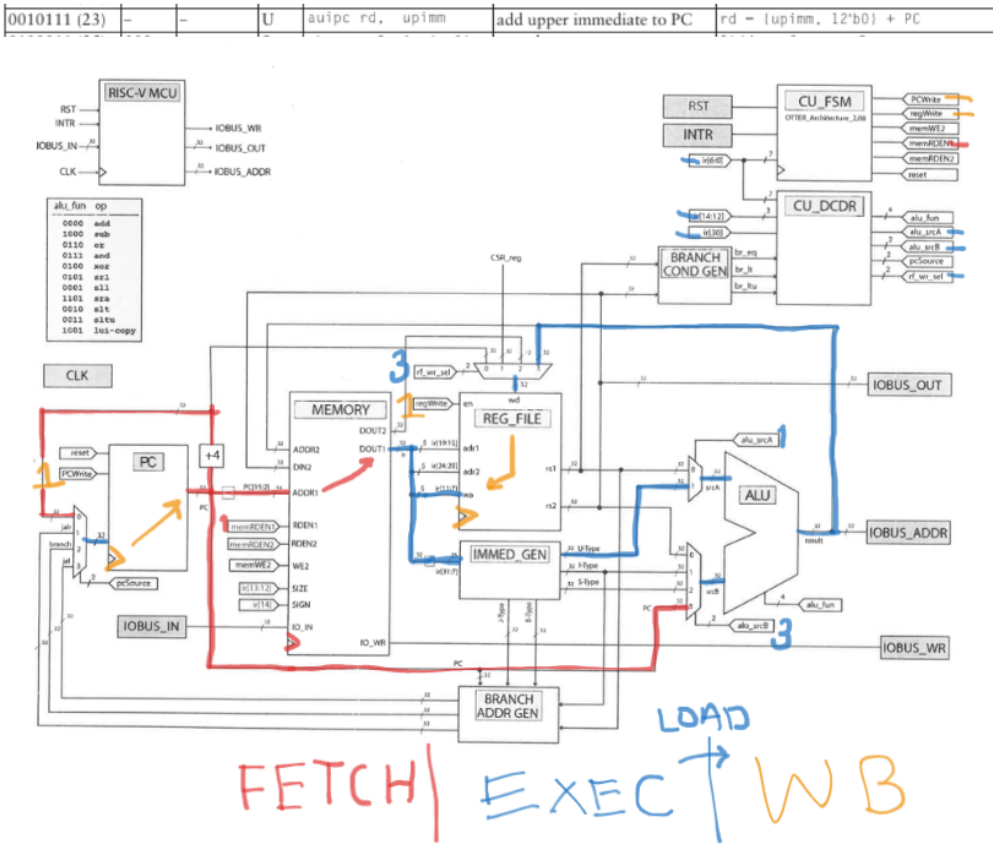


Figure 25: MultiCycle Otter Pathing for auipc

0100011 (35)	000	—	S	sb	rs2, imm(rs1)	store byte	[Address] _{7:0} = rs2 _{7:0}
0100011 (35)	001	—	S	sh	rs2, imm(rs1)	store half	[Address] _{15:0} = rs2 _{15:0}
0100011 (35)	010	—	S	sw	rs2, imm(rs1)	store word	[Address] _{31:0} = rs2

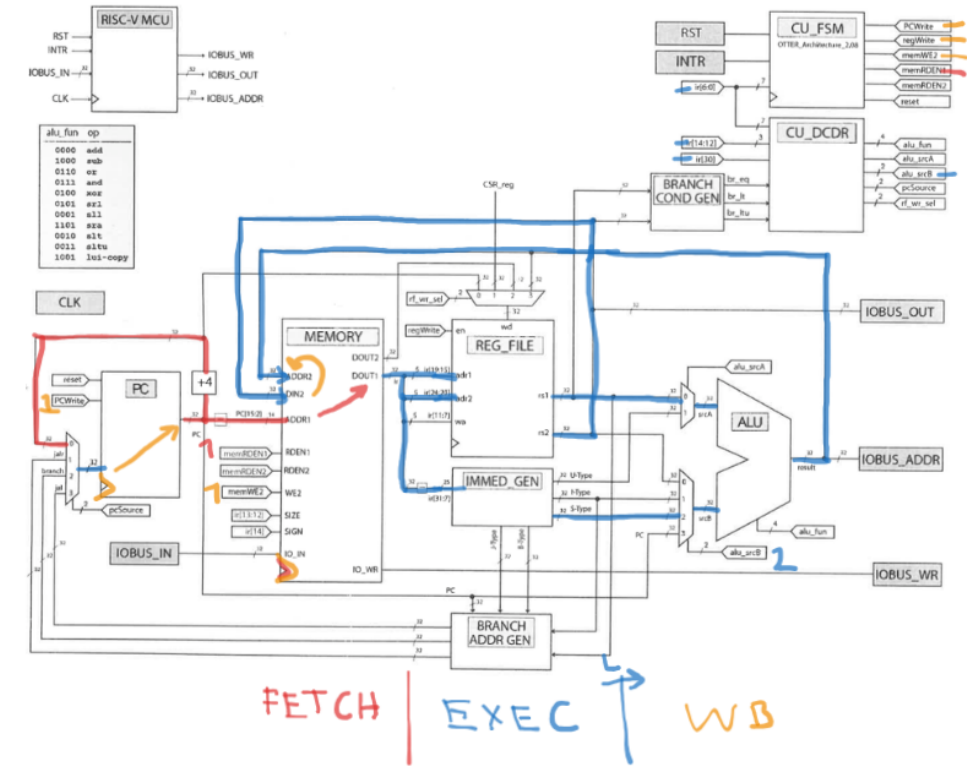


Figure 26: MultiCycle Otter Pathing for store

0110011 (51)	000	0000000	R	add	rd, rs1, rs2	add	rd = rs1 + rs2
0110011 (51)	000	0100000	R	sub	rd, rs1, rs2	sub	rd = rs1 - rs2
0110011 (51)	001	0000000	R	sll	rd, rs1, rs2	shift left logical	rd = rs1 << rs2 _{4:0}
0110011 (51)	010	0000000	R	slt	rd, rs1, rs2	set less than	rd = (rs1 < rs2)
0110011 (51)	011	0000000	R	sltu	rd, rs1, rs2	set less than unsigned	rd = (rs1 < rs2)
0110011 (51)	100	0000000	R	xor	rd, rs1, rs2	xor	rd = rs1 ^ rs2
0110011 (51)	101	0000000	R	srl	rd, rs1, rs2	shift right logical	rd = rs1 >> rs2 _{4:0}
0110011 (51)	101	0100000	R	sra	rd, rs1, rs2	shift right arithmetic	rd = rs1 >>> rs2 _{4:0}
0110011 (51)	110	0000000	R	or	rd, rs1, rs2	or	rd = rs1 rs2
0110011 (51)	111	0000000	R	and	rd, rs1, rs2	and	rd = rs1 & rs2

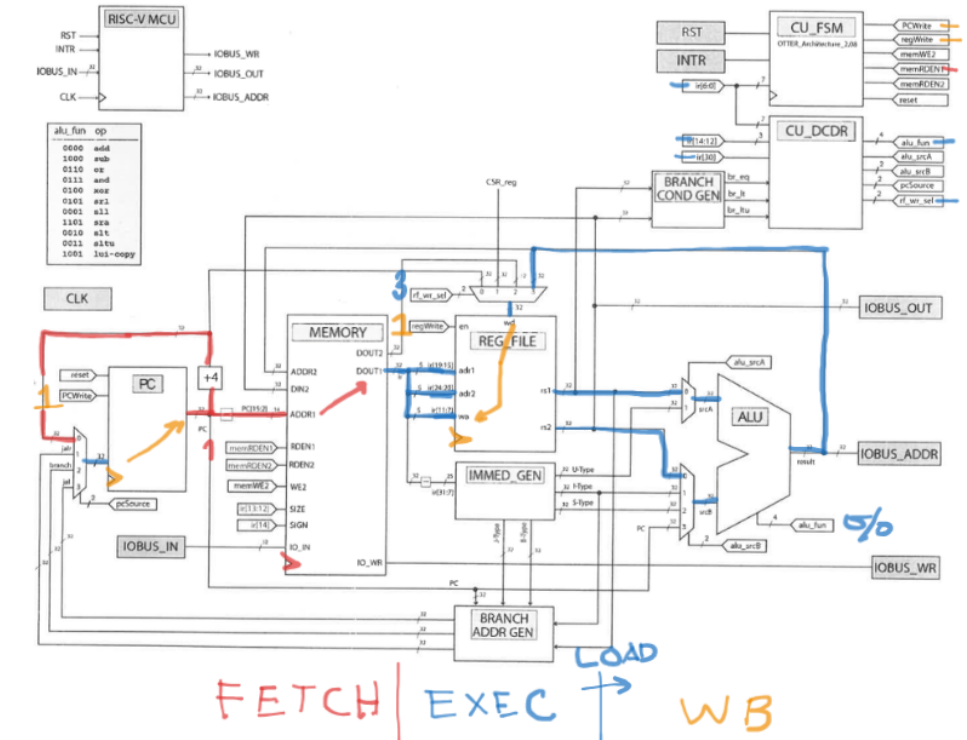


Figure 27: MultiCycle Otter Pathing for dual register ops

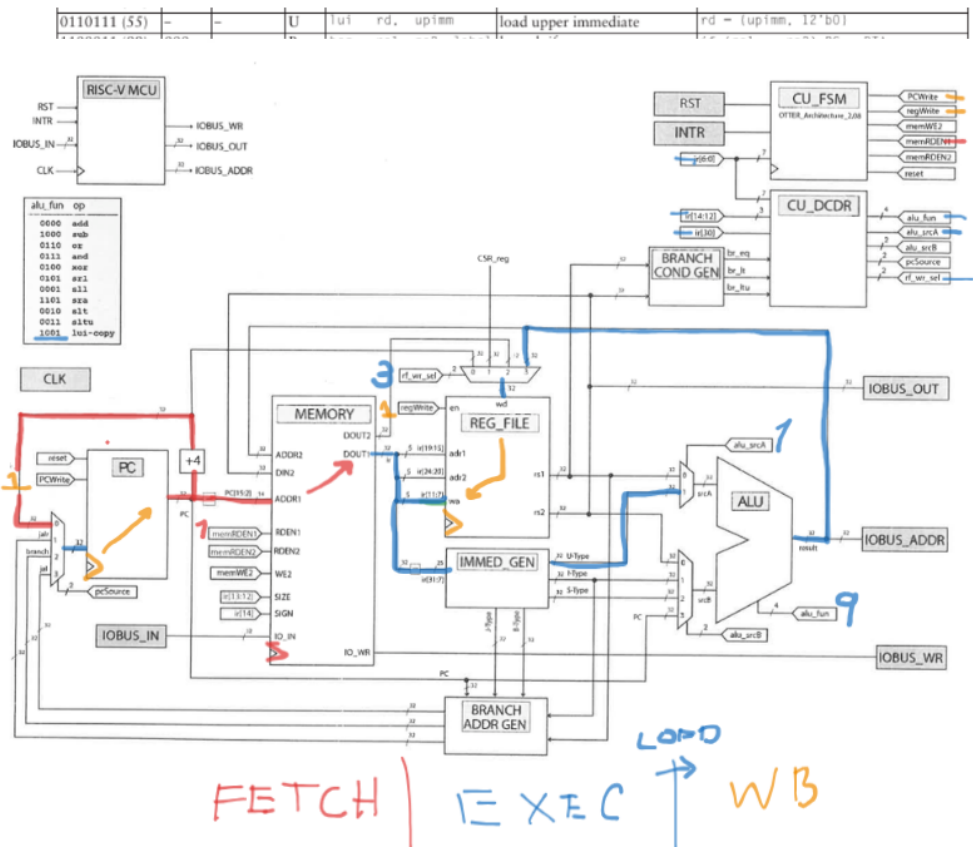


Figure 28: MultiCycle Otter Pathing for lui

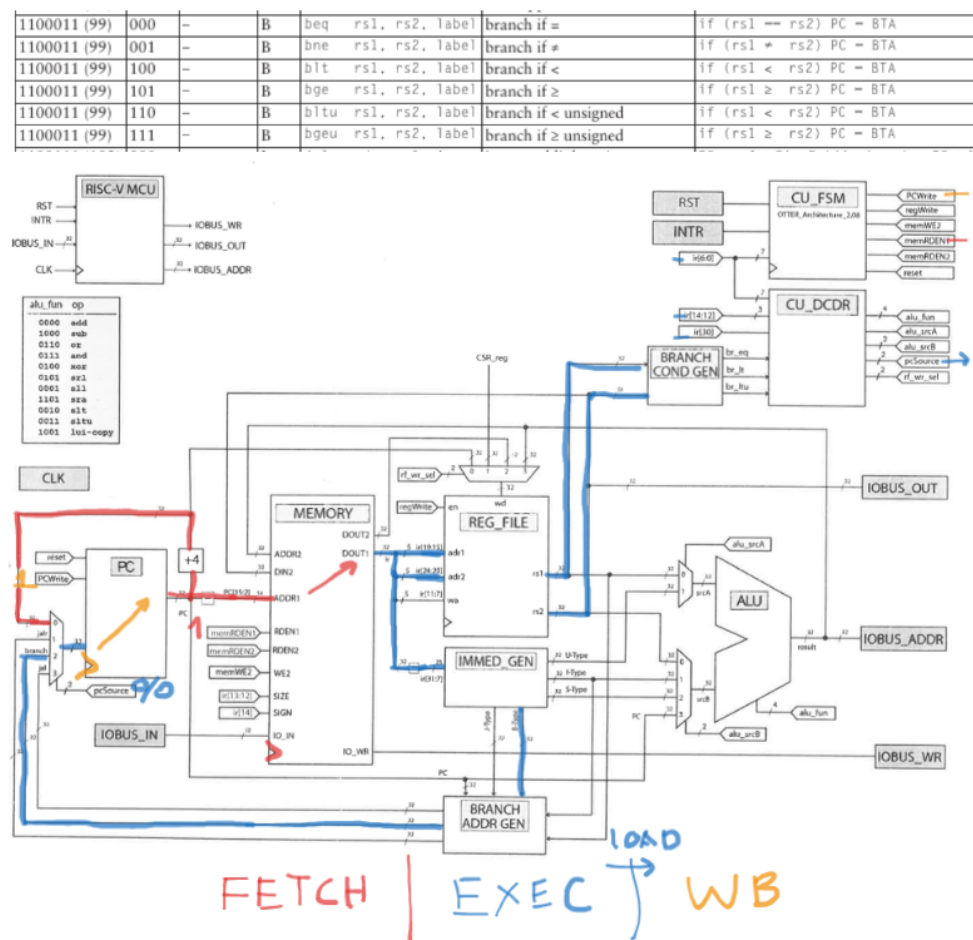


Figure 29: MultiCycle Otter Pathing for branch

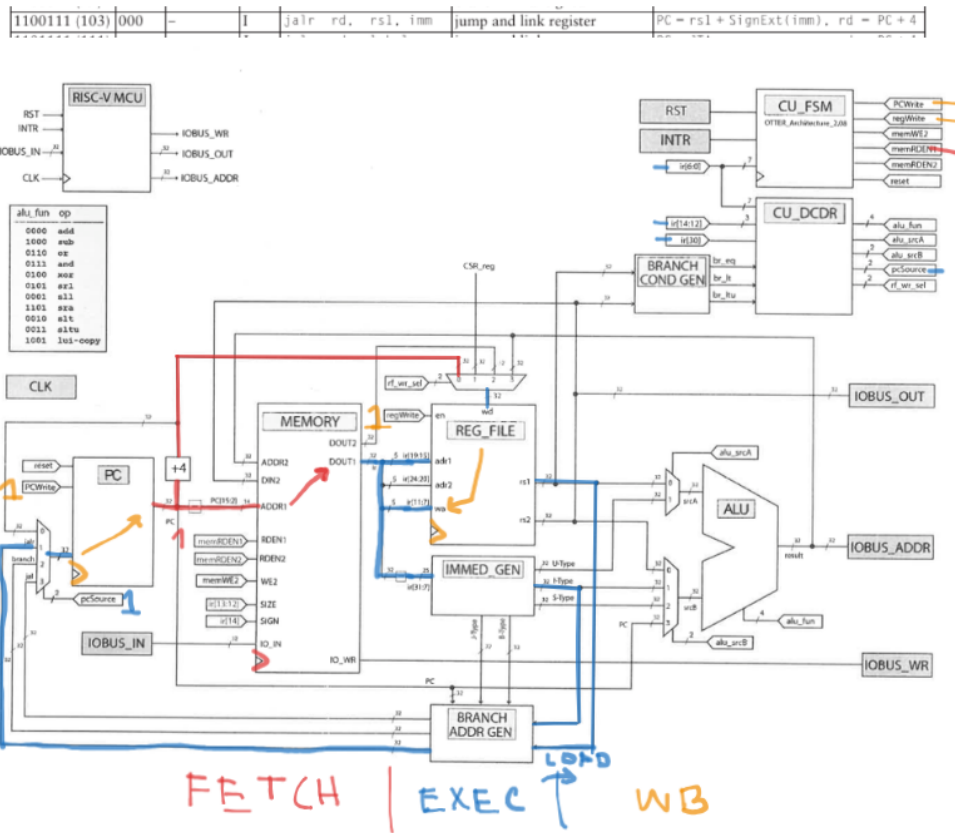


Figure 30: MultiCycle Otter Pathing for `jalr`

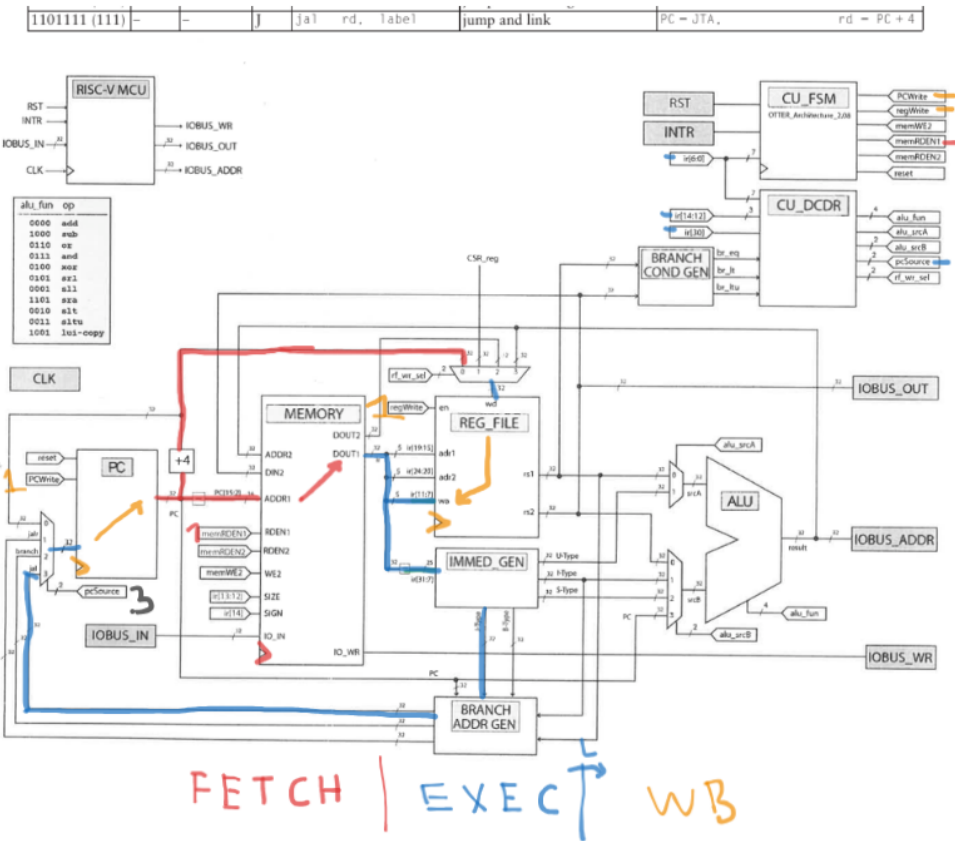


Figure 31: MultiCycle Otter Pathing for `jal`

Control Unit Decoder

After mapping the data and control signals, we need a way to control data flowing into the ALU, register file, program counter, as well as the ALU itself. We depend on the Decoder to do this, writing to the selectors for each mux. Note that the decoder can only make an intelligent switch during and after the EXEC cycle s.t. the opcode is available to process.

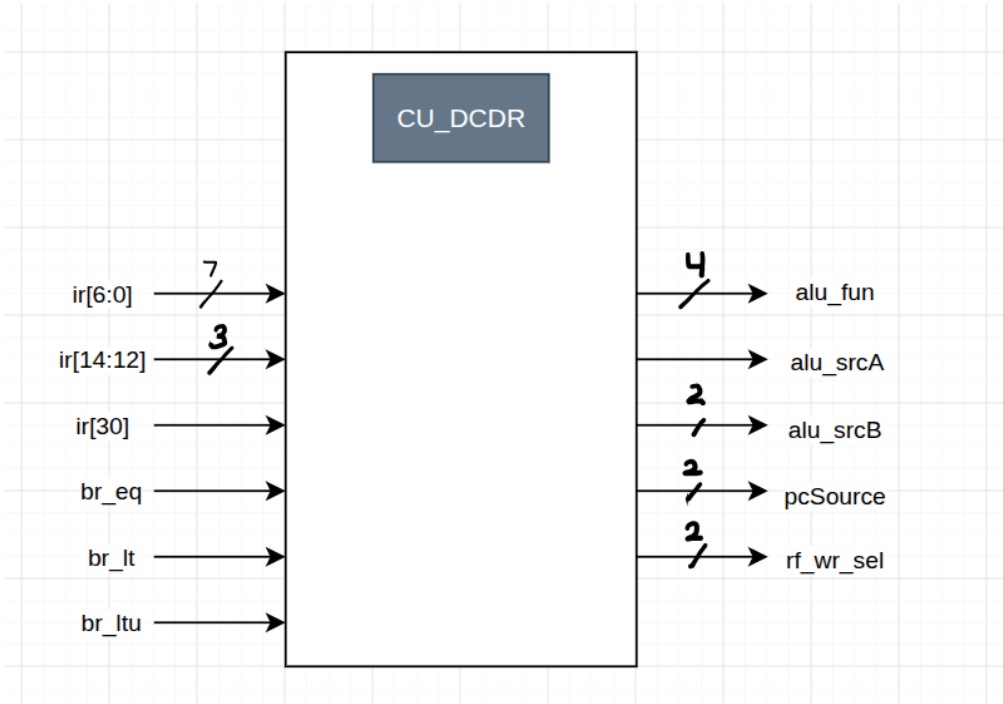


Figure 32: Control Unit Decoder Top Level

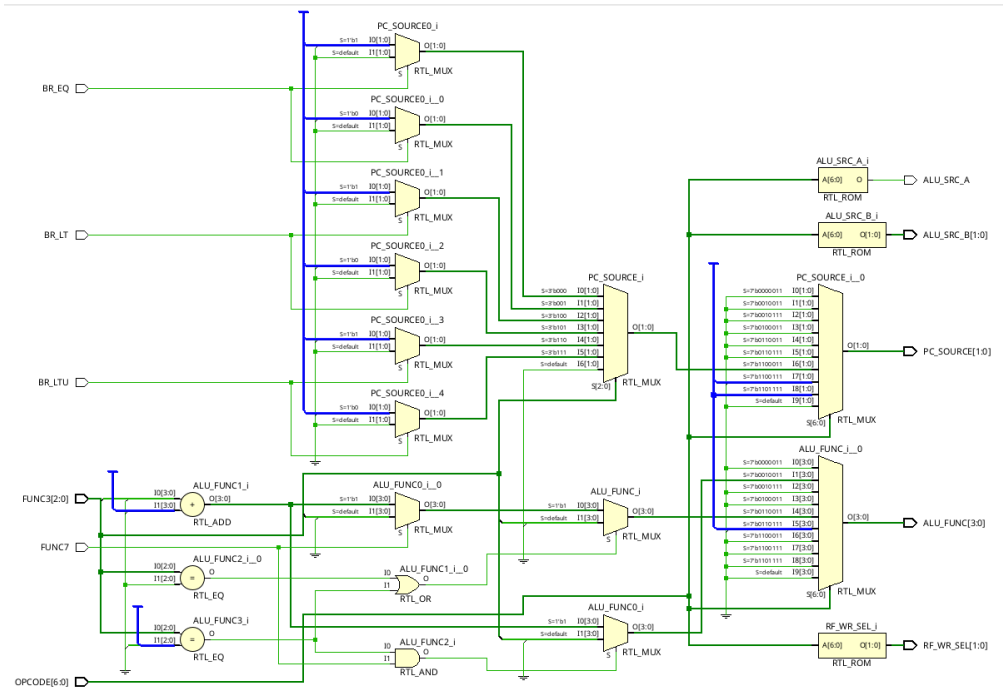


Figure 33: Control Unit Decoder RTL

See CUnit_Decoder.sv [35](#) and sim test cases CUnit_Decoder_tb.sv [36](#).

Control Unit FSM

With the decoder controlling the data/mux selectors, we need a module to issue the control signals for read/writing. The FSM allows us to do just that, toggling read signals for memory and PC while also controlling when to write contents in memory and registers. We’ve confined our FSM to 4 stages (INIT, FETCH, EXEC, WRITEBACK).

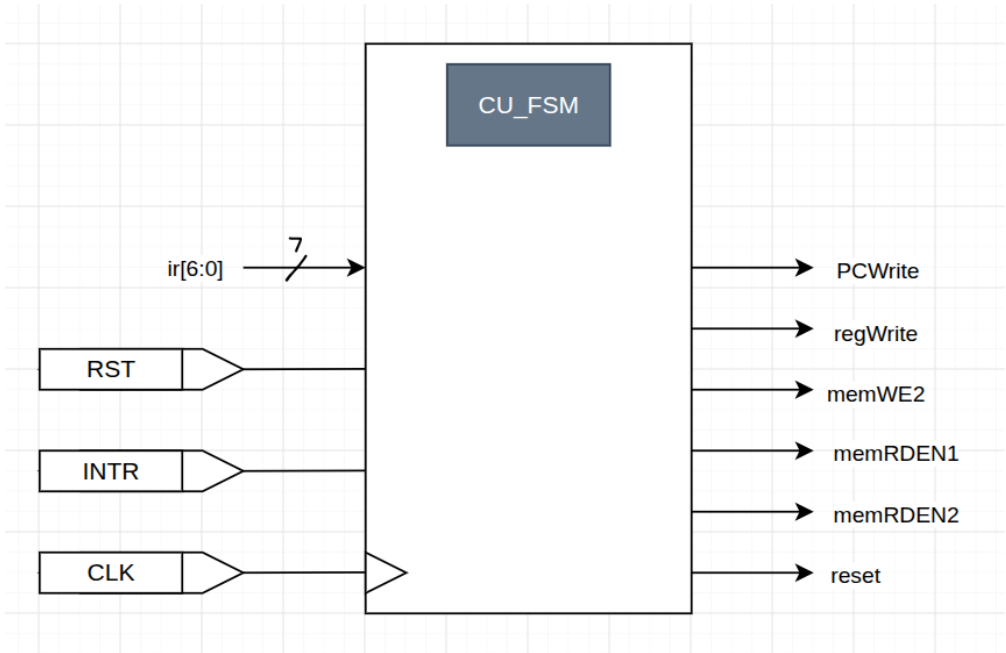


Figure 34: Control Unit FSM Top Level

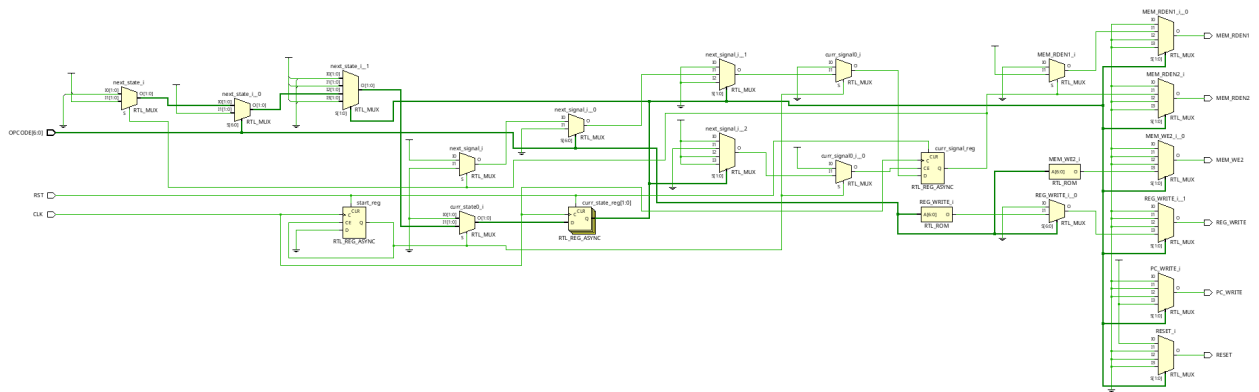


Figure 35: Control Unit FSM RTL

Stages :

- **INIT** : stage upon startup or reset that zeros out all signals.
- **FETCH** : PC is available and this stage processes the contents at that PC address, tucked away inside the memory module. By the end, DOUT1 should be available (and maybe DOUT2 depending on the instr).
- **EXEC** : stage is used to give instr operations runway to compute and arrive to their respective mux channels.
- **WRITEBACK** : stage is used to overwrite a desired PC, memory, or register.

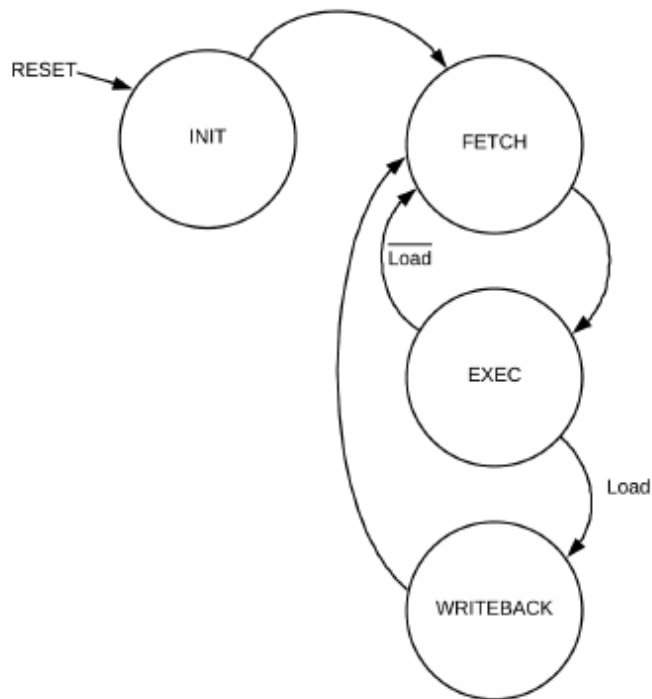


Figure 36: FSM Diagram

See CUnit_FSM.sv 37 and sim test cases CUnit_FSM_tb.sv 38.

Otter Micro-Controller

At last, we’ve reached a point where we can combine all created modules from HW2-6 and begin to process memory files with RISCv instructions.

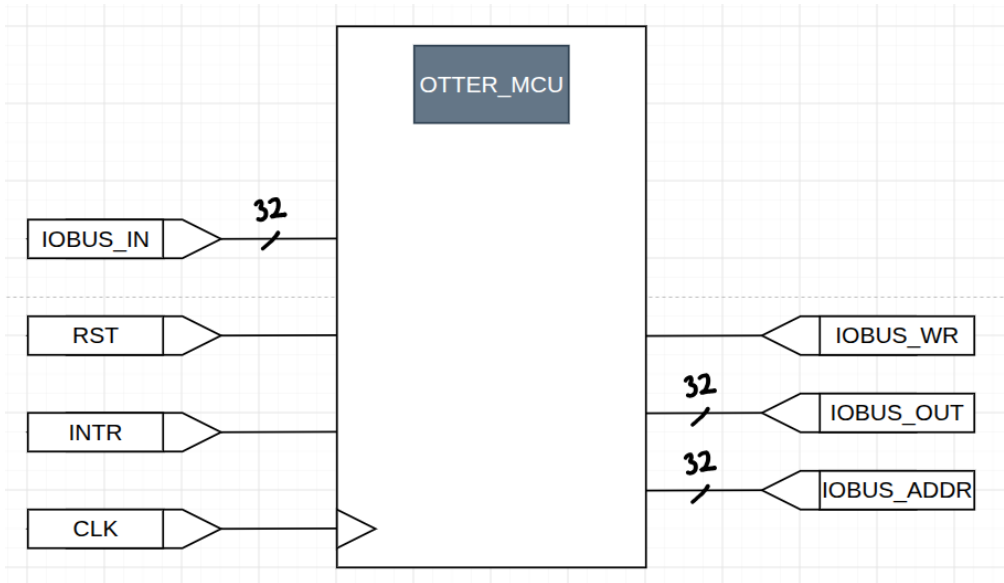


Figure 37: OTTER MCU : Non-Interrupt

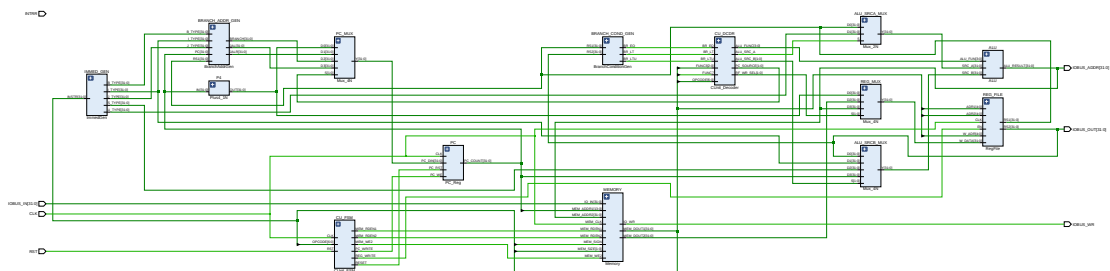


Figure 38: MCU RTL

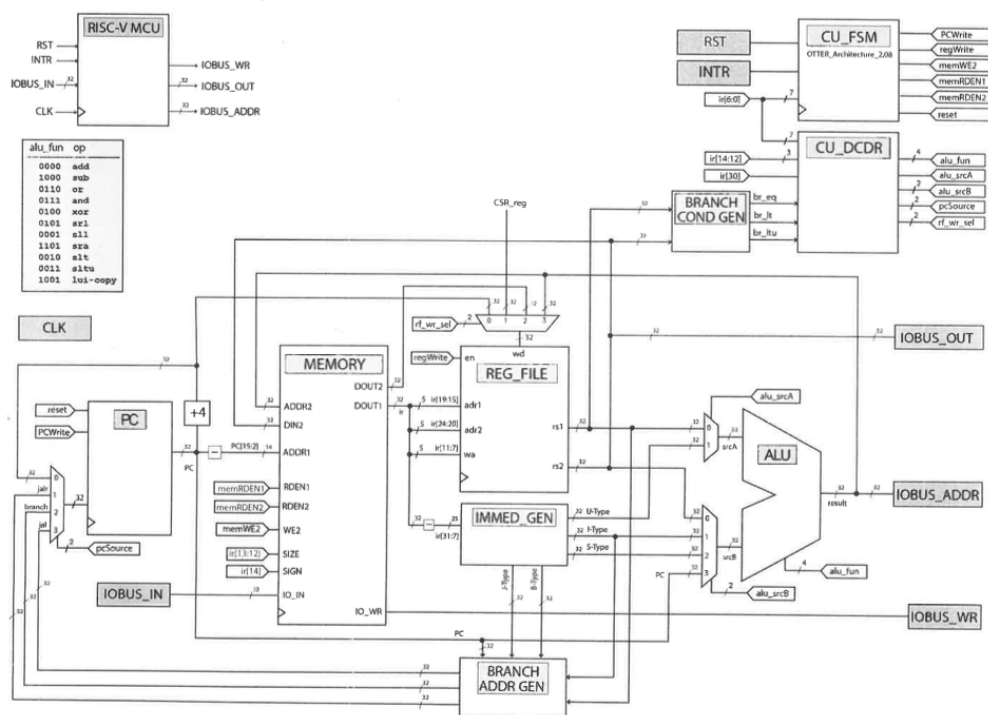


Figure 39: Non-Interrupt MultiCycle RISCv Otter

To perform a small validation for the MCU, a small assembly file was created, compiled under RARS tool for RISC-V compatibility, and the memory dump (`otter_hw6sample_test.mem`) injected into the memory portion of the OTTER MCU Memory.

Listing 4: Simple Test Script

```

1 # Author : WhisperingRock
2 #
3 # Purpose : machine code dump for which to compare results on OTTER MCU
4 #
5 # Input :
6 #
7 # Output :
8
9 main:
10     lui     x5,      0xAA055          # x5 = 0xAA05_5000
11     addi    x8,      x5,      0x765    # x8 = 0xAA05_5000 + 0x0000_0765 = 0xAA05_5765
12     slll    x10,     x8,      3        # x10 = 0x502A_BB28
13     slt     x12,     x5,      x8       # x12 = 1
14     xor     x13,     x8,      x10      # x13 = 0xFA2F_EC4D
15     beq     x0,      x0,      main     # 0 == 0, return to main

```

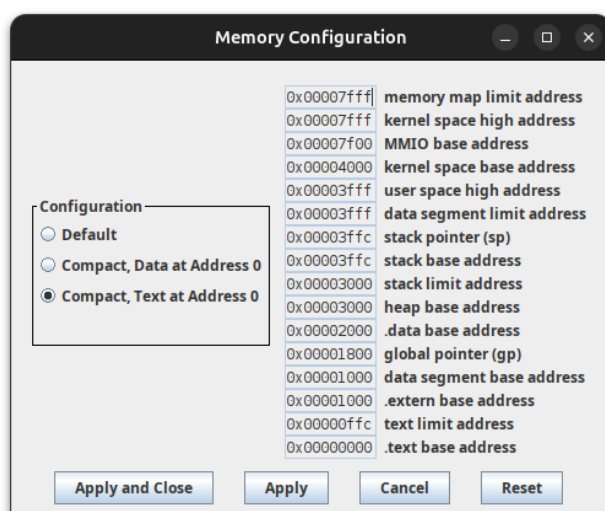


Figure 40: Memory config for RARS

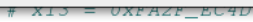


Figure 41: Memory dump config for RARS

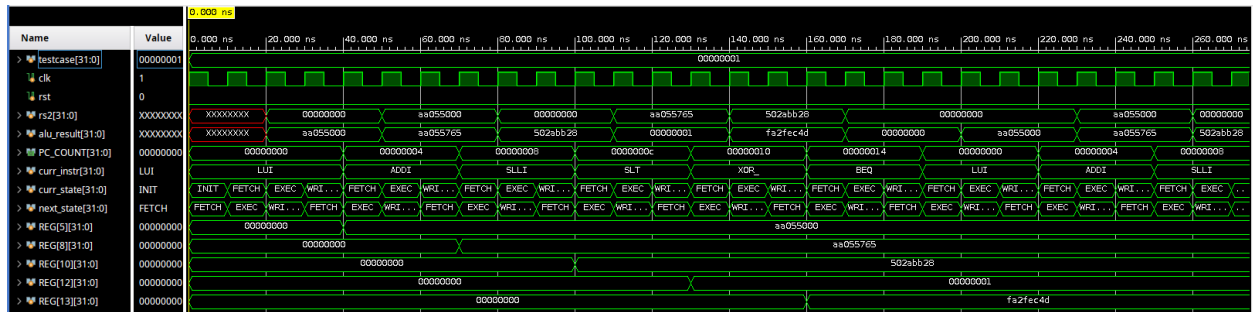


Figure 42: Verification Waveforms in simulation testing

See `Otter_MCU.sv` 39 and sim test cases `Otter_MCU_tb.sv` 40.

0.2 Software

0.2.1 SW1 - Intro to Assembly Lang

Note : Assume caller/callee does not need to preserve their respective registers in order to use them. KISS method applied for assignment 1.

Part 1

"Read three 16-bit unsigned values consecutively from SWITCHES (address 0x11000000), add them together, and output the result to LEDS (address 0x11000020). Assume the input values are 16-bit unsigned values. (Assume it is possible for the switches to change between each time they are read)"

Assuming the LEDS input value is limited to 16-bit unsigned as well.

Flowchart

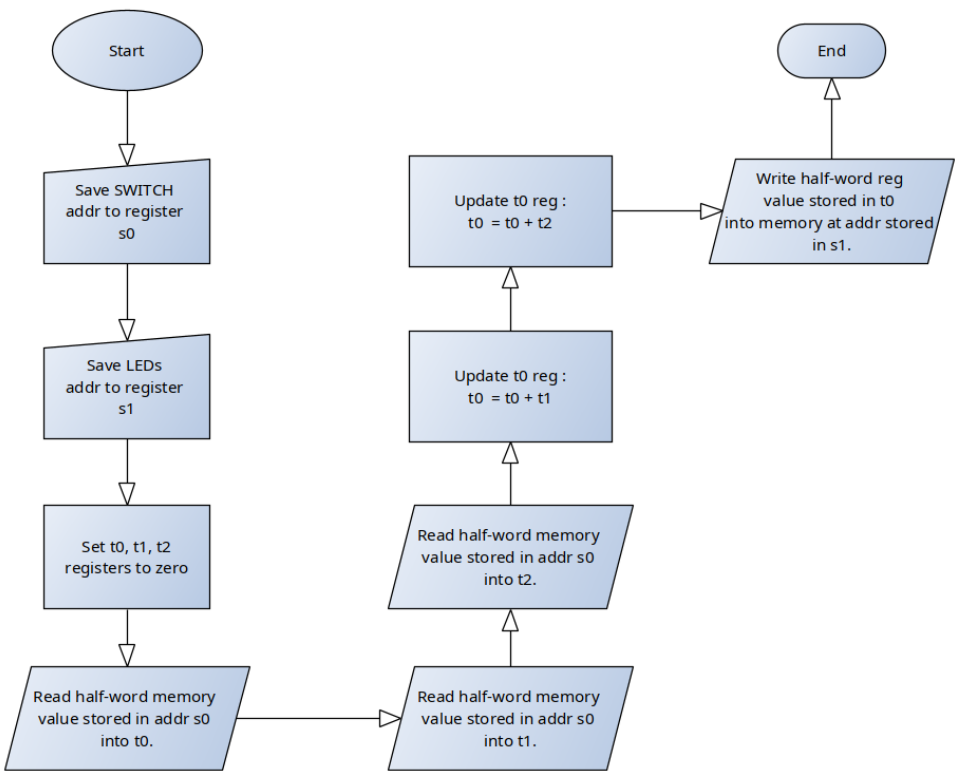


Figure 43: Flow Chart for Part 1

Verification Simulations

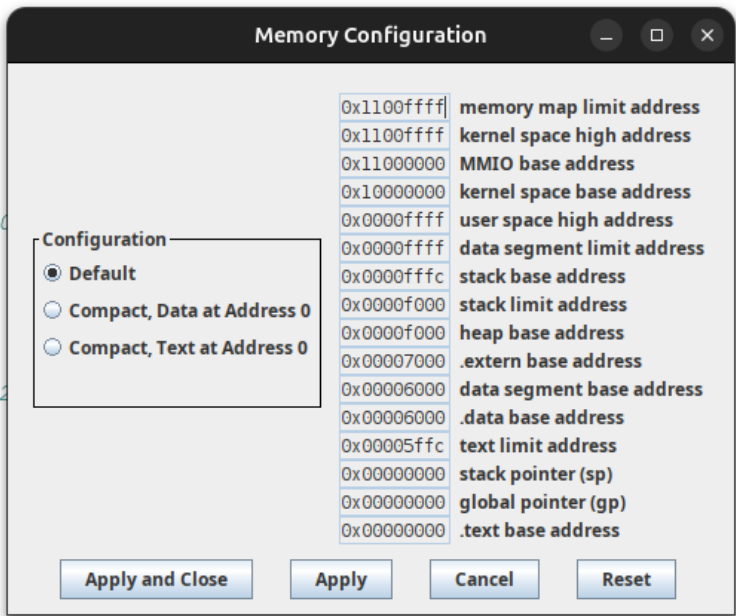


Figure 44: RARS Memory Configuration

```

20
21 #.equ SWITCHES_H,      0x11000 # upper 20 bits
22 #.equ SWITCHES_L,      0x000   # lower 12 bits
23 .equ SWITCHES_H,       0x00006 # upper 20 bits (data seg at 0x00006000)
24 .equ SWITCHES_L,       0x000   # lower 12 bits
25
26 #.equ LEDS_H,          0x11000 # upper 20 bits
27 #.equ LEDS_L,          0x020   # lower 12 bits
28 .equ LEDS_H,           0x00006 # upper 20 bits (data seg at 0x00006020)
29 .equ LEDS_L,           0x020   # lower 12 bits
30

```

Figure 45: Address Correction to work with RARS

- Test Case 1

```

33 # TC1 : Word Clip
34 .equ READ1_H,          0x00001 # switch first read (0x1234)
35 .equ READ1_L,          0x234
36
37
38 .equ READ2_H,          0x12345 # switch second read (0x5678)
39 .equ READ2_L,          0x678
40
41
42 .equ READ3_H,          0x11111 # switch third read (0x1111)
43 .equ READ3_L,          0x111
44 #   answer= 0x79BD

```

Figure 46: TC1 given inputs

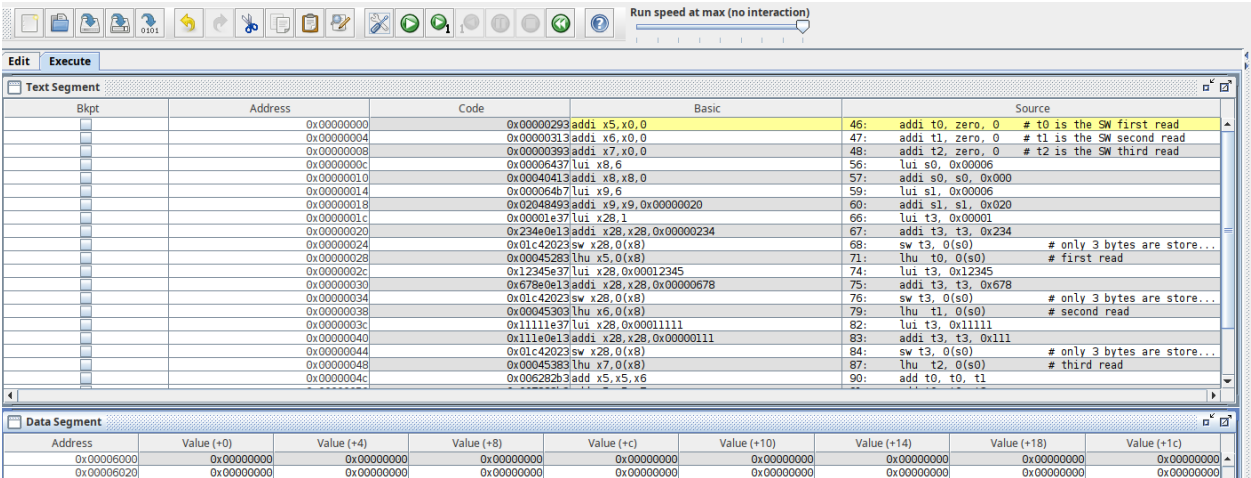


Figure 47: Data segment before executing instructions

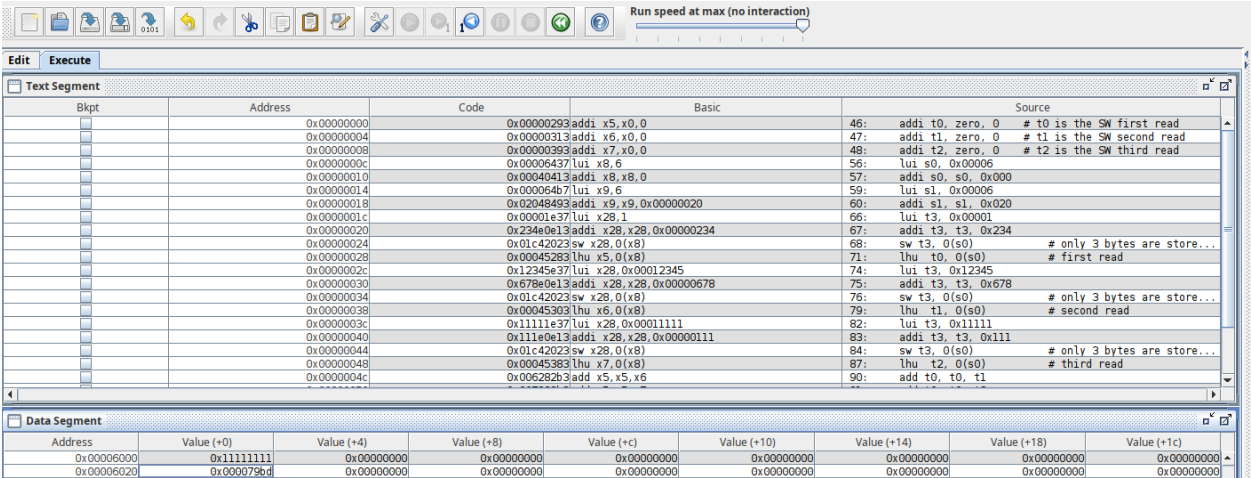


Figure 48: Data segment after executing instructions

Result at LEDS address in 48 matches the answer in 46.

- Test Case 2

```

46 # TC2 : Maxed
47 .equ READ1_H,      0x00005 # switch first read (0x5555)
48 .equ READ1_L,      0x555
49
50 .equ READ2_H,      0x00005 # switch second read (0x5555)
51 .equ READ2_L,      0x555 # sign extended
52
53
54 .equ READ3_H,      0x00005 # switch third read (0x5555)
55 .equ READ3_L,      0x555
56 # answer= 0xFFFF

```

Figure 49: TC2 given input

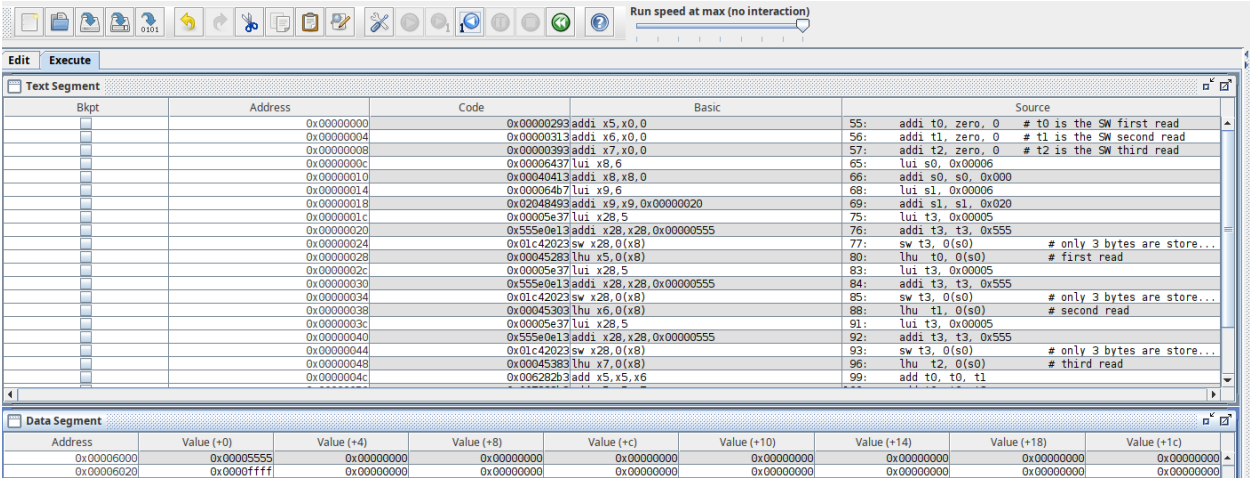


Figure 50: Data segment after executing instructions

Result at LEDS address in 50 matches the answer in 49.

- Test Case 3

```

59 # TC3 : Overflow
60 .equ READ1_H,      0x00005 # switch first read (0x5555)
61 .equ READ1_L,      0x555
62
63 .equ READ2_H,      0x00005 # switch second read (0x5555)
64 .equ READ2_L,      0x555 # sign extended
65
66
67 .equ READ3_H,      0x00005 # switch third read (0x5557)
68 .equ READ3_L,      0x557
69 # answer= 0x0001

```

Figure 51: TC3 given input

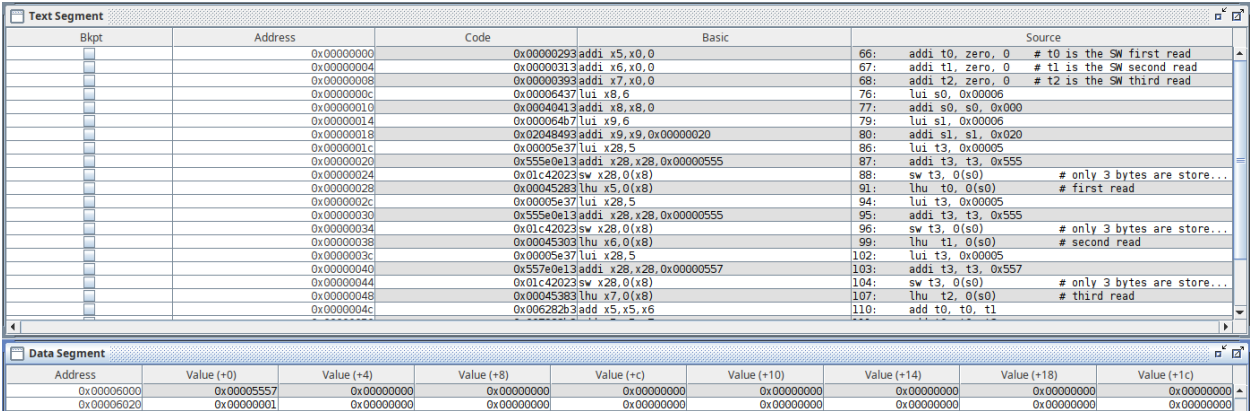


Figure 52: Data segment after executing instructions

Result at LEDS address in 52 matches the answer in 51.

- Test Case 4

```

71 # TC4 : zero
72 .equ READ1_H,      0x00000 # switch first read (0)
73 .equ READ1_L,      0x0
74
75
76 .equ READ2_H,      0x00000 # switch second read (0)
77 .equ READ2_L,      0x0 # sign extended
78
79 .equ READ3_H,      0x00000 # switch third read (0)
80 .equ READ3_L,      0x0
81 #   answer= 0x0000
82

```

Figure 53: TC4 given input

Text Segment									
Bkpt	Address	Code	Basic	Source					
	0x00000000	0x0000293	addi x5,x0,0	77:	addi t0, zero, 0	# t0 is the SW first read			
	0x00000004	0x0000313	addi x6,x0,0	78:	addi t1, zero, 0	# t1 is the SW second read			
	0x00000008	0x0000393	addi x7,x0,0	79:	addi t2, zero, 0	# t2 is the SW third read			
	0x0000000c	0x00006437	lui x8,6	87:	lui s0, 0x00006				
	0x00000010	0x0000413	addi x8,x8,0	88:	addi s0, s0, 0x000				
	0x00000014	0x000064b7	lui x9,6	90:	lui s1, 0x00006				
	0x00000018	0x02048493	addi x9,x9,0x00000020	91:	addi s1, s1, 0x020				
	0x0000001c	0x00000e37	lui x28,0	97:	lui t3, 0x00000				
	0x00000020	0x000e0e13	addi x28,x28,0	98:	addi t3, t3, 0x0				
	0x00000024	0x01c42023	sw x28,0(x8)	99:	sw t3, 0(s0)	# only 3 bytes are store...			
	0x00000028	0x00045283	lhu x5,0(x8)	102:	lhu t0, 0(s0)	# first read			
	0x0000002c	0x00000e37	lui x28,0	105:	lui t3, 0x00000				
	0x00000030	0x000e0e13	addi x28,x28,0	106:	addi t3, t3, 0x0				
	0x00000034	0x01c42023	sw x28,0(x8)	107:	sw t3, 0(s0)	# only 3 bytes are store...			
	0x00000038	0x00045303	lhu x6,0(x8)	110:	lhu t1, 0(s0)	# second read			
	0x0000003c	0x00000e37	lui x28,0	113:	lui t3, 0x00000				
	0x00000040	0x000e0e13	addi x28,x28,0	114:	addi t3, t3, 0x0				
	0x00000044	0x01c42023	sw x28,0(x8)	115:	sw t3, 0(s0)	# only 3 bytes are store...			
	0x00000048	0x00045383	lhu x7,0(x8)	118:	lhu t2, 0(s0)	# third read			
	0x0000004c	0x006282b3	add x5,x5,x6	121:	add t0, t0, t1				

Data Segment									
Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)	
0x00006000	0x00000000		0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	
0x00006020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	

Figure 54: Data segment after executing instructions

Result at LEDS address in 54 matches the answer in 53.

- Test Case 5

```

84 # TC5 : MAXXXED
85 .equ READ1_H,      0x0000F # switch first read (0xFFFF)
86 .equ READ1_L,      -1
87
88 .equ READ2_H,      0xFFFFF # switch second read (0xFFFF)
89 .equ READ2_L,      -1
90
91
92 .equ READ3_H,      0x00000 # switch third read (0)
93 .equ READ3_L,      0
94 #   answer= 0xFFFE

```

Figure 55: TC5 given input

Text Segment									
Bkpt	Address	Code	Basic	Source					
	0x00000004	0x0000313	addi x6,x0,0	89:	addi t1, zero, 0	# t1 is the SW second read			
	0x00000008	0x0000393	addi x7,x0,0	90:	addi t2, zero, 0	# t2 is the SW third read			
	0x0000000c	0x00006437	lui x8,6	98:	lui s0, 0x00006				
	0x00000010	0x0000413	addi x8,x8,0	99:	addi s0, s0, 0x000				
	0x00000014	0x000064b7	lui x9,6	101:	lui s1, 0x00006				
	0x00000018	0x02048493	addi x9,x9,0x00000020	102:	addi s1, s1, 0x020				
	0x0000001c	0x0000fe37	lui x28,15	108:	lui t3, 0x0000F				
	0x00000020	0xfffffe13	ori x28,x28,0xffffffff	109:	ori t3, t3, -1				
	0x00000024	0x01c42023	sw x28,0(x8)	110:	sw t3, 0(s0)	# only 3 bytes are store...			
	0x00000028	0x00045283	lhu x5,0(x8)	113:	lhu t0, 0(s0)	# first read			
	0x0000002c	0xfffffe37	lui x28,0xffffffff	116:	lui t3, 0xfffff				
	0x00000030	0xfffffe13	ori x28,x28,0xffffffff	117:	ori t3, t3, -1				
	0x00000034	0x01c42023	sw x28,0(x8)	118:	sw t3, 0(s0)	# only 3 bytes are store...			
	0x00000038	0x00045303	lhu x6,0(x8)	121:	lhu t1, 0(s0)	# second read			
	0x0000003c	0x00000e37	lui x28,0	124:	lui t3, 0x00000				
	0x00000040	0x000e0e13	ori x28,x28,0	125:	ori t3, t3, 0				
	0x00000044	0x01c42023	sw x28,0(x8)	126:	sw t3, 0(s0)	# only 3 bytes are store...			
	0x00000048	0x00045383	lhu x7,0(x8)	129:	lhu t2, 0(s0)	# third read			
	0x0000004c	0x006282b3	add x5,x5,x6	132:	add t0, t0, t1				
	0x00000050	0x007282b3	add x5,x5,x7	133:	add t0, t0, t2				

Data Segment									
Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)	
0x00006000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	
0x00006020	0x0000ffff	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	

Figure 56: Data segment after executing instructions

Result at LEDS address in 56 matches the answer in 55.

- Test Case 6

```

96
97 # TC6 : All highs into full
98 .equ READ1_H,      0x00008 # switch first read
99 .equ READ1_L,      0x000
100
101 .equ READ2_H,      0x00008 # switch second read
102 .equ READ2_L,      0x000
103
104 .equ READ3_H,      0x00008 # switch third read
105 .equ READ3_L,      0x000
106
107 #   answer= 0x8000 (0x18000 as full word though)

```

Figure 57: TC6 given input

Text Segment

Bkpt	Address	Code	Basic	Source
	0x00000000	0x00000293	addi x5,x0,0	99: addi t0, zero, 0 # t0 is the SW first read
	0x00000004	0x00000313	addi x6,x0,0	100: addi t1, zero, 0 # t1 is the SW second read
	0x00000008	0x00000363	addi x7,t0,0	101: addi t2, zero, 0 # t2 is the SW third read
	0x0000000c	0x00006437	lui x8,6	109: lui s0, 0x00006
	0x00000010	0x00040413	addi x8,x8,0	110: addi s0, s0, 0x000
	0x00000014	0x000064b7	lui x9,6	112: lui s1, 0x00006
	0x00000018	0x02048493	addi x9,x9,0x00000020	113: addi s1, s1, 0x020
	0x0000001c	0x00008e37	lui x28,8	119: lui t3, 0x00008
	0x00000020	0x000e6e13	ori x28,x28,0	120: ori t3, t3, 0x000
	0x00000024	0x01c42023	sw x28,0(x8)	121: sw t3, 0(s0) # only 3 bytes are store...
	0x00000028	0x00045283	lhu x5,0(x8)	124: lhu t0, 0(s0) # first read
	0x0000002c	0x00008e37	lui x28,8	127: lui t3, 0x00008
	0x00000030	0x000e6e13	ori x28,x28,0	128: ori t3, t3, 0x000
	0x00000034	0x01c42023	sw x28,0(x8)	129: sw t3, 0(s0) # only 3 bytes are store...
	0x00000038	0x00045303	lhu x6,0(x8)	132: lhu t1, 0(s0) # second read
	0x0000003c	0x00008e37	lui x28,8	135: lui t3, 0x00008
	0x00000040	0x000e6e13	ori x28,x28,0	136: ori t3, t3, 0x000
	0x00000044	0x01c42023	sw x28,0(x8)	137: sw t3, 0(s0) # only 3 bytes are store...
	0x00000048	0x00045383	lhu x7,0(x8)	140: lhu t2, 0(s0) # third read
	0x0000004c	0x006282b3	add x5,x5,x6	143: add t0, t0, t1

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x00006000	0x00008000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x00006020	0x00008000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Figure 58: Data segment after executing instructions

Result at LEDS address in 58 matches the answer in 57.

Assembly Source Code

Listing 5: Question 1 of SW1

```

1 # Author : WhisperingRock
2 #
3 # Purpose : Read from the SWITCHES addr in memory three consecutive
4 #           times, summing the unsigned half-word reads and writing
5 #           to the LEDS addr.
6 #
7 # Notes :
8 #   - Assume caller/calle has no need to preserve reg values.
9 #   - KISS method for this assignment
10 #
11 #   - Split 32-bit addresses according to instr imm handling.
12
13
14
15
16 .global main
17
18 .equ SWITCHES_H, 0x11000 # upper 20 bits
19 .equ SWITCHES_L, 0x000 # lower 12 bits
20 .equ SWITCHES_H, 0x00006 # upper 20 bits (data seg at 0x00006000)
21 .equ SWITCHES_L, 0x000 # lower 12 bits
22
23 .equ LEDS_H, 0x11000 # upper 20 bits
24 .equ LEDS_L, 0x020 # lower 12 bits
25 .equ LEDS_H, 0x00006 # upper 20 bits (data seg at 0x00006020)
26 .equ LEDS_L, 0x020 # lower 12 bits
27
28 # ~~~~~ Test Cases ~~~~~
29 # TC1 : Word Clip
30 .equ READ1_H, 0x00001 # switch first read (0x1234)
31 .equ READ1_L, 0x234
32
33 .equ READ2_H, 0x12345 # switch second read (0x5678)
34 .equ READ2_L, 0x678
35
36 .equ READ3_H, 0x11111 # switch third read (0x1111)
37 .equ READ3_L, 0x111
38 # answer= 0x79BD
39
40 # TC2 : Maxed
41 .equ READ1_H, 0x00005 # switch first read (0x5555)

```

```

42 #.equ READ1_L,      0x555
43
44 #.equ READ2_H,      0x00005 # switch second read (0x5555)
45 #.equ READ2_L,      0x555 # sign extended
46
47 #.equ READ3_H,      0x00005 # switch third read (0x5555)
48 #.equ READ3_L,      0x555
49 # answer= 0xFFFF
50
51 # TC3 : Overflow
52 #.equ READ1_H,      0x00005 # switch first read (0x5555)
53 #.equ READ1_L,      0x555
54
55 #.equ READ2_H,      0x00005 # switch second read (0x5555)
56 #.equ READ2_L,      0x555 # sign extended
57
58 #.equ READ3_H,      0x00005 # switch third read (0x5557)
59 #.equ READ3_L,      0x557
60 # answer= 0x0001
61
62 # TC4 : zero
63 #.equ READ1_H,      0x00000 # switch first read (0)
64 #.equ READ1_L,      0x0
65
66 #.equ READ2_H,      0x00000 # switch second read (0)
67 #.equ READ2_L,      0x0 # sign extended
68
69 #.equ READ3_H,      0x00000 # switch third read (0)
70 #.equ READ3_L,      0x0
71 # answer= 0x0000
72
73 # TC5 : MAXXXED
74 #.equ READ1_H,      0x0000F # switch first read (0xFFFF)
75 #.equ READ1_L,      -1
76
77 #.equ READ2_H,      0xFFFFF # switch second read (0xFFFF)
78 #.equ READ2_L,      -1
79
80 #.equ READ3_H,      0x00000 # switch third read (0)
81 #.equ READ3_L,      0
82 # answer= 0xFFFE
83
84 # TC6 : All highs into full
85 .equ READ1_H,      0x00008 # switch first read
86 .equ READ1_L,      0x000
87
88 .equ READ2_H,      0x00008 # switch second read
89 .equ READ2_L,      0x000
90
91 .equ READ3_H,      0x00008 # switch third read
92 .equ READ3_L,      0x000
93 # answer= 0x8000 (0x18000 as full word though)
94 # ~~~~~
95
96 main:
97
98 # ~~~~ register init ~~~~
99 addi t0, zero, 0 # t0 is the SW first read
100 addi t1, zero, 0 # t1 is the SW second read
101 addi t2, zero, 0 # t2 is the SW third read
102 # t3 is a dummy variable for testing
103
104 # s0 contains the SWITCH addr
105 # s1 contains the LEDS addr
106
107
108 # ~~~~ load addr into registers ~~~~
109 lui s0, SWITCHES_H
110 addi s0, s0, SWITCHES_L
111
112 lui s1, LEDS_H
113 addi s1, s1, LEDS_L
114
115
116 # ~~~~ read SWITCHES and sum ~~~~
117
118 # ~~ behind the scenes : switch mem changes value ~~
119 lui t3, READ1_H
120 ori t3, t3, READ1_L
121 sw t3, 0(s0) # only 3 bytes are stored (0x234)
122 # ~~~~~
123
124 lhu t0, 0(s0) # first read
125
126 # ~~ behind the scenes : switch mem changes value ~~
127 lui t3, READ2_H
128 ori t3, t3, READ2_L
129 sw t3, 0(s0) # only 3 bytes are stored (0x678)
130 # ~~~~~
131
132 lhu t1, 0(s0) # second read

```

```
133
134 # ~~~~ behind the scenes : switch mem changes value ~~~~
135 lui t3, READ3_H
136 ori t3, t3, READ3_L
137 sw t3, 0(s0)      # only 3 bytes are stored (0x111)
138 # ~~~~~
139
140 lhu t2, 0(s0)      # third read
141
142 # ~~~~ sum values ~~~~
143 add t0, t0, t1
144 add t0, t0, t2
145
146
147 # ~~~~ Write the unsigned half word to LEDS mem addr
148 sh t0, 0(s1)
149
150
151 # ~~~~ return with exit status ~~~~
152 li a7, 10 # load system call number for exit()
153 li a0, 0  # Load the exit status code ( EXIT_SUCCESS )
154 ecall    # Make the system call
```

Part 2

"Read an input from SWITCHES (address 0x11000000). Assume the input is a 16-bit signed value in 2's complement (RC). Change the sign of the input and output the result to LEDS (address 0x11000020). The result should also be a 16-bit signed value in 2's complement (RC)."

Flowchart

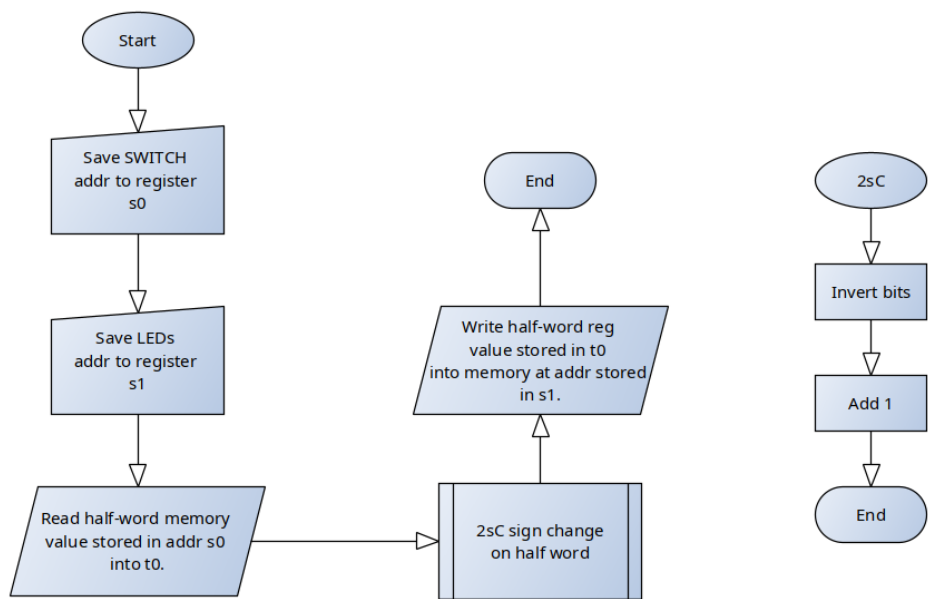


Figure 59: Flow Chart for Part 2

Verification Simulations

Test cases passed with definitions defined in code below.

```
16 # TC1 : zero
17 #A: .half 0
18 # answer= 0x0000
```

Data Segment	
Address	Value (+0)
0x00006000	0x00000000
0x00006020	0x00000000

Figure 60: IO and Data Segment of TC1

```
21 # TC2 : positive
22 #A: .half 0x5
23 # answer= 0xFFFFB (-5)
```

Data Segment	
Address	Value (+0)
0x00006000	0x00000005
0x00006020	0x0000ffffb

Figure 61: IO and Data Segment of TC2

25	# TC3 : negative		
26	#A: .half 0xFFFB	# -5	
27	# answer= 0x0005	# 5	
28			

Data Segment	
Address	Value (+0)
0x00006000	0x0000ffff
0x00006020	0x00000005

Figure 62: IO and Data Segment of TC3

30	# TC4 : positive		
31	#A: .half 0x5678	# 22136	
32	# answer= 0xA988	# -22136	

Data Segment	
Address	Value (+0)
0x00006000	0x00005678
0x00006020	0x0000a988

Figure 63: IO and Data Segment of TC4

34	# TC5 : reverse		
35	#A: .half 0xA988		
36	# answer= 0x5678		
37			

Data Segment	
Address	Value (+0)
0x00006000	0x0000a988
0x00006020	0x00005678

Figure 64: IO and Data Segment of TC5

39	# TC6 : negative		
40	#A : .half 0xFFFF #(-1)		
41	# answer= 1		
42			

Data Segment	
Address	Value (+0)
0x00006000	0x0000ffff
0x00006020	0x00000001

Figure 65: IO and Data Segment of TC6

43	# TC7 : negative 2		
44	#A : .half 0xFFFE #(-2)		
45	# answer= 2		
46			

Data Segment	
Address	Value (+0)
0x00006000	0x0000fffe
0x00006020	0x00000002

Figure 66: IO and Data Segment of TC7

48	# TC7 : negative 32 bit		
49	#A : .word 0x89ABCDEF	#(0xCDEF = -12817)	
50	# answer= 0x3211	# 12817	
51			

Data Segment	
Address	Value (+0)
0x00006000	0x89abcdef
0x00006020	0x00003211

Figure 67: IO and Data Segment of TC8

Assembly Code

Listing 6: Question 2 of SW1

```

1 # Author : WhisperingRock
2 #
3 # Purpose : Read from the SWITCHES addr in memory a unsigned half-word
4 #           changing the sign, then writing to the LEDS addr.
5 #
6 # Notes :
7 #     - Assume caller/callee has no need to preserve reg values.
8 #       KISS method for this assignment
9 #
10 #     - Split 32-bit addresses according to instr imm handling.
11
12 .data
13 # ~~~~~ Test Cases ~~~~~
14 # TC1 : zero
15 #A: .half 0
16 # answer= 0x0000
17
18 # TC2 : positive
19 #A: .half 0x5
20 # answer= 0xFFFB (-5)
21
22 # TC3 : negative
23 #A: .half 0xFFFB # -5
24 # answer= 0x0005 # 5
25
26 # TC4 : positive
27 #A: .half 0x5678 # 22136

```

```

28 # answer= 0xA988      # -22136
29
30 # TC5 : reverse
31 #A: .half 0xA988
32 # answer= 0x5678
33
34 # TC6 : negative
35 #A : .half 0xFFFF #(-1)
36 # answer= 1
37
38 # TC7 : negative 2
39 #A : .half 0xFFFE #(-2)
40 # answer= 2
41
42 # TC7 : negative 32 bit
43 #A : .word 0x89ABCDEF    #(0xCDEF = -12817)
44 # answer= 0x3211      # 12817
45
46 # ~~~~~
47
48
49 .global _start
50 .text
51 #.equ SWITCHES_H, 0x11000 # upper 20 bits
52 #.equ SWITCHES_L, 0x000 # lower 12 bits
53 #.equ SWITCHES_H, 0x00006 # upper 20 bits (data seg at 0x00006000)
54 #.equ SWITCHES_L, 0x000 # lower 12 bits
55
56 #.equ LEDS_H, 0x11000 # upper 20 bits
57 #.equ LEDS_L, 0x020 # lower 12 bits
58 #.equ LEDS_H, 0x00006 # upper 20 bits (data seg at 0x00006020)
59 #.equ LEDS_L, 0x020 # lower 12 bits
60
61
62
63 _start:
64
65 # ~~~~ register init ~~~~
66 addi t0, zero, 0 # t0 is the SW first read
67 # t3 is a dummy variable for testing
68
69 # s0 contains the SWITCH addr
70 # s1 contains the LEDS addr
71
72 # ~~~~ load addr into registers ~~~~
73 lui s0, SWITCHES_H
74 addi s0, s0, SWITCHES_L
75
76 lui s1, LEDS_H
77 addi s1, s1, LEDS_L
78
79
80 # ~~~~ read SWITCHES and sum ~~~~
81
82 # ~~ behind the scenes : switch value ~~
83 lw t3, A
84 sw t3, 0(s0)
85 # ~~~~~
86
87 lhu t0, 0(s0) # first read
88
89
90 # ~~~~ invert the sign (2sC) ~~~~
91 not t0, t0
92 addi t0, t0, 1
93
94 # ~~~~ Write the unsigned half word to LEDS mem addr
95 sh t0, 0(s1)
96
97
98 # ~~~~ return with exit status ~~~~
99 li a7, 10 # load system call number for exit()
100 li a0, 0 # Load the exit status code ( EXIT_SUCCESS )
101 ecall # Make the system call

```

0.2.2 SW2 - Conditional Statements

Assignment:
First, create a flowchart that will perform the following behavior. Then implement the flowchart with OTTER MCU assembly language using only the instructions in the OTTER assembly manual. Make sure your code is formatted properly including comments. Use the RARS simulator to ensure the program performs as desired.

- 1. Read a 32-bit unsigned value from SWITCHES (address 0x11000000). If the input is greater than or equal to 32,768, the value is divided by 4. You can ignore any remainder. If the value is less than 32,768, the value is multiplied by 2. The result should be written to the 7 SEGMENT (address 0x11000040).
- 2. Read a 32-bit unsigned value from SWITCHES (address 0x11000000). If the input value is a multiple of 4, all of the bits should be inverted, otherwise, if the input value is odd, add 4095 and divide the result by 2, otherwise subtract 1 from the value. The result should be written to the 7 SEGMENT (address 0x11000040).

Assume the BASYS3 SWITCHES are expanded to 32 individual switches and 7 SEGMENT is expanded to 8 digits.

Figure 68: Questions to address

Part 1

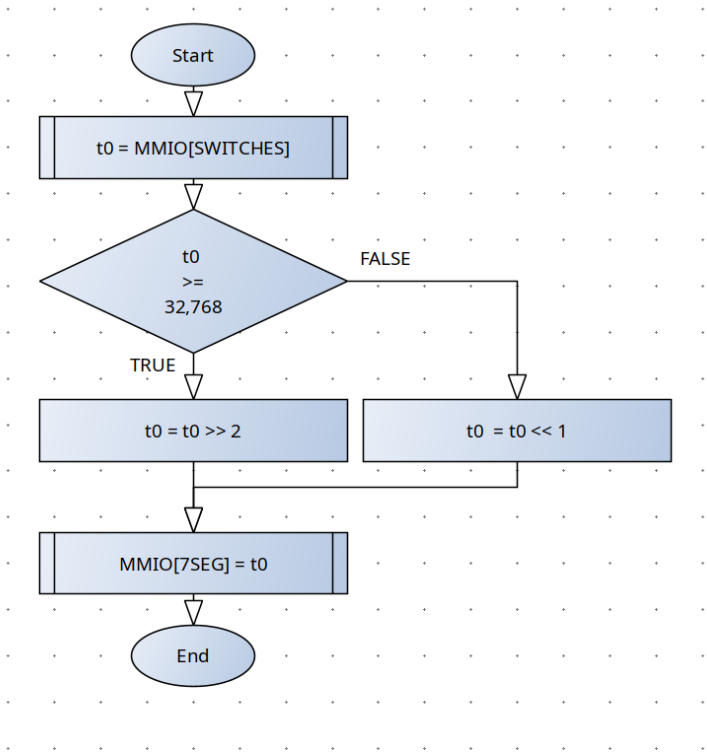


Figure 69: Flow chart for part 1

TC#	Desc	Given MMIO[SWITCHES]	Correct MMIO[7SEG]	Result
1	Lowest unsigned val	0	0	Passed
2	Below boundary	0xFFF	0x1FFE	Passed
3	1 before boundary	0x7FFF	0xFFFE	Passed
4	On boundary	0x8000	0x2000	Passed
5	1 above boundary	0x8001	0x2000	Passed
6	Above boundary	0x12345678	0x048D159E	Passed
7	32-bit unsigned ceiling	0xFFFFFFFF	0x3FFFFFFF	Passed

Table 4: Test cases for part 1

Listing 7: sw2_p1.asm

```

1 # Author : WhisperingRock
2 #
3 # Purpose : Conditionals, branching, jumping
4 # Input :
5 #     SWITCHES MMIO address to 32-bit unsigned value
6 #     7SEG MMIO address to 8-digits, each of 4-bits
7 #     constant CONST
8 #
9 # Output :
10 #     Write to 7SEG addr
11
12
13 .data
14 # ~~~~ Test Cases ~~~~
15
16 # ~~ TC1 : Zero ~~
17 #TC:      .word 0
18 # answer = 0 (verified)
19
20 # ~~ TC2 : Under ~~
21 #TC:      .word 0xFFFF
22 # answer = 0x1FFE (verified)
23
24 # ~~ TC3 : One before ~~
25 #TC:      .word 0x07FFF
26 # answer = 0xFFFE (verified)
27
28 # ~~ TC4 : On boundary ~~
29 #TC:      .word 0x08000
30 # answer = 0x2000 (verified)
31
32 # ~~ TC5 : Above boundary ~~
33 #TC:      .word 0x08001
34 # answer = 0x2000 (verified)
35
36 # ~~ TC6 : Above boundary 2 ~~
37 #TC:      .word 0x12345678
38 # answer = 0x048D159E (verified)
39
40 # ~~ TC7 : 32-bit unsigned ceiling ~~
41 #TC:      .word 0xFFFFFFFF
42 # answer = 0x3FFFFFFF (verified)
43
44 .global _start
45
46 .text
47 .equ MMIO_SWITCHES, 0x11000000
48 .equ MMIO_SEVSEG, 0x11000040
49 .equ CONST, 32768
50
51 _start:
52 # ~~~~ init reg ~~~~
53 li t1, MMIO_SWITCHES    # t1 is the addr for MMIO SWITCHES
54 li t2, CONST            # t2 is the constant 32768
55 li t3, MMIO_SEVSEG      # t3 is the addr for MMIO 7SEG
56                          # t4 is a test case variable
57
58 # ~~~~ TC injection ~~~~
59 lw t4, TC
60 sw t4, 0(t1)
61
62 # ~~~~ read SWITCHES ~~~~
63 lw t0, 0(t1)            # load SW value into t0
64
65 # ~~~~ unsigned branch ~~~~
66 bgeu t0, t2, Div4
67
68 # ~~ else (false) case ~~
69 slli t0, t0, 1          # multiply by 2
70 j End
71
72 # ~~ if (true) case ~~
73 Div4:
74 srli t0, t0, 2          # logical shift for unsigned
75
76 # ~~~~ write to 7SEG ~~~~
77 End:
78 sw t0, 0(t3)
79
80 # ~~~~ syscall exit status ~~~~
81 li a7, 10              # load syscall num for exit()
82 li a0, 0               # EXIT_SUCCESS code
83 ecall                 # execute sys call
84
85

```

Figure 70: Assembly code for SW2 part 1

Part 2

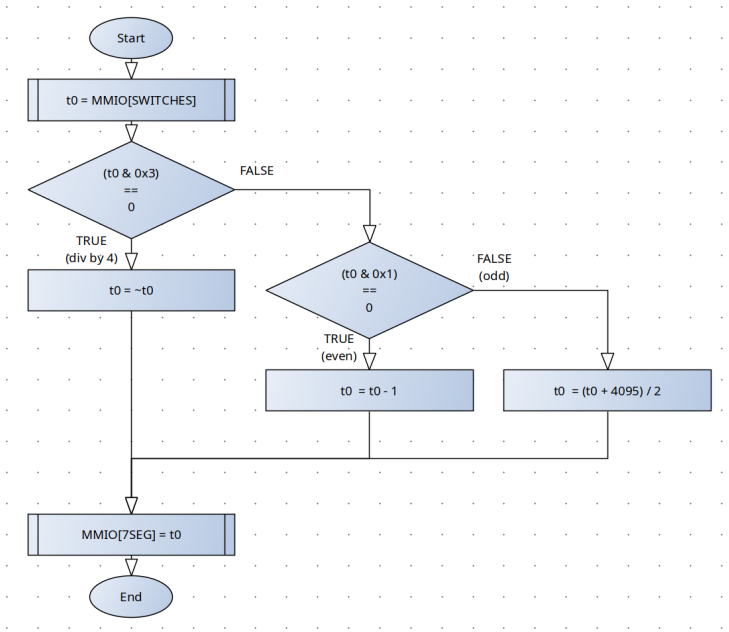


Figure 71: Flow chart for part 2

TC#	Desc	Given MMIO[SWITCHES]	Correct MMIO[7SEG]	Result
1	Zero is div by 4	0	0xFFFFFFFF	Passed
2	Two is even	0x2	0x1	Passed
3	One is odd	0x1	0x0800	Passed
4	32-bit unsinged ceiling	0xFFFFFFFF	0x07FF	Passed
5	Div by 4 near ceiling	0xFFFFFFFFC	0x3	Passed
6	Odd near ceiling	0xFFFFFFFDD	0x07FE	Passed
7	Even near ceiling	0xFFFFFFFEE	0xFFFFFFFDD	Passed

Table 5: Test cases for part 1

Listing 8: sw2_p2.asm

```

1  # Author : WhisperingRock
2  #
3  # Purpose : Conditionals, branching, jumping
4  # Input :
5  #     SWITCHES MMIO address to 32-bit unsigned value
6  #     7SEG MMIO address to 8-digits, each of 4-bits
7  #     constant CONST
8  #
9  # Output :
10 #     Write to 7SEG addr
11
12
13 .data
14 # ~~~~ Test Cases ~~~~
15
16 # ~~ TC1 : Zero (Div4) ~~
17 #TC:      .word 0
18 # answer = 0xFFFF_FFFF (verified)
19
20 # ~~ TC2 : Two (Div2 = even) ~~
21 #TC:      .word 2
22 # answer = 0x1 (verified)
23
24 # ~~ TC3 : 1 (odd) ~~
25 #TC:      .word 0x1
26 # answer = 0x0800 (verified)
27
28 # ~~ TC4 : 32-bit unsigned ceiling ~~
29 #TC:      .word 0xFFFFFFFF
30 # answer = 0x0000_07FF | 2047 (verified)
31
32 # ~~ TC5 : unsigned ceiling Div4 ~~
33 #TC:      .word 0xFFFF_FFC
34 # answer = 0x03 (verified)
35
36 # ~~ TC6 : unsigned ceiling odd ~~
37 #TC:      .word 0xFFFF_FFD
38 # answer = 0x07FE (verified)
39
40 # ~~ TC6 : unsigned ceiling even ~~
41 #TC:      .word 0xFFFF_FFE
42 # answer = 0xFFFF_FFD (verified)
43
44 .global _start
45
46 .text
47 .equ MMIO_SWITCHES, 0x11000000
48 .equ MMIO_SEVSEG, 0x11000040
49
50
51 _start:
52 # ~~~~ init reg ~~~~
53 li s1, MMIO_SWITCHES # s1 is the addr for MMIO SWITCHES
54 li s2, MMIO_SEVSEG   # s2 is the addr for MMIO 7SEG
55 # t0 is the working variable for SWITCHES value
56 # t1 is a logic case variable
57 li t2, 4095          # t2 is reserved for const
58
59 # ~~~~ TC injection ~~~~
60 lw t0, TC
61 sw t0, 0(s1)
62
63 # ~~~~ read SWITCHES ~~~~
64 lw t0, 0(s1) # load SW value into t0
65
66 # ~~~~ branch : div by 4 ~~~~
67 andi t1, t0, 0x3
68 beqz t1, Div4
69
70 # ~~~~ branch : div by 2 ~~~~
71 andi t1, t0, 0x1
72 beqz t1, Div2
73
74 # ~~ else case ~~
75 add t0, t0, t2 # add const 4095
76 srli t0, t0, 1 # then scale
77 j End
78
79 Div4:
80 not t0, t0 # invert bits
81 j End
82
83 Div2:
84 addi t0, t0, -1
85
86 End: # ~~~~ write to 7SEG ~~~~
87 sw t0, 0(s2)
88
89
90
91 # ~~~~ syscall exit status ~~~~
92 li a7, 10 # load syscall num for exit()
93 li a0, 0 # EXIT_SUCCESS code
94 ecalls # execute sys call
95
96

```

0.2.3 SW3 - Loops

Assignment:

First, create a flowchart that will perform the following behavior. Then implement the flowchart with OTTER MCU assembly language using only the instructions in the OTTER assembly manual. Make sure your code is formatted properly including comments. Use the RARS simulator to ensure the program performs as desired.

- 1. Read a 32-bit value from SWITCHES (address 0x11000000). Assume the 32-bit value is the combination of two 16-bit unsigned values. Split the 32-bit input into two 16-bit values (the upper and lower 16-bits). Assume both of the 16-bit values are unsigned. Multiply the two 16-bit values together and output the result as a 32-bit unsigned value to 7 SEGMENT (address 0x11000040).
- 2. Read a 32-bit value from SWITCHES (address 0x11000000), delay for 0.5 seconds, and then output the value to 7 SEGMENT (address 0x11000040). The OTTER MCU operates at 25 MHz so each instruction takes 40 ns to run. (Hint: Use loops.) **There is no method to verify the run time in RARS and RARS does not execute at the same speed as the OTTER.** You will need to verify your timing by showing your calculation for the number of instructions being performed between the load (reading from the switches) and store (writing to the 7 segment display) instructions.

Assume the BASYS3 SWITCHES are expanded to 32 individual switches and 7 SEGMENT is expanded to 8 digits.

Figure 73: Questions to address

Part 1

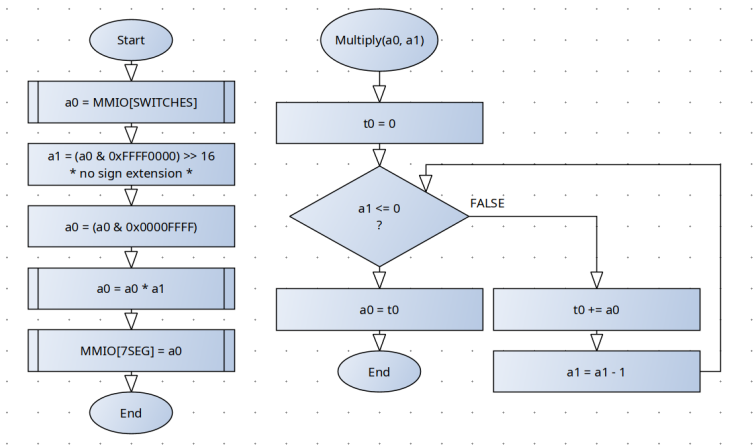


Figure 74: Flow chart for part 1

TC#	Desc	Given MMIO[SWITCHES]	Correct MMIO[7SEG]	Result
1	Zero	0x0000_0000	0x0000_0000	Passed
2	Zero + One	0x0000_0001	0x0000_0000	Passed
3	One + Zero	0x0001_0000	0x0000_0000	Passed
4	One + One	0x0001_0001	0x0000_0001	Passed
5	num1	0x0013_0004	0x0000_004C	Passed
6	num1 reverse	0x0004_0013	0x0000_004C	Passed
7	shift 1	0xFFFF_FFFF	0xFFFE_0001	Passed
8	shift 2	0x0010_FFFF	0x000F_FFF0	Passed
9	shift 3	0x0100_FFFF	0x00FF_FF00	Passed

Table 6: Test cases for part 1

Listing 9: sw3_p1.asm

```
1 # Author : WhisperingRock
2 #
3 # Purpose : Half word multiply
4 # Input :
5 #     SWITCHES MMIO address to 32-bit unsigned value
6 #     7SEG MMIO address to 8-digits, each of 4-bits
7 #
8 # Output :
9 #     Write to 7SEG addr
10
11
12 .data
13 # ~~~~ Test Cases ~~~~
14
15 # ~~ TC1 : Zero ~~
16 #TC:      .word 0
17 # answer = 0      (verified)
```

```

18
19 # ~~ TC2 : Zero + One ~~
20 #TC:      .word 0x00000001
21 # answer = 0      (verified)
22
23 # ~~ TC3 : One + Zero ~~
24 #TC:      .word 0x00010000
25 # answer = 0      (verified)
26
27 # ~~ TC4 : One ~~
28 #TC:      .word 0x00010001
29 # answer = 1      (verified)
30
31 # ~~ TC5 : num1 ~~
32 #TC:      .word 0x00130004
33 # answer = 0x0000_004C (verified)
34
35 # ~~ TC6 : num1 reverse ~~
36 #TC:      .word 0x00040013
37 # answer = 0x0000_004C (verified)
38
39 # ~~ TC7 : shift1 ~~
40 #TC:      .word 0xFFFFFFFF
41 # answer = 0xFFFE_0001 (verified)
42
43 # ~~ TC8 : shift2 ~~
44 #TC:      .word 0x0010FFFF
45 # answer = 0x000F_FFF0 (verified)
46
47 # ~~ TC9 : shift3 ~~
48 #TC:      .word 0x0100FFFF
49 # answer = 0x00FF_FF00 (verified)
50
51 .global _start
52
53 .text
54 #.equ MMIO_SWITCHES, 0x11000000 # moved to Compact(Text @ 0) config
55 #.equ MMIO_SEVSEG, 0x11000040 # sp moved tp 0x00003FFC
56 #.equ MMIO_SWITCHES, 0x00007F00
57 #.equ MMIO_SEVSEG, 0x00007F40
58 #.equ UPPER_WORD, 0xFFFF0000
59 #.equ LOWER_WORD, 0x0000FFFF
60
61 _start:
62 # ~~~~ init reg ~~~~
63 li s1, MMIO_SWITCHES # s1 is the addr for MMIO SWITCHES
64 li s2, MMIO_SEVSEG # s2 is the addr for MMIO 7SEG
65 # a0 is a working variable for input and return
66 # a1 is a working var for upper half of word
67 # a2 is a working var for lower half of word
68 # t0 is single use TC injection var
69
70 # ~~~~ TC injection ~~~~
71 lw t0, TC
72 sw t0, 0(s1)
73
74 # ~~~~ read SWITCHES ~~~~
75 lw a0, 0(s1)
76
77 # ~~~~ parse input val (and load args for func) ~~~~
78 # ~~ a1 = (a0 & 0xFFFF0000) >> 16 ~~
79 li t0, UPPER_WORD
80 and a1, a0, t0
81 srli a1, a1, 16
82 # ~~ a0 = a0 & 0x0000FFFF ~~
83 li t0, LOWER_WORD
84 and a0, a0, t0
85
86 # ~~~~ caller(_start) saves non-preserved ~~~~
87 # "Not needed b/c we dont need t0"
88 jal Multiply
89
90 # ~~~~ write return to 7SEG ~~~~
91 sw a0, 0(s2)
92
93 # ~~~~ syscall exit ~~~~
94 li a7, 10 # load syscall num for exit()
95 li a0, 0 # EXIT_SUCCESS code
96 ecall # execute sys call
97
98
99 # ~~~~~~
100 # Purpose : Multiply two unsigned 32-bit numbers
101 # Input :
102 #     a0 : 32-bit unsigned word
103 #     a1 : 32-bit unsigned word
104 # Temps :
105 #     t0 :
106 # Note :
107 #     - consumes a1
108 # Return:

```

```

109 #      a0 : product of a0 * a1
110 Multiply:
111
112 # ~~~~ calle(Multiply) saves preserved ~~~~
113 # ~~ 1. calle(Multiply) allocates space on stack ~~
114 addi sp, sp, -12
115
116 # ~~ 2. calle(Multiply) stores reg values on stack ~~
117 sw ra, 8(sp)
118 sw s1, 4(sp)
119 sw s2, 0(sp)
120
121 # ~~ 3. calle(Multiply) function execution ~~
122 # ~ sum(t0) init ~
123 addi t0, zero, 0
124 # ~ while a1 <= 0
125 While:
126 bleu a1, zero, EndWhile
127 add t0, t0, a0
128 addi a1, a1, -1
129 j While
130 EndWhile:
131 addi a0, t0, 0
132
133 # ~~ 4. calle(Multiply) restores preserved from stack ~~
134 lw ra, 8(sp)
135 lw s1, 4(sp)
136 lw s2, 0(sp)
137
138 # ~~ 5. calle(Multiply) deallocates space on stack ~~
139 addi sp, sp, 12
140
141 # ~~ calle(Multiply) return ~~
142 jr ra
143 # ~~~~~~

```

Part 2

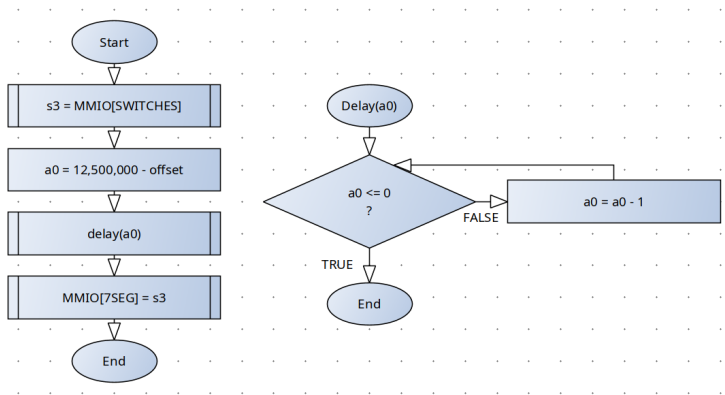


Figure 75: Flow chart for part 2

Consider the following calculations

$$\begin{aligned}
 12.5 \cdot 10^6 \text{ instr} &= 0.5s \cdot \frac{\text{instr}}{40ns} \cdot \frac{10^9 ns}{1s} \\
 &= 0x00BE_BC20
 \end{aligned}$$

- There are 13 instructions, after reading from SWITCHES, that are only executed once. Lets refer to these instructions as bias s.t. $b = 13$.
- There are 3 instruction in our delay loop s.t. $m = 3$.

Now lets relate our instructions to the instruction layout s.t. we identify how many times we actually need to loop.

$$\begin{aligned}
 3m + b &= 12.5 \cdot 10^9 \\
 m &= \frac{12,500,000 - 13}{3} \\
 &= 4,166,663
 \end{aligned}$$

Rather than use this number in directly, I opted to write the constant in terms of total instructions needed against how many loops we need.

$$\begin{aligned}
\text{offset} &= 12.5 \cdot 10^9 - m \\
&= 8,333,337 \\
&= 0x007F_2819
\end{aligned}$$

Listing 10: Assembly code for sw3_p2.asm

```

1 # Author : WhisperingRock
2 #
3 # Purpose : Delay
4 # Input :
5 #     SWITCHES MMIO address to 32-bit unsigned value
6 #     7SEG MMIO address to 8-digits, each of 4-bits
7 #
8 # Output :
9 #     Write to 7SEG addr
10
11
12 .data
13 # ~~~~ Test Cases ~~~~
14
15 # ~~ TC1 : simple ~~
16 TC:  .word 0x12345678
17 # answer = 0      (verified)
18
19 .global _start
20
21 .text
22 #.equ MMIO_SWITCHES, 0x11000000 # moved to Compact(Text @ 0) config
23 #.equ MMIO_SEVSEG, 0x11000040 # sp moved tp 0x00003FFC
24 .equ MMIO_SWITCHES, 0x00007F00
25 .equ MMIO_SEVSEG, 0x00007F40
26 .equ HALF_SEC_DELAY, 0x00BEB20 # 12,500,000 instr for 0.5s delay
27 .equ OFFSET, 0x007F2819 # loop and bias compensation (see notes)
28
29 _start:
30 # ~~~~ init reg ~~~~
31 li s1, MMIO_SWITCHES # s1 is the addr for MMIO SWITCHES
32 li s2, MMIO_SEVSEG # s2 is the addr for MMIO 7SEG
33 # s3 is a working variable for input and return
34 li a0, HALF_SEC_DELAY # a0 is reserved for const
35 li a1, OFFSET # a1 is reserved for offset
36 # t0 is single use TC injection var
37
38 # ~~~~ TC injection ~~~~
39 lw t0, TC
40 sw t0, 0(s1)
41
42 # ~~~~ read SWITCHES ~~~~
43 lw s3, 0(s1)
44
45 # ~~~~ caller(_start) saves non-preserved ~~~~
46 # "Not needed b/c we dont need t0"
47 sub a0, a0, a1 # a0 = delay - offset ; +1 bias
48 jal Delay # +1 bias
49
50 # ~~~~ write return to 7SEG ~~~~
51 sw s3, 0(s2)
52
53 # ~~~~ syscall exit ~~~~
54 li a7, 10 # load syscall num for exit()
55 li a0, 0 # EXIT_SUCCESS code
56 ecall # execute sys call
57
58
59 # ~~~~~~
60 # Purpose : delay(busy wait) of 0.5s
61 # Input :
62 #     a0 : [unsigned 32-bit] delay amount in instruction count
63 # Note :
64 #     - Consuming a0 in the process
65 # Return:
66 #
67
68 Delay:
69
70 # ~~~~ calle(Delay) saves preserved ~~~~
71 # ~~ 1. calle(Delay) allocates space on stack ~~
72 addi sp, sp, -16 # +1 bias
73
74 # ~~ 2. calle(Delay) stores reg values on stack ~~
75 sw ra, 12(sp) # +1 bias
76 sw s1, 8(sp) # +1 bias
77 sw s2, 4(sp) # +1 bias
78 sw s3, 0(sp) # +1 bias
79
80

```

```

81
82 # ~~ 3. calle(Delay) function execution ~~
83 # ~ while a0 <= 0
84 While: # loop = 3x instr
85     bleu a0, zero, EndWhile
86     addi a0, a0, -1
87     j While
88 EndWhile:
89
90 # ~~ 4. calle(Delay) restores preserved from stack ~~
91 lw ra, 12(sp)           # +1 bias
92 lw s1, 8(sp)            # +1 bias
93 lw s2, 4(sp)            # +1 bias
94 lw s3, 0(sp)            # +1 bias
95
96 # ~~ 5. calle(Delay) deallocates space on stack ~~
97 addi sp, sp, 16         # +1 bias
98
99 # ~~ calle(Delay) return ~~
100 jr ra                  # +1 bias
101 # ~~~~~~

```

0.2.4 SW4 - Arrays

Assignment:
First, create a flowchart that will perform the following behavior. Then implement the flowchart with OTTER MCU assembly language using only the instructions in the OTTER assembly manual. Use register alternative names (ie s0, t0) instead of register numbers (ie x5, x10). Make sure your code is formatted properly including comments. Use the RARS simulator to ensure the program performs as desired.

- 1. Using `.data`, define an array in the data segment that contains the first 25 values of the Fibonacci sequence (starting with 0 and ending with 46368). Write a program that progresses through the array and calculates the difference between values that are 3 spots away from each other. For example, the first value (0) and the 4th value (2) is a difference of 2. The next calculation would be between the 2nd (1) and 5th (3) values. When no item exists 3 spots away, no difference can be calculated. Each difference should be output to LEDS (address 0x11000020).
- 2. Create an array of 10 32-bit values. Read 10 32-bit values consecutively from SWITCHES (address 0x11000000) saving them in the array. After the 10th value has been saved, the array should be sorted from least to greatest. The sorted array should be output to LEDS (address 0x11000020) in order from least to greatest.

Assume the BASYS3 SWITCHES and LEDS are expanded to 32 individual switches and LEDs.

Figure 76: Questions to address

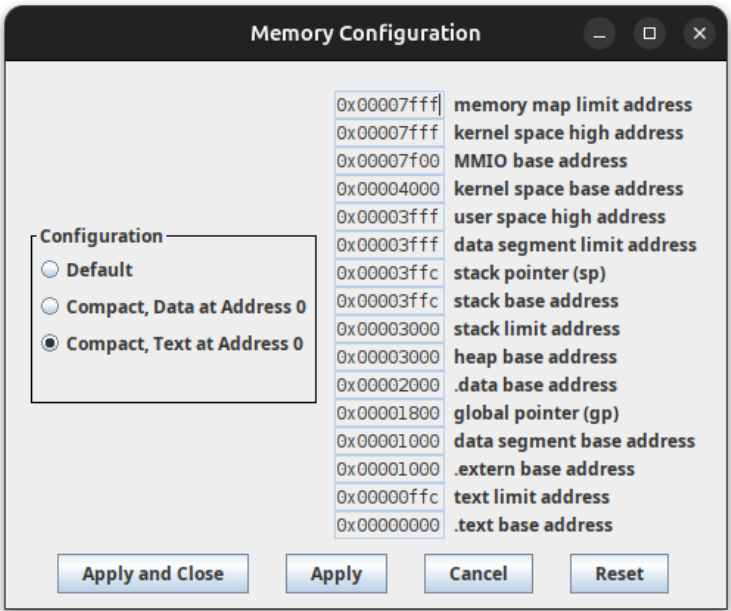


Figure 77: Memory Map for this Assignment

Part 1

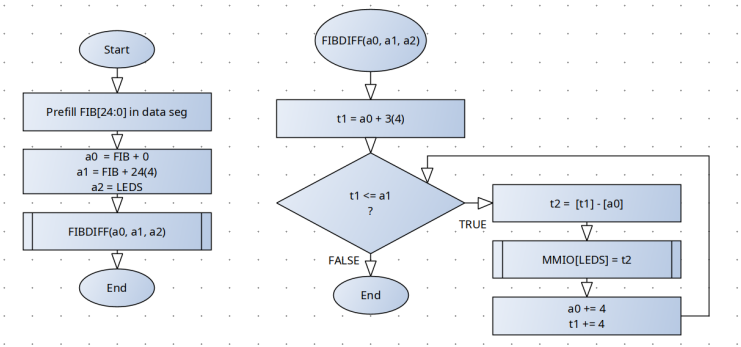


Figure 78: Flow chart for part 1

Listing 11: sw4_p1.asm

```
1 # Author : WhisperingRock
2 #
3 # Purpose : Difference between array elems
4 # Input :
5 #     Hardcoded values of fibonacci in array
6 #     LEDS MMIO address to 32 bits
7 #
8 # Output :
```

```

9  #       Write to LEDS addr
10
11
12 .data
13 FIB:  .word 0, 1, 1, 2, 3
14       .word 5, 8, 13, 21, 34
15       .word 55, 89, 144, 233, 377
16       .word 610, 987, 1597, 2584, 4181
17       .word 6765, 10946, 17711, 28657, 46368
18
19 # answer = 2, 2, 4, 6, 10, 16, 26, 42, 68, 110, 178,
20 #       288, 466, 754, 1220, 1974, 3194, 5168, 8362
21 #       13530, 21892, 35422
22 # Verified - values match and does not exceed bounds
23
24 .global _start
25
26 .text
27 #.equ MMIO_SWITCHES, 0x00007F00
28 #.equ MMIO_SEVSEG, 0x00007F40
29 .equ MMIO_LEDS, 0x00007F20
30 .equ LAST_ELEM, 96 # 25th elem, 4 bytes apart
31
32 _start:
33 # ~~~~ init reg ~~~~
34 la a0, FIB # a0 is the addr for head of FIB
35 addi a1, a0, LAST_ELEM # a1 is the addr for end of FIB
36 li a2, MMIO_LEDS # a2 is the addr for MMIO LEDS
37
38 # ~~~~ caller(_start) saves non-preserved ~~~~
39 # ~~ 1. caller(_start) allocates space on stack ~~
40 addi sp, sp, -24
41 # ~~ 2. caller(_start) stores reg values on stack ~~
42 sw t2, 20(sp)
43 sw t1, 16(sp)
44 sw t0, 12(sp)
45 sw a2, 8(sp)
46 sw a1, 4(sp)
47 sw a0, 0(sp)
48 # ~~ 3. caller(_start) function execution ~~
49 jal FIBDIFF
50 # ~~ 4. caller(_start) restores preserved from stack ~~
51 lw t2, 20(sp)
52 lw t1, 16(sp)
53 lw t0, 12(sp)
54 lw a2, 8(sp)
55 lw a1, 4(sp)
56 lw a0, 0(sp)
57 # ~~ 5. caller(_start) deallocates space on stack ~~
58 addi sp, sp, 24
59
60 # ~~~~ syscall exit ~~~~
61 li a7, 10 # load syscall num for exit()
62 li a0, 0 # EXIT_SUCCESS code
63 ecall # execute sys call
64
65
66 # ~~~~~~
67 # Purpose : For all elements of the array, write (TO LEDS) the difference
68 # between a elem and its left (3 over) neighbor, if they exist.
69 #
70 # Input :
71 # a0 : [unsigned 32-bit] head address of array
72 # a1 : [unsigned 32-bit] tail address of array
73 # a2 : [unsigned 32-bit] LEDS MMIO addr
74 #
75 # Return:
76 #
77
78 FIBDIFF:
79
80 # ~~~~ calle(FIBDIFF) saves preserved ~~~~
81 # ~~ 1. calle(FIBDIFF) allocates space on stack ~~
82 addi sp, sp, -4
83
84 # ~~ 2. calle(FIBDIFF) stores reg values on stack ~~
85 sw ra, 0(sp)
86
87 # ~~ 3. calle(FIBDIFF) function execution ~~
88
89 # ~ init vars ~
90 addi t1, a0, 12 # t1 is the reference index and right neighbor
91 addi t2, zero, 0 # t2 is the window difference
92 addi t3, zero, 0 # t3 is a working variable
93
94 # ~ while the sliding window hasn't hit the right limit ~
95 While: # while t1 <= a1
96 bgtu t1, a1, EndWhile
97
98 # t2 = *(t1) - *(t0)
99 lw t2, 0(t1) # t2 = *(t1) (right neighbor)

```

```

100 lw t3, 0(a0) # t3 = *(a0) (left neighbor)
101 sub t2, t2, t3 # t2 = t2 - t3
102
103 # MMIO[LEDS] = t2
104 sw t2, 0(a2)
105
106 # slide window right
107 addi a0, a0, 4
108 addi t1, t1, 4
109
110 j While
111 EndWhile:
112
113 # ~~ 4. calle(FIBDIFF) restores preserved from stack ~~
114 lw ra, 0(sp)
115
116 # ~~ 5. calle(FIBDIFF) deallocates space on stack ~~
117 addi sp, sp, 4
118
119 # ~~ calle(FIBDIFF) return ~~
120 jr ra
121 # ~~~~~

```

Part 2

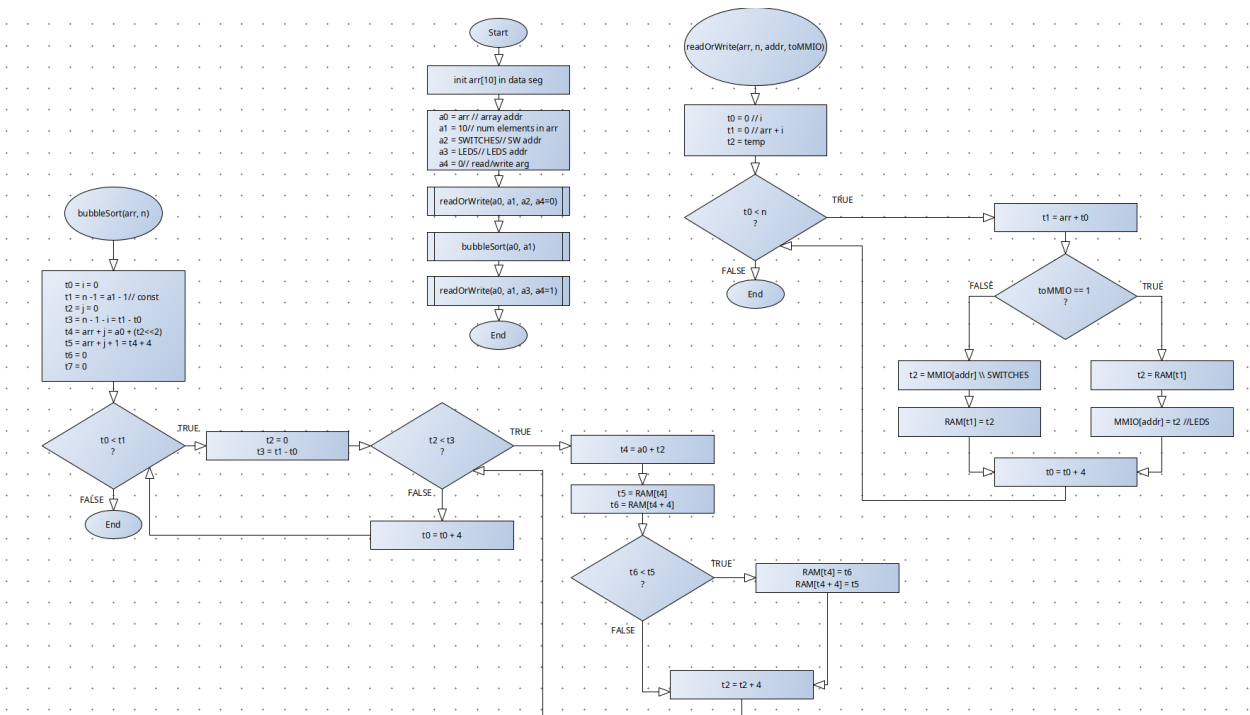


Figure 79: Flow chart for part 2

Listing 12: Assembly code for sw4_p2.asm

```

1 # Author : WhisperingRock
2 #
3 # Purpose : Read 10 conseq inputs from SWITCHES into array,
4 #           sort array in ASC order,
5 #           then write the the entire array to LEDS consecutively
6 # Input :
7 #         SWITCHES MMIO address to 32 bits
8 #         LEDS MMIO address to 32 bits
9 #
10 # Output :
11 #         Write 10 to LEDS addr
12
13
14 .data
15 ARR: .word -7:10
16
17 # ~~~~ Test Cases ~~~~
18 # ~~ TC1: worst bubble case ~~
19 #TC: .word 10, 9, 8, 7, 6, 5, 4, 3, 2, 1
20 # answer = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
21 # verified r/w and sort
22
23 # ~~ TC2: negatives ~~
24 #TC: .word 1, -5, 2, 8, 3, 3, -1000, 1000, 0, 10
25 # answer = -1000, -5, 0, 1, 2, 3, 3, 8, 10, 1000
26 # verified r/w and sort
27
28 # ~~ TC3: mirror ~~
29 TC: .word 0x7FFFFFFF:4
30 .word 0
31 .word 0xEFFFFFFF:5

```

```

32 # answer = 0xEFFFFFFF:5 , 0x7FFFFFFF:4
33 # verified r/w and sort
34
35 .global _start
36
37 .text
38 .equ MMIO_SWITCHES, 0x00007F00
39 #.equ MMIO_SEVSEG, 0x00007F40
40 .equ MMIO_LEDS, 0x00007F20
41 .equ ARR_N_ELEM, 10
42
43 _start:
44 # ~~~~ init reg ~~~~
45 la s0, ARR # s0 is the addr for head of ARR
46 addi s1, zero, ARR_N_ELEM # s1 is the const num elem in ARR
47 li s2, MMIO_SWITCHES # s2 is the addr for MMIO SWITCHES
48 li s3, MMIO_LEDS # s3 is the addr for MMIO LEDES
49
50
51 # ~~~~ caller(_start) saves non-preserved ~~~~
52 # ~~ 1. caller(_start) allocates space on stack ~~
53 # ~~ 2. caller(_start) stores reg values on stack ~~
54 # ~~ 3. caller(_start) function execution ~~
55 # ~ read from MMIO[SWITCHES] ~~
56 addi a0, s0, 0 # a0 is the base addr for ARR
57 addi a1, s1, 0 # a1 is the num of elem in ARR
58 addi a2, s2, 0 # a2 is the r/w base address (set to SW to read)
59 addi a3, zero, 0 # a3 is the "toMMIO" set to 0 for read
60 jal ReadOrWrite
61 # ~~ 4. caller(_start) restores preserved from stack ~~
62 # ~~ 5. caller(_start) deallocates space on stack ~~
63
64 # ~~~~ caller(_start) saves non-preserved ~~~~
65 # ~~ 1. caller(_start) allocates space on stack ~~
66 # ~~ 2. caller(_start) stores reg values on stack ~~
67 # ~~ 3. caller(_start) function execution ~~
68 # ~ sort array ~~
69 addi a0, s0, 0 # a0 is the base addr for ARR
70 addi a1, s1, 0 # a1 is the num of elem in ARR
71 jal BubbleSort
72 # ~~ 4. caller(_start) restores preserved from stack ~~
73 # ~~ 5. caller(_start) deallocates space on stack ~~
74
75 # ~~~~ caller(_start) saves non-preserved ~~~~
76 # ~~ 1. caller(_start) allocates space on stack ~~
77 # ~~ 2. caller(_start) stores reg values on stack ~~
78 # ~~ 3. caller(_start) function execution ~~
79 # ~ write to MMIO[LEDES] ~~
80 addi a0, s0, 0 # a0 is the base addr for ARR
81 addi a1, s1, 0 # a1 is the num of elem in ARR
82 addi a2, s3, 0 # a2 is the r/w base address (set to LED to write)
83 addi a3, zero, 1 # a3 is the "toMMIO" set to 1 for write
84 jal ReadOrWrite
85 # ~~ 4. caller(_start) restores preserved from stack ~~
86 # ~~ 5. caller(_start) deallocates space on stack ~~
87
88
89 # ~~~~ syscall exit ~~~~
90 li a7, 10 # load syscall num for exit()
91 li a0, 0 # EXIT_SUCCESS code
92 ecall # execute sys call
93
94
95 # ~~~~~~ReadOrWrite()~~~~~
96 # Purpose : Read from from the MMIO address n times and place into array
97 # or write the contents of array onto MMIO addr conseq n times.
98 #
99 #
100 # Input :
101 # a0 : [unsigned 32-bit] base address of array
102 # a1 : [unsigned 32-bit] num elem in array
103 # a2 : [unsigned 32-bit] base addres of MMIO
104 # a3 : [unsigned 32-bit] r/w command (0-read, 1-write)
105 # t3 : [unsigned 32-bit] base addr for test case array (if needed)
106 #
107 # Note :
108 # - modifying a1 to save a register (caller preserve)
109 # Return:
110 #
111
112 ReadOrWrite:
113
114 # ~~~~ calle(READORWRITE) saves preserved ~~~~
115 # ~~ 1. calle(READORWRITE) allocates space on stack ~~
116 addi sp, sp, -20
117
118 # ~~ 2. calle(READORWRITE) stores reg values on stack ~~
119 sw s3, 16(sp)
120 sw s2, 12(sp)
121 sw s1, 8(sp)
122 sw s0, 4(sp)

```

```

123     sw ra, 0(sp)
124
125     # ~~ 3. calle(READORWRITE) function execution ~~
126
127     # ~ init vars ~
128     addi t0, zero, 0      # t0 = i = 0
129     addi t1, zero, 0      # t1 = arr + i
130                             # t2 is local temp
131     la t3, TC             # t3 is a test case var (array base addr)
132                             # t4 is a test case var (dummy)
133
134     slli a1, a1, 2        # shift s.t. we dont use reg for i and 4i
135
136     # ~ for(i=0; i < 10; i++) ~
137 For_ROW:
138     bgeu t0, a1, End_For_ROW
139     add t1, a0, t0        # t1 = arr + 4i
140
141
142     beqz a3, ReadMMIO
143     # ~ write to MMIO addr ~
144     lw t2, 0(t1)          # t2 = *(arr + 4i)
145     sw t2, 0(a2)          # MMIO[LEDS]= t2
146     j End_ReadMMIO
147
148
149 ReadMMIO:
150     # ~~~~ TC injection point ~~~~
151     addi t4, t0, 0        # t4 = 4i
152     add t4, t4, t3        # t4 = TC + 4i
153     lw t4, 0(t4)          # t4 = *(TC + 4i)
154     sw t4, 0(a2)          # MMIO[SWITCHES] = t4
155     # ~~~~~~
156     # ~ read from MMIO addr ~
157     lw t2, 0(a2)          # read SW : t2 = MMIO[SWITCHES]
158     sw t2, 0(t1)          # store in arr : *(arr+4i) = t2
159 End_ReadMMIO:
160
161
162     addi t0, t0, 4        # i++
163     j For_ROW
164 End_For_ROW:
165
166     # ~~ 4. calle(READORWRITE) restores preserved from stack ~~
167     lw s3, 16(sp)
168     lw s2, 12(sp)
169     lw s1, 8(sp)
170     lw s0, 4(sp)
171     lw ra, 0(sp)
172
173     # ~~ 5. calle(READORWRITE) deallocates space on stack ~~
174     addi sp, sp, 20
175
176     # ~~ calle(READORWRITE) return ~~
177     jr ra
178 # ~~~~~~
179
180 # ~~~~~~BubbleSort()~~~~~
181 # Purpose : Sort array in ascending order
182 #
183 #
184 # Input :
185 #     a0 : [unsigned 32-bit] base address of array
186 #     a1 : [unsigned 32-bit] num elem in array
187 #
188 # Note :
189 #
190 # Return:
191 #
192
193 BubbleSort:
194
195     # ~~~~ calle(BubbleSort) saves preserved ~~~~
196     # ~~ 1. calle(BubbleSort) allocates space on stack ~~
197     addi sp, sp, -20
198
199     # ~~ 2. calle(BubbleSort) stores reg values on stack ~~
200     sw s3, 16(sp)
201     sw s2, 12(sp)
202     sw s1, 8(sp)
203     sw s0, 4(sp)
204     sw ra, 0(sp)
205
206     # ~~ 3. calle(BubbleSort) function execution ~~
207
208     slli a1, a1, 2        # shift s.t. we dont use reg for i and 4i
209
210     # ~ init vars ~
211     addi t0, zero, 0      # t0 = i = 0 (4i)
212     addi t1, a1, -4       # t1 = n - 1
213                             # t2 = j = 0

```

```

214             # t3 = n - 1 - i
215             # t4 = arr + j
216             # t5, t6 = dummy
217
218 For_i_Bubble: # for(int i = 0; i < n-1; i++)
219     bgeu t0, t1, End_For_i_Bubble
220
221
222     addi t2, zero, 0           # j = 0
223     sub t3, t1, t0            # n - 1 - i
224 For_j_Bubble: # for(int j = 0; j < n-1-i; j++)
225     bgeu t2, t3, End_For_j_Bubble
226
227     add t4, a0, t2            # t4 = arr + j
228     lw t5, 0(t4)              # t5 = *(arr + j)
229     lw t6, 4(t4)              # t6 = *(arr + j + 1)
230
231     # if(t6 < t5)
232     bgt t6, t5, NoSwap
233     sw t6, 0(t4)              # arr[j] = t6
234     sw t5, 4(t4)              # arr[j+1] = t5
235 NoSwap:
236
237     addi t2, t2, 4            # j++
238     j For_j_Bubble
239 End_For_j_Bubble:
240
241     addi t0, t0, 4            # i++
242     j For_i_Bubble
243 End_For_i_Bubble:
244
245 # ~~ 4. calle(BubbleSort) restores preserved from stack ~~
246 lw s3, 16(sp)
247 lw s2, 12(sp)
248 lw s1, 8(sp)
249 lw s0, 4(sp)
250 lw ra, 0(sp)
251
252 # ~~ 5. calle(BubbleSort) deallocates space on stack ~~
253 addi sp, sp, 20
254
255 # ~~ calle(BubbleSort) return ~~
256 jr ra
257 # ~~~~~

```

0.2.5 SW5 - Division and Stacks

- Assignment:**
First, create a flowchart that will perform the following behavior. Then implement the flowchart with OTTER MCU assembly language using only the instructions in the OTTER assembly manual. Use register alternative names (ie s0, t0) instead of register numbers (ie x5, x10). Make sure your code is formatted properly including comments. Use the RARS simulator to ensure the program performs as desired.
1. Read two 16-bit unsigned values from an array in the data segment. Divide the first value by the 2nd and output the quotient (result) to 7 SEGMENT (address 0x11000040) and the remainder to LEDS (address 0x11000020).
 2. Read 32-bit values repeatedly from SWITCHES (address 0x11000000). Each value should be added to the stack until the value 0xFFFFFFFF is read. When 0xFFFFFFFF is input the program will not add it to the stack but instead will output all of the saved values to LEDS (address 0x11000020) in the reverse order they were input. Specify the limits of your program. (Assume it is possible for the switches to change between each time they are read)

Assume the BASYS3 SWITCHES and LEDS are expanded to 32 individual switches and LEDs.

Figure 80: Questions to address

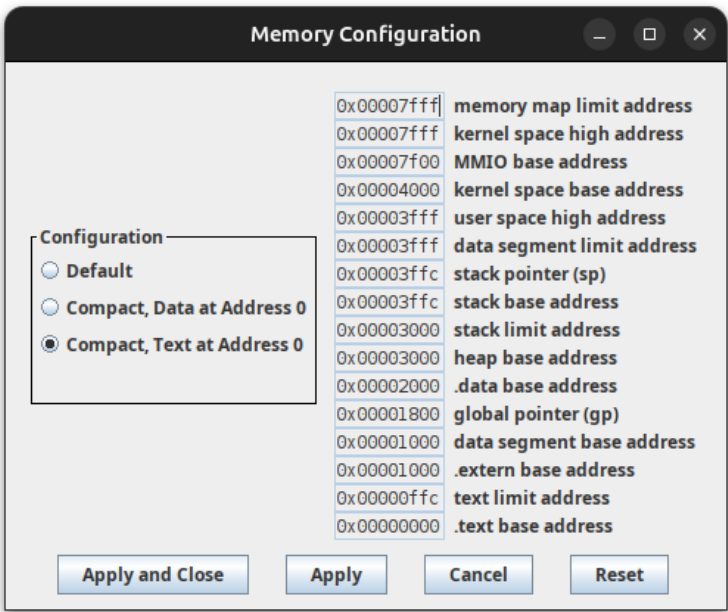


Figure 81: Memory Map for this Assignment

Part 1

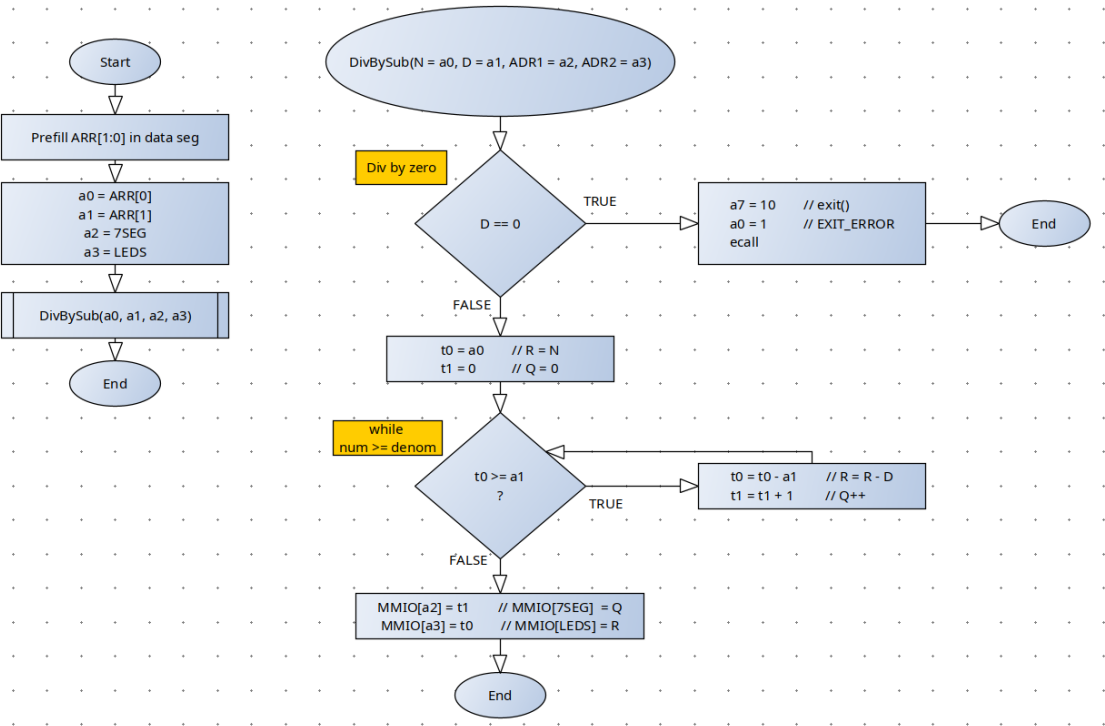


Figure 82: Flow chart for part 1

Listing 13: sw5_p1.asm

```

1 # Author : WhisperingRock
2 #
3 # Purpose : Division by repeat subtraction
4 # Input :
5 #     - array of two unsigned 16-bit elems
6 #     - 7SEG & LEDS MMIO address to 32 bits
7 #
8 # Output :
9 #     Write to 7SEG and LEDS addr
10
11
12 .data
13 # ~~~~ Test Cases ~~~~
14
15 # ~~ TC1 : 0 / 1 ~~
16 #TC:      .half 0, 1
17 # answer : *verified*
18 # Q = 0
19 # R = 0
20
21 # ~~ TC2 : 1 / 0 ~~
22 #TC:      .half 1, 0
23 # answer : *verified*
24 # RaiseError
25 #
26
27 # ~~ TC3 : 1 / 1 ~~
28 #TC:      .half 1, 1
29 # answer : *verified*
30 # Q = 1
31 # R = 0
32
33 # ~~ TC4 : 2 / 1 ~~
34 #TC:      .half 2, 1
35 # answer : *verified*
36 # Q = 2
37 # R = 0
38
39 # ~~ TC5 : 1 / 2 ~~
40 #TC:      .half 1, 2
41 # answer : *verified*
42 # Q = 0
43 # R = 1
44
45 # ~~ TC6 : half max'd ~~
46 #TC:      .half 0xFFFF, 0xFFFF
47 # answer : *verified*
48 # Q = 1
49 # R = 0
50
51 # ~~ TC7 : quotient max'd ~~
52 #TC:      .half 0xFFFF, 1
53 # answer : *verified*
54 # Q = 0xFFFF
55 # R = 0
56
57 # ~~ TC8 : prime1 ~~
58 #TC:      .half 7919, 13
59 # answer : *verified*
60 # Q = 609
61 # R = 2
62
63 # ~~ TC9 : prime2 ~~
64 #TC:      .half 9973, 241
65 # answer : *verified*
66 # Q = 41
67 # R = 92
68
69 # ~~ TC10 : prime3 ~~
70 #TC:      .half 0xFFFF, 149
71 # answer : *verified*
72 # Q = 439
73 # R = 124
74
75 # ~~ TC11 : 0 / 0 ~~
76 #TC:      .half 0, 0
77 # answer : *verified*
78 # RaiseError
79 #
80
81 # ~~ TC12 : denom max'd ~~
82 #TC:      .half 1, 0xFFFF
83 # answer : *verified*
84 # Q = 0
85 # R = 1
86
87
88 .global _start
89

```

```

90 .text
91 #.equ MMIO_SWITCHES, 0x00007F00
92 #.equ MMIO_SEVSEG, 0x00007F40
93 #.equ MMIO_LEDS, 0x00007F20
94
95 _start:
96 # ~~~~ init reg + TC injection ~~~~
97 la s0, TC # s0 is the addr for ARR
98 li s1, MMIO_SEVSEG # a2 is the addr for MMIO 7SEG
99 li s2, MMIO_LEDS # a2 is the addr for MMIO LEDES
100
101 # ~~~~ caller(_start) saves non-preserved ~~~~
102 # ~~ 1. caller(_start) allocates space on stack ~~
103 # ~~ 2. caller(_start) stores reg values on stack ~~
104 # ~~ 3. caller(_start) function execution ~~
105 lhu a0, 0(s0) # a0 is N
106 lhu a1, 2(s0) # a1 is D; NOTE: halfword spacing
107 addi a2, s1, 0 # a2 is the addr for MMIO 7SEG
108 addi a3, s2, 0 # a3 is the addr for MMIO LEDES
109 jal DivBySub
110 # ~~ 4. caller(_start) restores preserved from stack ~~
111 # ~~ 5. caller(_start) deallocates space on stack ~~
112
113 # ~~~~ syscall exit ~~~~
114 li a7, 10 # load syscall num for exit()
115 li a0, 0 # EXIT_SUCCESS code
116 ecall # execute sys call
117
118
119 # ~~~~~~
120 # Purpose : Perform division by subtraction, writing the quotient to
121 # addr1 and the remainder to addr2
122 #
123 #
124 # Input :
125 # a0 : [unsigned 16-bit] Numerator variable N
126 # a1 : [unsigned 16-bit] Denominator variable D
127 # a2 : [unsigned 32-bit] addr1
128 # a3 : [unsigned 32-bit] addr2
129 #
130 # Return:
131 #
132
133 DivBySub:
134
135 # ~~~~ calle(DivBySub) saves preserved ~~~~
136 # ~~ 1. calle(DivBySub) allocates space on stack ~~
137 addi sp, sp, -16
138
139 # ~~ 2. calle(DivBySub) stores reg values on stack ~~
140 sw s2, 12(sp)
141 sw s1, 8(sp)
142 sw s0, 4(sp)
143 sw ra, 0(sp)
144
145 # ~~ 3. calle(DivBySub) function execution ~~
146
147 # ~ error ~
148 blez a1, RaiseError
149
150 # ~ init ~
151 addi t0, a0, 0 # R = t0 = N
152 addi t1, zero, 0 # Q = t1 = 0
153
154 While: # ~ while div produces integers s.t. (R >= D) ~
155 # (R>= D) = (D > R)
156 bgtu a1, t0, EndWhile
157 sub t0, t0, a1 # R = R - D
158 addi t1, t1, 1 # Q++
159 j While
160 EndWhile:
161 # ~ write to MMIOs ~
162 sw t1, 0(a2) # Q goes to addr1
163 sw t0, 0(a3) # R goes to addr2
164
165 # ~~ 4. calle(DivBySub) restores preserved from stack ~~
166 lw s2, 12(sp)
167 lw s1, 8(sp)
168 lw s0, 4(sp)
169 lw ra, 0(sp)
170
171 # ~~ 5. calle(DivBySub) deallocates space on stack ~~
172 addi sp, sp, 16
173
174 # ~~ calle return ~~
175 jr ra
176 # ~~~~~~
177
178 RaiseError:
179 # ~~~~ syscall exit ~~~~
180 li a7, 10 # load syscall num for exit()

```

```

181  li a0, 1      # EXIT_FAILURE code
182  ecall        # execute sys call

```

Part 2

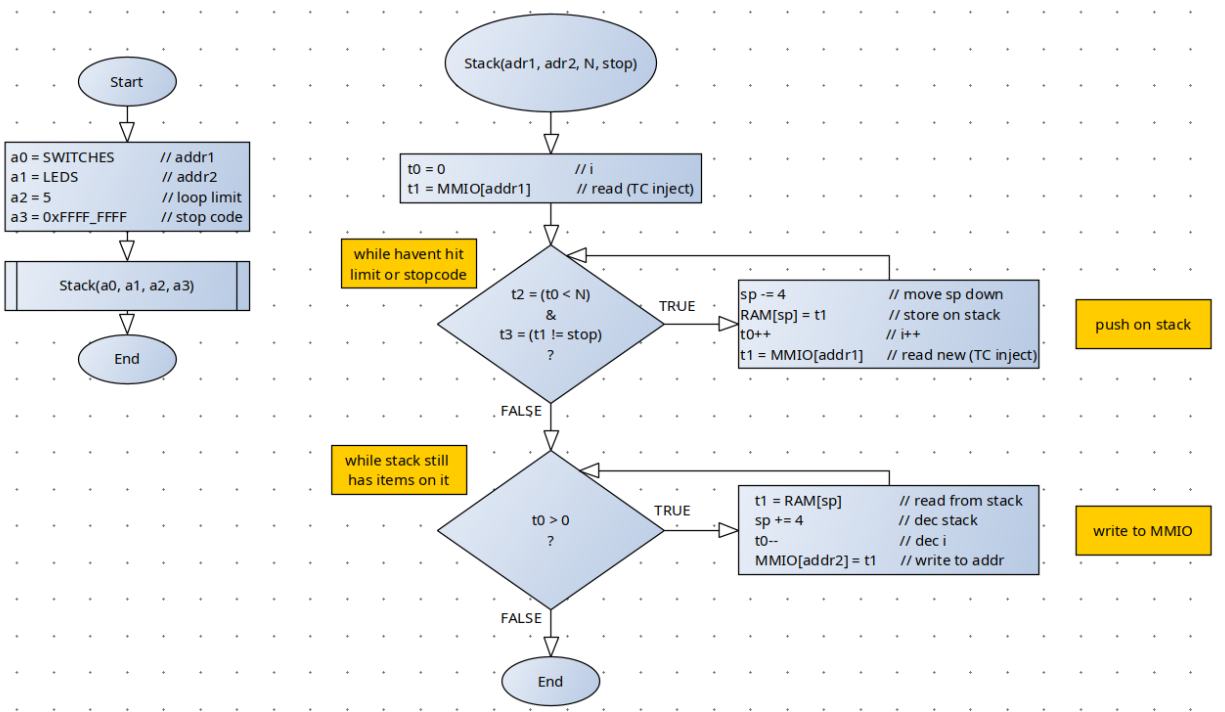


Figure 83: Flow chart for part 2

Listing 14: Assembly code for sw5_p2.asm

```

1 # Author : WhisperingRock
2 #
3 # Purpose : Stack using the stack
4 # Input :
5 #     - SWITCHES & LEDES MMIO address to 32 bits
6 #
7 # Output :
8 #     - conseq write to LEDES addr
9
10
11 .data
12 # ~~~~ Test Cases ~~~~
13
14 # ~~ TC1 : loop limited 1 ~~
15 #TC:      .word 5, 4, 3, 2, 1
16 # answer : 1, 2, 3, 4, 5 (conseq writes)
17 # verified!
18
19 # ~~ TC2 : stopcode limited 1 ~~
20 #TC:      .word 5, 4, 0xFFFFFFFF, 2, 1
21 # answer : 4, 5 (conseq writes)
22 # verified!
23
24 # ~~ TC3 : immediate stopcode ~~
25 #TC:      .word 0xFFFFFFFF, 5, 4, 3, 2, 1
26 # answer : *no writes*
27 # verified!
28
29 # ~~ TC4 : loop limit is zero ~~
30 # ~ NOTE : change LIMIT to 0 ~
31 #TC:      .word 5, 4, 3, 2, 1
32 # answer : *no writes*
33 # verified!
34
35 # ~~ TC5 : near ceiling ~~
36 #TC:      .word 0xFFFFFFF, 0xFFFFFFF, 0xFFFFFFF, 0xFFFFFFF, 0xFFFFFFF, 0
37 # answer : 0xFFFFFFF, 0xFFFFFFF, 0xFFFFFFF, 0xFFFFFFF, (conseq writes)
38 # verified!
39
40 # ~~ TC6 : stopcode and loop limits at same time ~~
41 #TC:      .word 5, 4, 3, 2, 1, 0xFFFFFFFF, 999
42 # answer : 1, 2, 3, 4, 5 (conseq writes)
43 # verified!
44
45 # ~~ TC7 : beyond loop limit ~~
46 #TC:      .word 10, 9, 8, 7, 6, 5, 4, 3, 2, 1
47 # answer : 6, 7, 8, 9, 10 (conseq writes)
48 # verified!
49
50 .global _start

```

```

51
52 .text
53
54 .equ MMIO_SWITCHES, 0x00007F00
55 #.equ MMIO_SEVSEG, 0x00007F40
56 .equ MMIO_LEDS, 0x00007F20
57
58 .equ LIMIT, 5
59 .equ STOPCODE, 0xFFFFFFFF
60
61 _start:
62 # ~~~~ init reg + TC injection ~~~~
63 la s0, TC # s0 is the addr for TC array
64 li s1, MMIO_SWITCHES # a2 is the addr for MMIO SWITCHES
65 li s2, MMIO_LEDS # a2 is the addr for MMIO LEDs
66
67 # ~~~~ caller(_start) saves non-preserved ~~~~
68 # ~~ 1. caller(_start) allocates space on stack ~~
69 # ~~ 2. caller(_start) stores reg values on stack ~~
70 # ~~ 3. caller(_start) function execution ~~
71 addi a0, s1, 0 # addr1
72 addi a1, s2, 0 # addr2
73 li a2, LIMIT # loop limit
74 li a3, STOPCODE # stop value
75 jal Stack
76 # ~~ 4. caller(_start) restores preserved from stack ~~
77 # ~~ 5. caller(_start) deallocates space on stack ~~
78
79 # ~~~~ syscall exit ~~~~
80 li a7, 10 # load syscall num for exit()
81 li a0, 0 # EXIT_SUCCESS code
82 ecall # execute sys call
83
84
85 # ~~~~~~
86 # Purpose : Read values from addr1 onto the stack until the loop limit
87 # or stop code is reached. Then write the values, from the stack
88 # to addr2.
89 #
90 #
91 # Input :
92 # a0 : [unsigned 32-bit] read address
93 # a1 : [unsigned 32-bit] write address
94 # a2 : [unsigned 32-bit] loop limit
95 # a3 : [unsigned 32-bit] stop code
96 #
97 # Note :
98 # - using s0 as a global for TC injection
99 # Return:
100 #
101
102 Stack:
103
104 # ~~~~ calle(Stack) saves preserved ~~~~
105 # ~~ 1. calle(Stack) allocates space on stack ~~
106 addi sp, sp, -16
107
108 # ~~ 2. calle(Stack) stores reg values on stack ~~
109 sw s2, 12(sp)
110 sw s1, 8(sp)
111 sw s0, 4(sp)
112 sw ra, 0(sp)
113
114 # ~~ 3. calle(Stack) function execution ~~
115 # ~ init ~
116 addi t0, zero, 0 # t0 = i = 0
117 # t1, t2, t3 = read and logic temps
118 # t4 = TC arr index variable
119 # ---- TC Injection ----
120 slli t4, t0, 2 # t4 = 4i (TC)
121 add t4, t4, s0 # t4 = TC + 4i
122 lw t4, 0(t4) # t4 = *(TC + 4i)
123 sw t4, 0(a0) # MMIO[addr1] = t4
124 # -----
125 lw t1, 0(a0) # t1 = MMIO[addr1]
126 ReadWhile:
127 sltu t2, t0, a2 # t2 = (i < limit)
128 seqz t2, t2
129 xor t3, t1, a3 # t3 = (t1 != stopcode)
130 seqz t3, t3
131 or t3, t3, t2 # t3 = t3 && t2
132 bgt t3, zero, End_ReadWhile # while (i < limit) && (t1 != stopcode)
133 addi sp, sp, -4 # sp -= 4
134 sw t1, 0(sp) # RAM[sp] = read
135 addi t0, t0, 1 # i++
136 # ---- TC Injection ----
137 slli t4, t0, 2 # t4 = 4i (TC)
138 add t4, t4, s0 # t4 = TC + 4i
139 lw t4, 0(t4) # t4 = *(TC + 4i)
140 sw t4, 0(a0) # MMIO[addr1] = t4
141 # -----

```

```

142     lw t1, 0(a0)           # t1 = MMIO[addr1]
143     j ReadWhile
144 End_ReadWhile:
145 WriteWhile:
146     beqz t0, End_WriteWhile # while(i > 0)
147     lw t1, 0(sp)           # t1 = RAM[sp]
148     addi sp, sp, 4         # sp += 4
149     addi t0, t0, -1        # i--
150     sw t1, 0(a1)          # MMIO[addr2] = t1
151     j WriteWhile
152 End_WriteWhile:
153
154 # ~~ 4. calle(Stack) restores preserved from stack ~~
155 lw s2, 12(sp)
156 lw s1, 8(sp)
157 lw s0, 4(sp)
158 lw ra, 0(sp)
159
160 # ~~ 5. calle(Stack) deallocates space on stack ~~
161 addi sp, sp, 16
162
163 # ~~ calle return ~~
164 jr ra
165 # ~~~~~~
166
167 RaiseError:
168 # ~~~~ syscall exit ~~~~
169 li a7, 10      # load syscall num for exit()
170 li a0, 1       # EXIT_FAILURE code
171 ecall         # execute sys call

```

0.2.6 SW6 - Subroutines

- 1. Create a subroutine that will divide a value by 10. The subroutine should utilize appropriate registers for passing parameters, returning results, and only affect temporaries. Write a main program that reads a 16-bit unsigned value from an array in the data segment and converts the value into a 5 digit, BCD (binary coded decimal) value using the divide by 10 subroutine. For example, if the input value is 0x3A6C (14956 decimal), it should be split into 5 nibbles (4-bits) for 1, 4, 9, 5, and 6, so the final answer would be 0x14956. The resulting 20-bit BCD can be written as a 32-bit word to another array in the data segment. The arrays can be of any length.
- 2. Create a subroutine that implements the following greatest common denominator function given below. The subroutine should utilize appropriate registers for passing parameters, returning results, and only affect temporaries. Write a main program that reads two 16-bit unsigned values from an array in the data segment, uses the subroutine to find the GCD, and saves the result to another array in the data segment. *Do not replace the subroutine call with a loop.* You can assume all values are unsigned and greater than 0. Specify any limits of your program.

```
int gcd(int a0, int a1){
    if (a0 > a1)
        a0 = gcd(a0-a1,a1);
    else if (a0 < a1)
        a0 = gcd(a0,a1-a0);
    return a0;
}
```

Code S6.1: GCD Subroutine Code

Figure 84: Questions to address

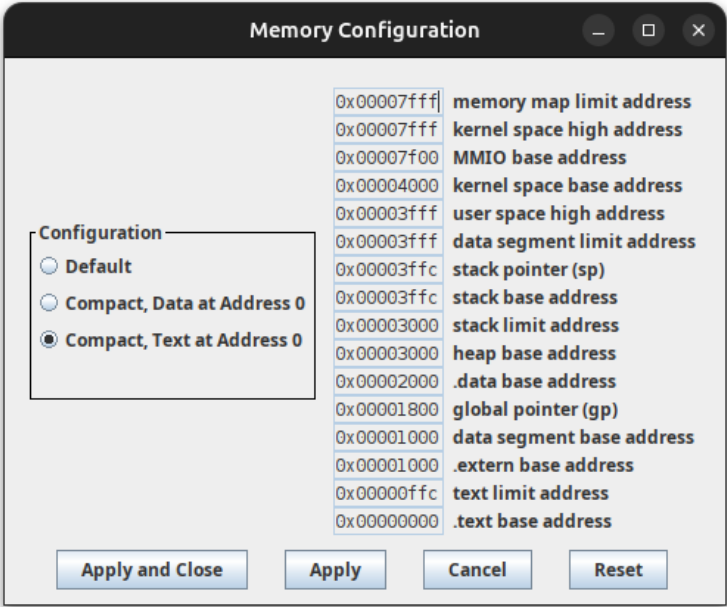


Figure 85: Memory Map for this Assignment

Part 1

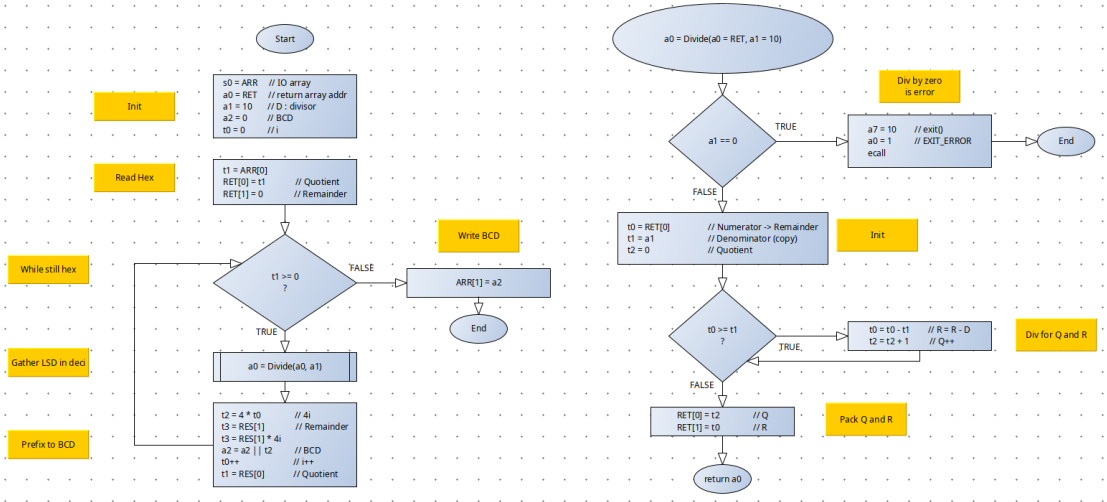


Figure 86: Flow chart for part 1

Listing 15: sw6_p1.asm

```

1 # Author : WhisperingRock
2 #
3 # Purpose : Convert an unsigned hex to BCD, where each hex char
4 #           in the result is actually base 10 decimal.
5 # Input :
6 #         - A single read from an unsigned 16-bit array
7 #
8 # Output :
9 #         - A single 32-bit write to a 32-bit array
10
11
12 .data
13 DIVSET:    .word 0, 0      # Quotient[0] and Remainder[1]
14
15 # ~~~~ Test Cases ~~~~
16 # ~~ TC1 : one decimal digit ~~
17 #TC:      .half    0x7
18 #RES:     .word 0
19 # answer : 0x00007    *verified*
20
21 # ~~ TC2 : two decimal digit ~~
22 #TC:      .half    0xA
23 #RES:     .word 0
24 # answer : 0x00010    *verified*
25
26 # ~~ TC3 : given ~~
27 #TC:      .half    0x3A6C
28 #RES:     .word 0
29 # answer : 0x14956    *verified*
30
31 # ~~ TC4 : max input for 16 unsign bit~~
32 #TC:      .half    0xFFFF
33 #RES:     .word 0
34 # answer : 0x65535    *verified*
35
36 # ~~ TC5 : zero ~~
37 #TC:      .half    0x0
38 #RES:     .word 0
39 # answer : 0x0        *verified*
40
41 # ~~ TC6 : lucky ~~
42 #TC:      .half    0x309
43 #RES:     .word 0
44 # answer : 0x777      *verified*
45
46 .global _start
47
48 .text
49 #.equ MMIO_SWITCHES, 0x00007F00
50 #.equ MMIO_SEVSEG,   0x00007F40
51 #.equ MMIO_LEDS,     0x00007F20
52 .equ DIVISOR, 10
53
54 _start:
55 # ~~~~ init reg + TC injection ~~~~
56 # ~~ preserved ~~
57 la s0, TC      # s0 is the addr for read array
58 la s1, RES     # s1 is the addr for write array
59 # ~~ non-preserved ~~
60 la a0, DIVSET  # a0 is an array for passing args
61 addi a1, zero, DIVISOR # a1 is the divisor arg (const)
62 addi a2, zero, 0 # a2 is working version of BCD
63 addi t0, zero, 0 # t0 = i = 0
64 # t1 is working var
65
66 # ~~~~ read and load hex from array ~~~~
67 lhu t1, 0(s0)
68 sw t1, 0(a0)    # Numerator is initial quotient
69 sw zero, 4(a0)  # Remainder
70
71 # ~~~~ while (numerator >= 0) s.t. hex can be further converted ~~~~
72 MainWhile:
73 blez t1, End_MainWhile
74
75 # ~~~~ caller(_start) saves non-preserved ~~~~
76 # ~~ 1. caller(_start) allocates space on stack ~~
77 addi sp, sp, -20
78 # ~~ 2. caller(_start) stores reg values on stack ~~
79 sw t1, 16(sp)
80 sw a0, 12(sp)
81 sw a1, 8(sp)
82 sw a2, 4(sp)
83 sw t0, 0(sp)
84 # ~~ 3. caller(_start) function execution ~~
85 # a0 remains addr (const)
86 # a1 remains const (10)
87 jal DivBySub
88 # return a0 should remain as above
89 # ~~ 4. caller(_start) restores preserved from stack ~~

```

```

90  lw t1, 16(sp)
91  lw a0, 12(sp)
92  lw a1, 8(sp)
93  lw a2, 4(sp)
94  lw t0, 0(sp)
95  # ~~ 5. caller(_start) deallocates space on stack ~~
96  addi sp, sp, 20
97
98  # ~~~~ prefix hex to BCD ~~~~
99  slli t2, t0, 2          # t2 = 4i
100 lw t3, 4(a0)           # t3 = RES[1] = Remainder
101 sll t3, t3, t2          # remainder shifted to correct hex spot
102 or a2, a2, t3          # prefix remainder to BCD
103 addi t0, t0, 1         # i++
104 lw t1, 0(a0)           # t1 = RES[0] = Quotinet
105
106 j MainWhile
107 End_MainWhile:
108
109 sw a2, 0(s1)           # write BCD to result array
110
111 # ~~~~ syscall exit ~~~~
112 li a7, 10              # load syscall num for exit()
113 li a0, 0                # EXIT_SUCCESS code
114 ecall                  # execute sys call
115
116
117 # ~~~~~~
118 # Purpose : Perform division by subtraction, writing the quotient and
119 #           remainder to a single array.
120 #
121 # Input :
122 #   a0 : [signed 32-bit] Array of two elements
123 #       a[0] is Numerator
124 #   a1 : [signed 32-bit] Denominator
125 #
126 # Return:
127 #   address to array stored in a0, with
128 #       arr[0] storing the quotient
129 #       arr[1] storing the remainder
130
131 DivBySub:
132
133 # ~~~~ calle(DivBySub) saves preserved ~~~~
134 # ~~ 1. calle(DivBySub) allocates space on stack ~~
135 addi sp, sp, -12
136
137 # ~~ 2. calle(DivBySub) stores reg values on stack ~~
138 sw s1, 8(sp)
139 sw s0, 4(sp)
140 sw ra, 0(sp)
141
142 # ~~ 3. calle(DivBySub) function execution ~~
143
144 # ~ Edge case : div by zero ~
145 blez a1, RaiseError
146
147 # ~ init ~
148 lw t0, 0(a0)           # R = N = RES[0]
149 addi t1, a1, 0         # D = a1 = 10
150 addi t2, zero, 0       # Q = 0
151
152 DivWhile: # ~ while div produces integers s.t. (R >= D) ~
153 # (R>= D) = (D > R)
154 bgtu t1, t0, End_DivWhile
155 sub t0, t0, t1          # R = R - D
156 addi t2, t2, 1         # Q++
157
158 j DivWhile
159 End_DivWhile:
160
161 # ~ write to return array ~
162 sw t2, 0(a0)           # Q saved in a0[0]
163 sw t0, 4(a0)           # R saved in a0[1]
164
165 # ~~ 4. calle(DivBySub) restores preserved from stack ~~
166 lw s1, 8(sp)
167 lw s0, 4(sp)
168 lw ra, 0(sp)
169
170 # ~~ 5. calle(DivBySub) deallocates space on stack ~~
171 addi sp, sp, 12
172
173 # ~~ calle return ~~
174 jr ra
175 # ~~~~~~
176
177 RaiseError:
178 # ~~~~ syscall exit ~~~~
179 li a7, 10              # load syscall num for exit()
180 li a0, 1                # EXIT_FAILURE code

```

Part 2

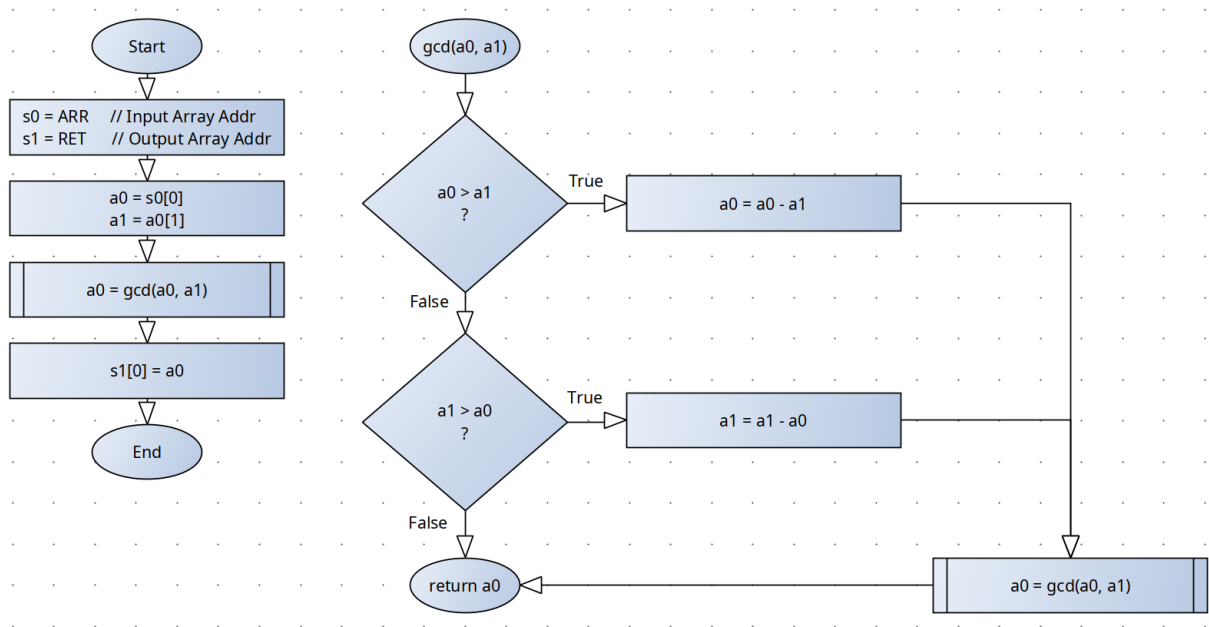


Figure 87: Flow chart for part 2

Listing 16: Assembly code for sw6_p2.asm

```
1 # Author : WhisperingRock
2 #
3 # Purpose : Find the greatest common denominator using recursion
4 #
5 # Assumptions : All input unsigned and greater than zero.
6 #
7 # Input :
8 #   - TC : [16-bit unsigned] array with at least 2 values
9 #
10 # Output :
11 #   - the gcd between the two numbers, written to a result array
12
13
14 .data
15
16 # ~~~~ Test Cases ~~~~
17 # ~~ TC1 : one decimal digit 1 ~~
18 #TC:    .half    8, 16
19 #RES: .word 0
20 # answer : 8 *verified*
21
22 # ~~ TC2 : one decimal digit rev ~~
23 #TC:    .half    16, 8
24 #RES: .word 0
25 # answer : 8 *verified*
26
27 # ~~ TC3 : same | floor ~~
28 #TC:    .half    1, 1
29 #RES: .word 0
30 # answer : 1 *verified*
31
32 # ~~ TC4 : same | ceiling ~~
33 #TC:    .half    0xFF, 0xFF
34 #RES: .word 0
35 # answer : 0xFF *verified*
36
37 # ~~ TC5 : not input 1 ~~
38 #TC:    .half    255, 102
39 #RES: .word 0
40 # answer : 51 *verified*
41
42 # ~~ TC6 : not input 1 rev ~~
43 #TC:    .half    102, 255
44 #RES: .word 0
45 # answer : 51 *verified*
46
47 # ~~ TC7 : one #1~~
48 #TC:    .half    1, 255
49 #RES: .word 0
50 # answer : 1 *verified*
51
52 # ~~ TC8 : one rev #2 ~~
53 #TC:    .half    255, 1
54 #RES: .word 0
```

```

55 # answer : 1 *verified*
56
57 .global _start
58
59 .text
60 #.equ MMIO_SWITCHES, 0x00007F00
61 #.equ MMIO_SEVSEG, 0x00007F40
62 #.equ MMIO_LEDS, 0x00007F20
63
64 _start:
65 # ~~~~ init reg + TC injection ~~~~
66 # ~~ preserved ~~
67 la s0, TC # s0 is the addr for read array
68 la s1, RES # s1 is the addr for write array
69 # ~~ non-preserved ~~
70 # a0 first arg (consumed)
71 # a1 second arg (consumed)
72
73 # ~~~~ read from array (TC injection) ~~~~
74 lhu a0, 0(s0)
75 lhu a1, 2(s0)
76
77 # ~~~~ caller(_start) saves non-preserved ~~~~
78 # ~~ 1. caller(_start) allocates space on stack ~~
79 # ~~ 2. caller(_start) stores reg values on stack ~~
80 # ~~ 3. caller(_start) function execution ~~
81 jal GCD
82 # ~~ 4. caller(_start) restores preserved from stack ~~
83 # ~~ 5. caller(_start) deallocates space on stack ~~
84
85 sw a0, 0(s1) # write GCD to result array
86
87 # ~~~~ syscall exit ~~~~
88 li a7, 10 # load syscall num for exit()
89 li a0, 0 # EXIT_SUCCESS code
90 ecall # execute sys call
91
92
93 # ~~~~~~
94 # Purpose : Find the GCD by recursively subtracting til equal
95 #
96 #
97 # Input :
98 # a0 : [unsigned 16-bit]
99 # a1 : [unsigned 16-bit]
100 #
101 # Return:
102 # a0 : [unsigned 16-bit] gcd between a0 and a1
103
104 GCD:
105
106 # ~~~~ calle(GCD) saves preserved ~~~~
107 # ~~ 1. calle(GCD) allocates space on stack ~~
108 addi sp, sp, -4
109 # ~~ 2. calle(GCD) stores reg values on stack ~~
110 sw ra, 0(sp)
111 # ~~ 3. calle(GCD) function execution ~~
112
113 bltu a1, a0, GCD_If
114 bltu a0, a1, GCD_ElseIf
115 j GCD_IfEnd
116
117 GCD_If:
118 sub a0, a0, a1
119
120 # a0 and a1 don't need to be preserved
121 # ~~~~ caller(GCD) saves non-preserved ~~~~
122 # ~~ 1. caller(GCD) allocates space on stack ~~
123 # ~~ 2. caller(GCD) stores reg values on stack ~~
124 # ~~ 3. caller(GCD) function execution ~~
125 jal GCD
126 # ~~ 4. caller(GCD) restores preserved from stack ~~
127 # ~~ 5. caller(GCD) deallocates space on stack ~~
128
129 j GCD_IfEnd
130
131 GCD_ElseIf:
132 sub a1, a1, a0
133 # a0 and a1 don't need to be preserved
134 # ~~~~ caller(GCD) saves non-preserved ~~~~
135 # ~~ 1. caller(GCD) allocates space on stack ~~
136 # ~~ 2. caller(GCD) stores reg values on stack ~~
137 # ~~ 3. caller(GCD) function execution ~~
138 jal GCD
139 # ~~ 4. caller(GCD) restores preserved from stack ~~
140 # ~~ 5. caller(GCD) deallocates space on stack ~~
141
142 GCD_IfEnd:
143
144 # ~~ return a0 as itself ~~
145

```

```

146 # ~~ 4. calle() restores preserved from stack ~~
147 lw ra, 0(sp)
148 # ~~ 5. calle() deallocates space on stack ~~
149 addi sp, sp, 4
150
151 # ~~ calle return ~~
152 jr ra
153 # ~~~~~
154
155 RaiseError:
156 # ~~~~ syscall exit ~~~~
157 li a7, 10      # load syscall num for exit()
158 li a0, 1      # EXIT_FAILURE code
159 ecall         # execute sys call

```

0.3 Otter Code & Test Sims

0.3.1 Mux(s)

Listing 17: 2 input Mux Mux_2N.sv

```
1  `timescale 1ns / 1ps
2  //////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 08/19/2026 10:35:06 AM
7  // Design Name:
8  // Module Name: Mux_2N
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //////////////////////////////////////////////////
21 module Mux_2N
22 #(parameter width = 32)
23 (
24     input logic[width-1:0] D0, D1,
25     input logic S,
26     output logic[width-1:0] Y
27 );
28
29     always_comb begin
30         Y = (S == 1'b1) ? D1 : D0;
31     end
32
33 endmodule
```

Listing 18: 4 input Mux Mux_4N.sv

```
1  `timescale 1ns / 1ps
2  //////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 08/19/2026 09:15:04 AM
7  // Design Name:
8  // Module Name: Mux_4N
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //////////////////////////////////////////////////
21 module Mux_4N
22 #(parameter width = 32)
23 (
24     input logic[width-1:0] D0, D1, D2, D3,
25     input logic[1:0] S,
26     output logic[width-1:0] Y
27 );
28
29     logic[width-1:0] ERR = 0;
30
31     always_comb
32     begin
33         case(S)
34             0: Y = D0;
35             1: Y = D1;
36             2: Y = D2;
37             3: Y = D3;
38             default Y = (~ERR) -13 ;
39         endcase
40     end
41 endmodule
```

Listing 19: 6 input Mux Mux_6N.sv

```
1  `timescale 1ns / 1ps
2  //////////////////////////////////////////////////
3  // Company:
4  // Engineer:
```

```

5 //
6 // Create Date: 07/17/2026 04:24:06 PM
7 // Design Name:
8 // Module Name: Mux_6N
9 // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module Mux_6N
24 #(parameter width = 32)                                // default bits
25 (
26     input logic[width-1:0] D0, D1, D2, D3, D4, D5,      // data channels
27     input logic[2:0] S,                                  // selector
28     output logic[width-1:0] Y
29 );
30
31     logic[width-1:0] ERR = 0;
32
33     always_comb
34         case(S)
35             0: Y = D0;
36             1: Y = D1;
37             2: Y = D2;
38             3: Y = D3;
39             4: Y = D4;
40             5: Y = D5;
41             default Y = (~ERR) -13 ;
42         endcase
43
44 endmodule

```

Listing 20: TestSim for 6 input Mux Mux_6N_tb.sv

```

1 `timescale 1ns / 1ps
2 ///////////////////////////////////////////////////////////////////
3 // Company:
4 // Engineer:
5 //
6 // Create Date: 07/17/2026 04:26:27 PM
7 // Design Name:
8 // Module Name: Mux_6N_tb
9 // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module Mux_6N_tb();
24
25     // ~~ local variables ~~
26     logic CLK;
27     logic[31:0] d0, d1, d2, d3, d4, d5;
28     logic[3:0] sel;
29     logic[31:0] out;
30
31
32     // ~~ high-level module instances (and hooks) ~~
33     Mux_6N #(32) UUT (
34         .D0(d0),
35         .D1(d1),
36         .D2(d2),
37         .D3(d3),
38         .D4(d4),
39         .D5(d5),
40         .S(sel),
41         .Y(out)
42     );
43
44     // ~~ action : CLK (10ns period) ~~
45     always begin
46         #5;
47         CLK <= !CLK;
48     end

```

```

49
50 // ~~ action : Data ~~
51 initial begin
52
53     // ~ init ~
54     CLK = 0;
55     d0 = 32'h0000_0000;
56     d1 = 32'h1234_0000;
57     d2 = 32'h0000_5678;
58     d3 = 32'h1111_1111;
59     d4 = 32'hFFFF_FFFF;
60     d5 = 32'hDEAD_BEEF;
61
62     // ~ selector test ~
63     sel = 0; #10
64     assert(out === d0) else $error("sel = 0 failed");
65
66     sel = 1; #10
67     assert(out === d1) else $error("sel = 1 failed");
68
69     sel = 2; #10
70     assert(out === d2) else $error("sel = 2 failed");
71
72     sel = 3; #10
73     assert(out === d3) else $error("sel = 3 failed");
74
75     sel = 4; #10
76     assert(out === d4) else $error("sel = 4 failed");
77
78     sel = 5; #10
79     assert(out === d5) else $error("sel = 5 failed");
80
81     sel = 0; #10
82     assert(out === d0) else $error("sel = 0 failed");
83
84     sel = 2; #10
85     assert(out === d2) else $error("sel = 2 failed");
86
87     sel = 0; #10
88     assert(out === d0) else $error("sel = 0 failed");
89
90     sel = 4; #10
91     assert(out === d4) else $error("sel = 4 failed");
92
93     // ~ edge case : input changes ~
94     sel = 3; #10
95     assert(out === d3) else $error("sel = 3 failed");
96
97     d3 = 32'hF000_0000; #10
98     assert(out === 32'hF000_0000) else $error("h7 failed");
99
100    d3 = 32'h0F00_0000; #10
101    assert(out === 32'h0F00_0000) else $error("h6 failed");
102
103    d3 = 32'h00F0_0000; #10
104    assert(out === 32'h00F0_0000) else $error("h5 failed");
105
106    d3 = 32'h000F_0000; #10
107    assert(out === 32'h000F_0000) else $error("h4 failed");
108
109    d3 = 32'h0000_F000; #10
110    assert(out === 32'h0000_F000) else $error("h3 failed");
111
112    d3 = 32'h0000_0F00; #10
113    assert(out === 32'h0000_0F00) else $error("h2 failed");
114
115    d3 = 32'h0000_00F0; #10
116    assert(out === 32'h0000_00F0) else $error("h1 failed");
117
118    d3 = 32'h0000_000F; #10
119    assert(out === 32'h0000_000F) else $error("h0 failed");
120
121    // ~ default case ~
122    sel = 6;
123    while(sel < 15) begin
124        #10;
125        assert(out === (32'hFFFF_FFFF - 13)) else $error("default %d failed", sel);
126        sel = sel + 1;
127    end
128
129 end
130
131
132 endmodule

```

0.3.2 Program Counter and Additives

Listing 21: Program Counter PC_Reg.sv

```
1  `timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer: WhisperingRock
5  //
6  // Create Date: 07/17/2026 03:36:51 PM
7  // Design Name:
8  // Module Name: PC_Reg
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description: Program Counter (register) holds PC_COUNT
13 //                constant unless a reset has occurred or
14 //                a write enable overwrites PC_COUNT with
15 //                PC_DIN
16 //
17 // Dependencies:
18 //
19 // Revision:
20 // Revision 0.01 - File Created
21 // Additional Comments:
22 //
23 ///////////////////////////////////////////////////////////////////
24
25
26 module PC_Reg(
27     input CLK,
28     input PC_RST,                // (highest-priority) active high synch reset s.t. 1 -> PC=0
29     input PC_WE,                // active high synch load value from PC_DIN into PC
30     input[31:0] PC_DIN,         // potential unsigned 32-bit PC
31     output logic [31:0] PC_COUNT // current value of PC (used as addr for ProgROM)
32 );
33
34     always_ff@(posedge CLK)      // changes on clk demand sequential logic (use nonblocking)
35     begin
36         if(PC_RST) PC_COUNT <= 0;
37         else if (PC_WE) PC_COUNT <= PC_DIN;
38     end
39
40 endmodule
```

Listing 22: TestSim for Program Counter PC_Reg_tb.sv

```
1  `timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 07/17/2026 06:36:03 PM
7  // Design Name:
8  // Module Name: PC_Reg_tb
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module PC_Reg_tb();
24
25     // ~~ local variables ~~
26     logic clk, reset, write;
27     logic[31:0] din, dout;
28     logic[31:0] answer;
29
30
31     // ~~ high-level module instances (and hooks) ~~
32     PC_Reg uut (
33         .CLK(clk),
34         .PC_RST(reset),
35         .PC_WE(write),
36         .PC_DIN(din),
37         .PC_COUNT(dout)
38     );
39
40     // ~~ action : CLK (10ns period) ~~
41     always begin
42         #5;
43         clk <= !clk;
```

```

44     end
45
46     // ~~ action : Data ~~
47     initial begin
48
49         // ~ init ~
50         clk = 1; reset = 0; write = 0;
51         reset = 1; #10;                                // zero out register
52
53         // ~ test cases ~
54         answer = 32'hDEAD_BEEF;
55         din = answer;
56
57         // ~ TC1 : write ~
58         reset = 0; #10;
59         assert(dout === 0) else $error("TC1: init failed");
60         write = 1; #10;
61         assert(dout === answer) else $error("TC1: write failed");
62
63         // ~ TC2: reset after write ~
64         reset = 1; #10;
65         assert(dout === 0) else $error("TC2: reset failed");
66
67         // ~ TC3: reset priority over write ~
68         write = 1; reset = 1; #10;
69         assert(dout === 0) else $error("TC3: reset failed");
70
71
72         // ~ TC4 : write within range ~
73         write = 1; reset = 0; #10;
74         answer = 32'h0246_8ACE;
75         for (int i = 0; i < 8; i++) begin
76             din = answer; #10;
77             assert(dout === answer) else $error("TC4: write failed");
78             answer = answer << 4;
79         end
80
81         // ~ TC5 : Rollover ~
82         din = 32'hFFFF_FFFF; #10;
83         assert(dout === din) else $error("TC5: write failed");
84         din = din + 1; #10;
85         assert(dout === 0) else $error("TC5: rollover failed");
86
87
88
89     end
90
91
92
93 endmodule

```

Listing 23: 4+ PC Incrementer Plus4_1N.sv

```

1  `timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 07/17/2026 07:28:53 PM
7  // Design Name:
8  // Module Name: Plus4_1N
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module Plus4_1N
24     #(parameter width = 32)                                // default bits
25     (
26         input logic[width-1:0] IN,
27         output logic[width-1:0] OUT
28     );
29
30     always_comb
31         OUT = IN + 4;
32
33 endmodule

```

0.3.3 Register File

Listing 24: Register File RegFile.sv

```
1  `timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer: WhisperingRock
5  //
6  // Create Date: 07/21/2026 03:41:49 PM
7  // Design Name:
8  // Module Name: RegFile
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //           A register file is a block of RAM, not ROM,
14 //           that contains the values of all 32 32-bit
15 //           registers x0:x31, with x0 held constant to
16 //           zero.
17 //
18 //           This register is a dual asynch fetch s.t.
19 //           the values at ADR1 and ADR2 are fetched
20 //           and forwarded immediately to RS1, RS2.
21 //
22 //           This register is single synch write s.t.
23 //           if en is high, W_DATA will overwrite the
24 //           value in W_ADR (excluding x0)
25 //
26 // Dependencies:
27 //
28 // Revision:
29 // Revision 0.01 - File Created
30 // Additional Comments:
31 //
32 ///////////////////////////////////////////////////////////////////
33 parameter BITWIDTH      = 32;
34
35 parameter N_REGS        = 32;
36 parameter REG_BUSWIDTH  = 5;
37
38 module RegFile(
39     input EN, CLK,
40     input[REG_BUSWIDTH-1:0] ADR1, ADR2,    // asynch fetch addr(s)
41     input[REG_BUSWIDTH-1:0] W_ADR,        // synch write addr
42     input[BITWIDTH-1:0] W_DATA,          // synch write value
43     output[BITWIDTH-1:0] RS1, RS2        // asynch fetch'd values
44 );
45
46
47 // ~~~~ alloc mem ~~~~
48 logic [BITWIDTH-1:0] ram [0:N_REGS-1];    // memory uses big-endian
49
50 // ~~~~ init mem ~~~~
51 initial
52 begin
53     for(int i = 0; i < N_REGS; i++)
54     begin
55         ram[i] = 0;
56     end
57 end
58
59 // ~~~~ asynch fetch ~~~~
60 assign RS1 = ram[ADR1];
61 assign RS2 = ram[ADR2];
62
63
64 // ~~~~ synch write ~~~~
65 always_ff@(posedge CLK)                // changes on clk demand sequential logic (use nonblocking)
66 begin
67     if((EN == 1'b1) && (W_ADR != 5'b00000))
68     begin
69         ram[W_ADR] <= W_DATA;
70     end
71 end
72
73 endmodule
```

Listing 25: TestSim for Register File RegFile_tb.sv

```
1  `timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer: WhisperingRock
5  //
6  // Create Date: 07/21/2026 03:42:37 PM
7  // Design Name:
8  // Module Name: RegFile_tb
9  // Project Name:
10 // Target Devices:
```

```

11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22 module RegFile_tb();
23
24 // ~~~~ init local vars ~~~~
25 logic enable, clk; // input
26 logic[4:0] addr1, addr2, w_addr; // input
27 logic[31:0] w_val; // input
28 logic[31:0] out1, out2; // output
29
30
31 // ~~~~ instances ~~~~
32 RegFile UUT(
33     .EN(enable),
34     .CLK(clk),
35     .ADR1(addr1),
36     .ADR2(addr2),
37     .W_ADR(w_addr),
38     .W_DATA(w_val),
39     .RS1(out1),
40     .RS2(out2)
41 );
42
43 // ~~ predefined tasks ~~
44 task fill_ram(input int bias);
45     enable = 1;
46     for(int i = 0; i < N_REGS; i++)
47     begin
48         w_addr = i;
49         w_val = i + bias;
50         #10;
51     end
52     enable = 0;
53 endtask
54
55 // ~~ action : CLK (10ns period) ~~
56 always begin
57     #5;
58     clk <= !clk;
59 end
60
61 // ~~~~ action : data ~~~~
62 initial begin
63
64     // ~~ init ~~
65     clk = 1; enable = 0;
66     addr1 = 0; addr2 = 0;
67     w_addr = 0; w_val = 0;
68     #10;
69     fill_ram(0);
70     #10
71
72     // ~~ TC1 : same asynch fetch ~~
73     for(int i = 0; i < 32; i++)
74     begin
75         addr1 = i;
76         addr2 = i;
77         #10;
78         assert(out1 == i) else $error("TC1: rs1 failed");
79         assert(out2 == i) else $error("TC1: rs2 failed");
80     end
81
82     // ~~ TC2 : prevent write to x0 ~~
83     addr1 = 0; enable = 1;
84     #10;
85     for(int i = 0; i < 100; i++)
86     begin
87         w_addr = 0;
88         w_val = $random();
89         #10;
90         assert(out1 == 0) else $error("TC2: x0 prevent failed");
91     end
92
93     // ~~ TC3 : write not enabled on same addr ~~
94     fill_ram(0); // cheap sanitizer
95     addr1 = 1; addr2 = 1; w_addr = 1; // vals should be 1 in reg
96     enable = 0;
97     #10;
98     for(int i = 0; i < 100; i++)
99     begin
100         w_val = $random();
101         #10;

```

```

102         assert(out1 === 1) else $error("TC3: write prevent failed");
103         assert(out2 === 1) else $error("TC3: write prevent failed");
104     end
105
106
107     // ~~ TC4 : reading the register being written to ~~
108     addr1 = 14; addr2 = 14; w_addr = 14;
109     enable = 1;
110     #10;
111     for(int i = 0; i < 500; i++)
112     begin
113         w_val = $random();
114         #10;
115         assert(out1 === w_val) else $error("TC4: r/w failed");
116         assert(out2 === w_val) else $error("TC4: r/w failed");
117     end
118
119 end
120
121
122 endmodule

```

0.3.4 Arithmetic Logic Unit

Listing 26: ALU ALU.sv

```
1  `timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer: WhisperingRock
5  //
6  // Create Date: 07/24/2026 09:28:51 PM
7  // Design Name:
8  // Module Name: ALU
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //     A Purely combinational operation execution(er).
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21 //parameter BITWIDTH      = 32;
22 parameter FUNWIDTH        = 4;
23
24
25 module ALU(
26     input logic [FUNWIDTH-1:0] ALU_FUN,
27     input logic [BITWIDTH-1:0] SRC_A, SRC_B,
28     output logic[BITWIDTH-1:0] ALU_RESULT
29 );
30
31     // ~~ local vars ~~
32     logic[BITWIDTH-1:0] ERR = 32'hDEAD_BEEF;
33
34     always_comb
35     begin
36         case(ALU_FUN)
37             // ~~ arithmetic operators~~
38             4'b0000: begin ALU_RESULT = SRC_A + SRC_B; end           // ADD
39             4'b1000: begin ALU_RESULT = SRC_A - SRC_B; end           // SUB
40
41             // ~~ bitwise operators ~~
42             4'b0111: begin ALU_RESULT = SRC_A & SRC_B; end           // AND
43             4'b0110: begin ALU_RESULT = SRC_A | SRC_B; end           // OR
44             4'b0100: begin ALU_RESULT = SRC_A ^ SRC_B; end           // XOR
45
46             // ~~ shift operators ~~
47             // Note : shifting only uses the 5 LSBs of SRC_B (see manual)
48             // Note : arithmetic shifting has its own systemverilog operator
49             4'b0101: begin ALU_RESULT = SRC_A >> SRC_B[4:0]; end       // SRL
50             4'b0001: begin ALU_RESULT = SRC_A << SRC_B[4:0]; end       // SLL
51             4'b1101: begin ALU_RESULT = $signed(SRC_A) >>> $signed(SRC_B[4:0]); end // SRA
52
53             // ~~ set operators ~~
54             4'b0010: begin ALU_RESULT = $signed(SRC_A) < $signed(SRC_B); end // SLT
55             4'b0011: begin ALU_RESULT = SRC_A < SRC_B; end           // SLTU
56
57             // ~~ copy operator ~~
58             4'b1001: begin ALU_RESULT = SRC_A; end                   // LUI-COPY (MV?)
59
60             // ~~ error ~~
61             default: begin ALU_RESULT = ERR; end
62         endcase
63     end
64
65 endmodule
```

Listing 27: TestSim for ALU ALU_tb.sv

```
1  `timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 07/24/2026 10:13:51 PM
7  // Design Name:
8  // Module Name: ALU_tb
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
```

```

19 //
20 //////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
21
22
23 module ALU_tb();
24
25     // ~~~~ init local vars ~~~~
26     logic[4:0] sel;
27     logic[31:0] a, b;
28     logic[31:0] result;
29
30     logic[31:0] testcase;    // testing var
31
32     // ~~~~ instances ~~~~
33     ALU UUT
34     (
35         .ALU_FUN(sel),
36         .SRC_A(a),
37         .SRC_B(b),
38         .ALU_RESULT(result)
39     );
40
41     // ~~~~ testing ~~~~
42     initial
43     begin
44
45         // ~~~~ ADD ~~~~
46         sel = 4'b0000;
47
48         // ~~ TC1 : ADD1 ~~
49         #5;
50         a = 32'hA50F_96C3;
51         b = 32'h5AF0_693C;
52         testcase = 1;
53         #5;
54         assert(result == 32'hFFFF_FFFF) else $error("TC1 : ADD failed");
55
56         // ~~ TC2 : ADD2 ~~
57         #5;
58         a = 32'h8410_5F21;
59         b = 32'h7B10_5FDE;
60         testcase++;
61         #5;
62         assert(result == 32'hFF20_BEFF) else $error("TC2 : ADD failed");
63
64         // ~~ TC3 : ADD3 ~~
65         #5;
66         a = 32'hFFFF_FFFF;
67         b = 32'h0000_0001;
68         testcase++;
69         #5;
70         assert(result == 32'h0000_0000) else $error("TC3 : ADD failed");
71
72
73
74         // ~~~~ SUB ~~~~
75         sel = 4'b1000;
76
77         // ~~ TC4 : SUB4 ~~
78         #5;
79         a = 32'h0000_0000;
80         b = 32'h0000_0001;
81         testcase++;
82         #5;
83         assert(result == 32'hFFFF_FFFF) else $error("TC4 : SUB failed");
84
85         // ~~ TC5 : SUB5 ~~
86         #5;
87         a = 32'hAA80_6355;
88         b = 32'h5501_62AA;
89         testcase++;
90         #5;
91         assert(result == 32'h557F_00AB) else $error("TC5 : SUB failed");
92
93         // ~~ TC6 : SUB6 ~~
94         #5;
95         a = 32'h5501_62AA;
96         b = 32'hAA80_6355;
97         testcase++;
98         #5;
99         assert(result == 32'hAA80_FF55) else $error("TC6 : SUB failed");
100
101
102
103         // ~~~~ AND ~~~~
104         sel = 4'b0111;
105
106         // ~~ TC7 : AND7 ~~
107         #5;
108         a = 32'hA55A_00FF;
109         b = 32'h5A5A_FFFF;

```

```

110     testcase++;
111     #5;
112     assert(result === 32'h005A_00FF) else $error("TC7 : AND failed");
113
114     // ~~ TC8 : AND8 ~~
115     #5;
116     a = 32'hC3C3_F966;
117     b = 32'hFF66_9F5A;
118     testcase++;
119     #5;
120     assert(result === 32'hC342_9942) else $error("TC8 : AND failed");
121
122
123     // ~~~~ OR ~~~~
124     sel = 4'b0110;
125
126     // ~~ TC9 ~~
127     #5;
128     a = 32'h9A9A_C300;
129     b = 32'h65A3_CC0F;
130     testcase++;
131     #5;
132     assert(result === 32'hFFBB_CF0F) else $error("TC9 : OR failed");
133
134     // ~~ TC10 ~~
135     #5;
136     a = 32'hC3C3_F966;
137     b = 32'hFF66_9F5A;
138     testcase++;
139     #5;
140     assert(result === 32'hFFE7_FF7E) else $error("TC10 : OR failed");
141
142
143     // ~~~~ XOR ~~~~
144     sel = 4'b0100;
145
146     // ~~ TC11 ~~
147     #5;
148     a = 32'hAA55_00FF;
149     b = 32'h5AA5_0FF0;
150     testcase++;
151     #5;
152     assert(result === 32'hF0F0_0F0F) else $error("TC11 : XOR failed");
153
154     // ~~ TC12 ~~
155     #5;
156     a = 32'hA5A5_6C6C;
157     b = 32'hFF00_C6FF;
158     testcase++;
159     #5;
160     assert(result === 32'h5AA5_AA93) else $error("TC12 : XOR failed");
161
162
163     // ~~~~ SRL ~~~~
164     sel = 4'b0101;
165
166     // ~~ TC13 ~~
167     #5;
168     a = 32'h805A_6CF3;
169     b = 32'h0000_0010;
170     testcase++;
171     #5;
172     assert(result === 32'h0000_805A) else $error("TC13 : SRL failed");
173
174     // ~~ TC14 ~~
175     #5;
176     a = 32'h705A_6CF3;
177     b = 32'h0000_0005;
178     testcase++;
179     #5;
180     assert(result === 32'h0382_D367) else $error("TC14 : SRL failed");
181
182     // ~~ TC15 ~~
183     #5;
184     a = 32'h805A_6CF3;
185     b = 32'h0000_0000;
186     testcase++;
187     #5;
188     assert(result === 32'h805A_6CF3) else $error("TC15 : SRL failed");
189
190     // ~~ TC16 ~~
191     #5;
192     a = 32'h805A_6CF3;
193     b = 32'h0000_0100;
194     testcase++;
195     #5;
196     assert(result === 32'h805A_6CF3) else $error("TC16 : SRL failed");
197
198
199
200

```

```

201 // ~~~~ SLL ~~~~
202 sel = 4'b0001;
203
204 // ~~ TC17 ~~
205 #5;
206 a = 32'h805A_6CF3;
207 b = 32'h0000_0010;
208 testcase++;
209 #5;
210 assert(result === 32'h6CF3_0000) else $error("TC17 : SLL failed");
211
212 // ~~ TC18 ~~
213 #5;
214 a = 32'h805A_6CF3;
215 b = 32'h0000_0005;
216 testcase++;
217 #5;
218 assert(result === 32'h0B4D_9E60) else $error("TC18 : SLL failed");
219
220 // ~~ TC19 ~~
221 #5;
222 a = 32'h805A_6CF3;
223 b = 32'h0000_0100;
224 testcase++;
225 #5;
226 assert(result === 32'h805A_6CF3) else $error("TC19 : SLL failed");
227
228
229 // ~~~~ SRA ~~~~
230 sel = 4'b1101;
231
232 // ~~ TC20 ~~
233 #5;
234 a = 32'h805A_6CF3;
235 b = 32'h0000_0010;
236 testcase++;
237 #5;
238 assert(result === 32'hFFFF_805A) else $error("TC20 : SRA failed");
239
240 // ~~ TC21 ~~
241 #5;
242 a = 32'h705A_6CF3;
243 b = 32'h0000_0005;
244 testcase++;
245 #5;
246 assert(result === 32'h0382_D367) else $error("TC21 : SRA failed");
247
248 // ~~ TC22 ~~
249 #5;
250 a = 32'h805A_6CF3;
251 b = 32'h0000_0000;
252 testcase++;
253 #5;
254 assert(result === 32'h805A_6CF3) else $error("TC22 : SRA failed");
255
256 // ~~ TC23 ~~
257 #5;
258 a = 32'h805A_6CF3;
259 b = 32'h0000_0100;
260 testcase++;
261 #5;
262 assert(result === 32'h805A_6CF3) else $error("TC23 : SRA failed");
263
264
265 // ~~~~ SLT ~~~~
266 sel = 4'b0010;
267
268 // ~~ TC24 ~~
269 #5;
270 a = 32'h7FFF_FFFF;
271 b = 32'h8000_0000;
272 testcase++;
273 #5;
274 assert(result === 32'h0000_0000) else $error("TC24 : SLT failed");
275
276 // ~~ TC25 ~~
277 #5;
278 a = 32'h8000_0000;
279 b = 32'h0000_0001;
280 testcase++;
281 #5;
282 assert(result === 32'h0000_0001) else $error("TC25 : SLT failed");
283
284 // ~~ TC26 ~~
285 #5;
286 a = 32'h0000_0000;
287 b = 32'h0000_0000;
288 testcase++;
289
290
291

```

```

292     #5;
293     assert(result === 32'h0000_0000) else $error("TC26 : SLT failed");
294
295     // ~~ TC27 ~~
296     #5;
297     a = 32'h5555_5555;
298     b = 32'h5555_5555;
299     testcase++;
300     #5;
301     assert(result === 32'h0000_0000) else $error("TC27 : SLT failed");
302
303
304     // ~~~~ SLTU ~~~~
305     sel = 4'b0011;
306
307     // ~~ TC28 ~~
308     #5;
309     a = 32'h7FFF_FFFF;
310     b = 32'h8000_0000;
311     testcase++;
312     #5;
313     assert(result === 32'h0000_0001) else $error("TC28 : SLTU failed");
314
315     // ~~ TC29 ~~
316     #5;
317     a = 32'h8000_0000;
318     b = 32'h0000_0001;
319     testcase++;
320     #5;
321     assert(result === 32'h0000_0000) else $error("TC29 : SLTU failed");
322
323     // ~~ TC30 ~~
324     #5;
325     a = 32'h0000_0000;
326     b = 32'h0000_0000;
327     testcase++;
328     #5;
329     assert(result === 32'h0000_0000) else $error("TC30 : SLTU failed");
330
331     // ~~ TC31 ~~
332     #5;
333     a = 32'h55AA_55AA;
334     b = 32'h55AA_55AA;
335     testcase++;
336     #5;
337     assert(result === 32'h0000_0000) else $error("TC31 : SLTU failed");
338
339
340     // ~~~~ LUI-COPY (MV?) ~~~~
341     sel = 4'b1001;
342
343     // ~~ TC32 ~~
344     #5;
345     a = 32'h0123_4567;
346     b = 32'h7654_3210;
347     testcase++;
348     #5;
349     assert(result === 32'h0123_4567) else $error("TC32 : LUI-COPY failed");
350
351     // ~~ TC33 ~~
352     #5;
353     a = 32'hFEDC_BA98;
354     b = 32'h89AB_CDEF;
355     testcase++;
356     #5;
357     assert(result === 32'hFEDC_BA98) else $error("TC33 : LUI-COPY failed");
358
359     end
360
361 endmodule

```

0.3.5 Immediate Generator

Listing 28: Immediate Gen ImmedGen.sv

```
1  `timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 07/25/2026 10:59:49 AM
7  // Design Name:
8  // Module Name: ImmedGen
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module ImmedGen(
24     input logic [BITWIDTH-1:0] INSTR,
25     output logic[BITWIDTH-1:0] U_TYPE, I_TYPE, S_TYPE, J_TYPE, B_TYPE
26 );
27
28     //~~~~~ immediates are immediate! ~~~~~
29     assign U_TYPE = { INSTR[31:12], 12'b0 };
30
31     assign I_TYPE = { {21{INSTR[31]}}, INSTR[30:20] };
32
33     assign S_TYPE = { {21{INSTR[31]}}, INSTR[30:25], INSTR[11:7] };
34
35     assign B_TYPE =
36     {
37         {20{INSTR[31]}},
38         INSTR[7],
39         INSTR[30:25],
40         INSTR[11:8],
41         1'b0
42     };
43
44     assign J_TYPE =
45     {
46         {12{INSTR[31]}},
47         INSTR[19:12],
48         INSTR[20],
49         INSTR[30:21],
50         1'b0
51     };
52
53
54
55 endmodule
```

Listing 29: TestSim for ImmedGen ImmedGen_tb.sv

```
1  `timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 07/25/2026 11:41:08 AM
7  // Design Name:
8  // Module Name: ImmedGen_tb
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module ImmedGen_tb();
24
25     // ~~~~~ init local vars ~~~~~
26     logic[31:0] instr;
27     logic[31:0] u, i, s, b, j;
28
```

```

29 logic[31:0] testcase; // testing var
30
31 // ~~~~ instances ~~~~
32 ImmedGen UUT
33 (
34     .INSTR(instr),
35     .U_TYPE(u),
36     .I_TYPE(i),
37     .S_TYPE(s),
38     .J_TYPE(j),
39     .B_TYPE(b)
40 );
41
42 // ~~~~ testing ~~~~
43 initial
44 begin
45
46     testcase = 1;
47
48     // ~~ TC1 : all zero ~~
49     #5;
50     testcase++;
51     instr = 0;
52     #5;
53     assert(u === 0) else $error("TC1 : U failed");
54     assert(i === 0) else $error("TC1 : I failed");
55     assert(s === 0) else $error("TC1 : S failed");
56     assert(b === 0) else $error("TC1 : B failed");
57     assert(j === 0) else $error("TC1 : J failed");
58
59
60     // ~~ TC2 : all one ~~
61     #5;
62     testcase++;
63     instr = 32'hFFFF_FFFF;
64     #5;
65     assert(u === 32'hFFFF_F000) else $error("TC2 : U failed");
66     assert(i === 32'hFFFF_FFFF) else $error("TC2 : I failed");
67     assert(s === 32'hFFFF_FFFF) else $error("TC2 : S failed");
68     assert(b === 32'hFFFF_FFFE) else $error("TC2 : B failed");
69     assert(j === 32'hFFFF_FFFE) else $error("TC2 : J failed");
70
71
72     // ~~ TC3 : hex1 ~~
73     #5;
74     testcase++;
75     instr = 32'h5A5A_5A5A;
76     #5;
77     assert(u === 32'h5A5A_5000) else $error("TC3 : U failed");
78     assert(i === 32'h0000_05A5) else $error("TC3 : I failed");
79     assert(s === 32'h0000_05B4) else $error("TC3 : S failed");
80     assert(b === 32'h0000_05B4) else $error("TC3 : B failed");
81     assert(j === 32'h000A_5DA4) else $error("TC3 : J failed");
82
83
84     // ~~ TC4 : hex2 ~~
85     #5;
86     testcase++;
87     instr = 32'hF0F0_F0F0;
88     #5;
89     assert(u === 32'hF0F0_F000) else $error("TC4 : U failed");
90     assert(i === 32'hFFFF_FF0F) else $error("TC4 : I failed");
91     assert(s === 32'hFFFF_FF01) else $error("TC4 : S failed");
92     assert(b === 32'hFFFF_FF00) else $error("TC4 : B failed");
93     assert(j === 32'hFFF0_FF0E) else $error("TC4 : J failed");
94
95
96     // ~~ TC5 : hex3 ~~
97     #5;
98     testcase++;
99     instr = 32'h3CA5_5A3C;
100     #5;
101     assert(u === 32'h3CA5_5000) else $error("TC5 : U failed");
102     assert(i === 32'h0000_03CA) else $error("TC5 : I failed");
103     assert(s === 32'h0000_03D4) else $error("TC5 : S failed");
104     assert(b === 32'h0000_03D4) else $error("TC5 : B failed");
105     assert(j === 32'h0005_53CA) else $error("TC5 : J failed");
106
107
108     end
109
110
111 endmodule

```

0.3.6 Memory

Listing 30: Memory module otter_memory_v1_07.sv

```

1  `timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer: J. Callenes, P. Hummel
5  //
6  // Create Date: 01/27/2019 08:37:11 AM
7  // Module Name: OTTER_mem
8  // Project Name: Memory for OTTER RV32I RISC-V
9  // Tool Versions: Xilinx Vivado 2019.2
10 // Description: 64k Memory, dual access read single access write. Designed to
11 //              purposely utilize BRAM which requires synchronous reads and write
12 //              ADDR1 used for Program Memory Instruction. Word addressable so it
13 //              must be adapted from byte addresses in connection from PC
14 //              ADDR2 used for data access, both internal and external memory
15 //              mapped IO. ADDR2 is byte addressable.
16 //              RDEN1 and EDEN2 are read enables for ADDR1 and ADDR2. These are
17 //              needed due to synchronous reading
18 //              MEM_SIZE used to specify reads as byte (0), half (1), or word (2)
19 //              MEM_SIGN used to specify unsigned (1) vs signed (0) extension
20 //              IO_IN is data from external IO and synchronously buffered
21 //
22 // Memory OTTER_MEMORY (
23 //     .MEM_CLK    (),
24 //     .MEM_RDEN1  (),
25 //     .MEM_RDEN2  (),
26 //     .MEM_WE2    (),
27 //     .MEM_ADDR1  (),
28 //     .MEM_ADDR2  (),
29 //     .MEM_DIN2   (),
30 //     .MEM_SIZE   (),
31 //     .MEM_SIGN   (),
32 //     .IO_IN      (),
33 //     .IO_WR      (),
34 //     .MEM_DOUT1  (),
35 //     .MEM_DOUT2  () );
36 //
37 // Revision:
38 // Revision 0.01 - Original by J. Callenes
39 // Revision 1.02 - Rewrite to simplify logic by P. Hummel
40 // Revision 1.03 - changed signal names, added instantiation template
41 // Revision 1.04 - added default to write case statement
42 // Revision 1.05 - changed MEM_WD to MEM_DIN2, changed default to save nothing
43 // Revision 1.06 - removed typo in instantiation template
44 // Revision 1.07 - remove unused wordAddr1 signal
45 //
46 ///////////////////////////////////////////////////////////////////
47
48 module Memory (
49     input          MEM_CLK,
50     input          MEM_RDEN1, // read enable : Instruction    (addr1)
51     input          MEM_RDEN2, // read enable : Data      (addr2)
52     input          MEM_WE2,   // write enable.
53     input [13:0]   MEM_ADDR1, // Addr : Instruction Memory (Connect to PC[15:2])
54     input [31:0]   MEM_ADDR2, // Addr : Data Memory
55     input [31:0]   MEM_DIN2,  // Data to save
56     input [1:0]    MEM_SIZE,   // {0-Byte, 1-Half, 2-Word}
57     input          MEM_SIGN,   // 1-unsigned 0-signed
58     input [31:0]   IO_IN,      // Data from IO
59     output logic   IO_WR,      // IO 1-write 0-read
60     output logic [31:0] MEM_DOUT1, // Instruction
61     output logic [31:0] MEM_DOUT2 // Data
62     //output ERR              // only used for testing
63 );
64
65 // ~~~~ local variables ~~~~
66 logic [13:0] wordAddr2;
67 logic [31:0] memReadWord, ioBuffer, memReadSized;
68 logic [1:0] byteOffset;
69 logic weAddrValid; // active when saving (WE) to valid memory address
70
71 // ~~~~ attributes (for memory) ~~~~
72 (* rom_style="{distributed | block}" *)
73 (* ram_decomp = "power" *) logic [31:0] memory [0:16383];
74
75 // ~~~~ memory injection ~~~~
76 initial
77 begin
78     //$readmemh("tc3x3.mem", memory, 0, 16383);
79     //$readmemh("tc10x10.mem", memory, 0, 16383);
80     //$readmemh("tc16x16.mem", memory, 0, 16383);
81     //$readmemh("Test_All.mem", memory, 0, 16383);
82     $readmemh("otter_hw6sample_test.mem", memory, 0, 16383);
83 end
84
85 // ~~~~ todo ~~~~
86 assign wordAddr2 = MEM_ADDR2[15:2];
87 assign byteOffset = MEM_ADDR2[1:0]; // byte offset of memory address

```

```

88
89 // NOT USED IN OTTER
90 //Check for misalligned or out of bounds memory accesses
91 //assign ERR = ((MEM_ADDR1 >= 2**ACTUAL_WIDTH)|| (MEM_ADDR2 >= 2**ACTUAL_WIDTH)
92 //          || MEM_ADDR1[1:0] != 2'b0 || MEM_ADDR2[1:0] !=2'b0)? 1 : 0;
93
94 // ~~~~ buffer the IO input for reading ~~~~
95 always_ff @(posedge MEM_CLK)
96 begin
97     if(MEM_RDEN2)
98         ioBuffer <= IO_IN;
99 end
100
101 // ~~~~ BRAM requires all reads and writes to occur synchronously ~~~~
102 always_ff @(posedge MEM_CLK)
103 begin
104
105     // ~~ save data (WD) to memory (ADDR2) ~~
106     if (weAddrValid == 1) // write enable and valid address space (not past sta
107     begin
108         case({MEM_SIZE,byteOffset})
109             // ~~ byte(00) ~~
110             4'b00_00: memory[wordAddr2][7:0] <= MEM_DIN2[7:0]; // sb instr
111             4'b00_01: memory[wordAddr2][15:8] <= MEM_DIN2[7:0];
112             4'b00_10: memory[wordAddr2][23:16] <= MEM_DIN2[7:0];
113             4'b00_11: memory[wordAddr2][31:24] <= MEM_DIN2[7:0];
114             // ~~ half(01) ~~
115             4'b01_00: memory[wordAddr2][15:0] <= MEM_DIN2[15:0]; // sh instr
116             4'b01_01: memory[wordAddr2][23:8] <= MEM_DIN2[15:0];
117             4'b01_10: memory[wordAddr2][31:16] <= MEM_DIN2[15:0];
118             // ~~ word(10) ~~
119             4'b10_00: memory[wordAddr2] <= MEM_DIN2; // sw instr
120
121             //default: memory[wordAddr2] <= 32'b0 // unsupported size, byte offset
122             // removed to avoid mistakes causing memory to be zeroed.
123         endcase
124     end
125
126     // ~~ read all data synchronously required for BRAM ~~
127     if (MEM_RDEN1) // need EN for extra load cycle to not change instr
128         MEM_DOUT1 <= memory[MEM_ADDR1];
129
130     if (MEM_RDEN2) // Read word from memory
131         memReadWord <= memory[wordAddr2];
132 end
133
134 // ~~~~ Change the data word, from ADR2, into sized bytes and sign extend ~~~~
135 always_comb
136 begin
137     // ~~~ sign(1 bit), size(2 bits), offset(2 bits) ~~~
138     case({MEM_SIGN,MEM_SIZE,byteOffset})
139
140         // ~~ signed (0) : extend extensions ~~~
141         // ~ byte(00) : extend for 3 bytes and 1 byte of data ~
142         5'b0_00_11: memReadSized = {{24{memReadWord[31]}},memReadWord[31:24]};
143         5'b0_00_10: memReadSized = {{24{memReadWord[23]}},memReadWord[23:16]};
144         5'b0_00_01: memReadSized = {{24{memReadWord[15]}},memReadWord[15:8]};
145         5'b0_00_00: memReadSized = {{24{memReadWord[7]}},memReadWord[7:0]};
146         // ~ half (01) : extend for 2 bytes and 2 bytes of data ~
147         5'b0_01_10: memReadSized = {{16{memReadWord[31]}},memReadWord[31:16]};
148         5'b0_01_01: memReadSized = {{16{memReadWord[23]}},memReadWord[23:8]};
149         5'b0_01_00: memReadSized = {{16{memReadWord[15]}},memReadWord[15:0]};
150         // ~ word (10) : no extension ~~
151         5'b0_10_00: memReadSized = memReadWord;
152
153
154         // ~~ unsigned (1) : no sign extensions ~~
155         // ~ byte (00) : 3 bytes of zero and 1 byte of data ~
156         5'b1_00_11: memReadSized = {24'd0,memReadWord[31:24]};
157         5'b1_00_10: memReadSized = {24'd0,memReadWord[23:16]};
158         5'b1_00_01: memReadSized = {24'd0,memReadWord[15:8]};
159         5'b1_00_00: memReadSized = {24'd0,memReadWord[7:0]};
160         // ~ half (01) : 2 bytes of zero and 2 bytes of data ~
161         5'b1_01_10: memReadSized = {16'd0,memReadWord[31:16]};
162         5'b1_01_01: memReadSized = {16'd0,memReadWord[23:8]};
163         5'b1_01_00: memReadSized = {16'd0,memReadWord[15:0]};
164         // ~ unsigned word is missing? ~
165
166         default: memReadSized = 32'b0; // unsupported size, byte offset combination
167     endcase
168 end
169
170 // ~~~~ MemoryMapped IO ~~~~
171 always_comb
172 begin
173     if(MEM_ADDR2 >= 32'h0001_0000) // external address range (beyond stack)
174     begin
175         IO_WR = MEM_WE2; // IO Write
176         MEM_DOUT2 = ioBuffer; // IO read from buffer
177         weAddrValid = 0; // address beyond memory range
178     end

```

```

179
180     else
181     begin
182         IO_WR = 0; // not MMIO
183         MEM_DOUT2 = memReadSized; // output sized and sign extended data
184         weAddrValid = MEM_WE2; // address in valid memory range
185     end
186 end
187
188 endmodule

```

0.3.7 Branch Address Generator

Listing 31: Branch Address Generator BranchAddrGen.sv

```
1  `timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 07/28/2026 07:35:34 PM
7  // Design Name:
8  // Module Name: BranchAddrGen
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module BranchAddrGen(
24     input  logic [31:0] RS1,
25     input  logic [31:0] I_TYPE, J_TYPE, B_TYPE,
26     input  logic [31:0] PC,
27     output logic [31:0] JAL, JALR, BRANCH
28 );
29     // ~~~~ combinational logic ~~~~
30     assign JAL      = PC      + J_TYPE;
31     assign JALR     = RS1     + I_TYPE;
32     assign BRANCH   = PC      + B_TYPE;
33
34 endmodule
```

Listing 32: TestSim for Branch Addr Gen BranchAddrGen_tb.sv

```
1  `timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 08/05/2026 08:23:37 AM
7  // Design Name:
8  // Module Name: BranchAddrGen_tb
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module BranchAddrGen_tb();
24
25     // ~~~~ init local vars ~~~~
26     logic[31:0] rs, i, j, b, pc;    // input
27     logic[31:0] jal, jalr, bran;    // output
28
29     logic[31:0] testcase;           // testing var
30
31     // ~~~~ instances ~~~~
32     BranchAddrGen UUT
33     (
34         // ~~ inputs ~~
35         .RS1(rs),
36         .I_TYPE(i),
37         .J_TYPE(j),
38         .B_TYPE(b),
39         .PC(pc),
40         // ~~ outputs ~~
41         .JAL(jal),                // = pc + j
42         .JALR(jalr),              // = rs1 + i
43         .BRANCH(bran)             // = pc + b
44     );
45
46     // ~~~~ testing ~~~~
47     initial
48     begin
49
```

```

50 // ~~ TC1 : all zero ~~
51 #5;
52 rs = 32'h0000_0000;
53 i = 32'h0000_0000;
54 j = 32'h0000_0000;
55 b = 32'h0000_0000;
56 pc = 32'h0000_0000;
57 testcase = 1;
58 #5;
59 $display("TC%d:", testcase);
60 assert(jalr === 32'h0000_0000) else $error("JALR failed"); // = rs + i
61 assert(jal === 32'h0000_0000) else $error("JAL failed"); // = pc + j
62 assert(bran === 32'h0000_0000) else $error("BRANCH failed"); // = pc + b
63
64
65 // ~~ TC2 : only types ~~
66 #5;
67 rs = 32'h0000_0000;
68 i = 32'h0101_0101;
69 j = 32'hF0F0_A050;
70 b = 32'h1234_5678;
71 pc = 32'h0000_0000;
72 testcase++;
73 #5;
74 $display("TC%d:", testcase);
75 assert(jalr === 32'h0101_0101) else $error("JALR failed"); // = rs + i
76 assert(jal === 32'hF0F0_A050) else $error("JAL failed"); // = pc + j
77 assert(bran === 32'h1234_5678) else $error("BRANCH failed"); // = pc + b
78
79
80 // ~~ TC3 : only pc ~~
81 #5;
82 rs = 32'h0000_0000;
83 i = 32'h0000_0000;
84 j = 32'h0000_0000;
85 b = 32'h0000_0000;
86 pc = 32'h5EC3_25AF;
87 testcase++;
88 #5;
89 $display("TC%d:", testcase);
90 assert(jalr === 32'h0000_0000) else $error("JALR failed"); // = rs + i
91 assert(jal === 32'h5EC3_25AF) else $error("JAL failed"); // = pc + j
92 assert(bran === 32'h5EC3_25AF) else $error("BRANCH failed"); // = pc + b
93
94
95
96 // ~~ TC4 : only rs ~~
97 #5;
98 rs = 32'h9876_6789;
99 i = 32'h0000_0000;
100 j = 32'h0000_0000;
101 b = 32'h0000_0000;
102 pc = 32'h0000_0000;
103 testcase++;
104 #5;
105 $display("TC%d:", testcase);
106 assert(jalr === 32'h9876_6789) else $error("JALR failed"); // = rs + i
107 assert(jal === 32'h0000_0000) else $error("JAL failed"); // = pc + j
108 assert(bran === 32'h0000_0000) else $error("BRANCH failed"); // = pc + b
109
110
111
112 // ~~ TC5 : rollover + ceiling 1 ~~
113 #5;
114 rs = 32'hFFFF_FFFE;
115 i = 32'h0000_0001;
116 j = 32'h0000_0001;
117 b = 32'h0000_0001;
118 pc = 32'hFFFF_FFFF;
119 testcase++;
120 #5;
121 $display("TC%d:", testcase);
122 assert(jalr === 32'hFFFF_FFFF) else $error("JALR failed"); // = rs + i
123 assert(jal === 32'h0000_0000) else $error("JAL failed"); // = pc + j
124 assert(bran === 32'h0000_0000) else $error("BRANCH failed"); // = pc + b
125
126
127 // ~~ TC6 : rollover + ceiling 2 ~~
128 #5;
129 rs = 32'h0000_0002;
130 i = 32'hFFFF_FFFE;
131 j = 32'hFFFF_FFFF;
132 b = 32'hFFFF_FFFE;
133 pc = 32'h0000_0003;
134 testcase++;
135 #5;
136 $display("TC%d:", testcase);
137 assert(jalr === 32'h0000_0000) else $error("JALR failed"); // = rs + i
138 assert(jal === 32'h0000_0002) else $error("JAL failed"); // = pc + j
139 assert(bran === 32'h0000_0001) else $error("BRANCH failed"); // = pc + b
140

```

```
141
142     end
143
144 endmodule
```

0.3.8 Branch Condition Generator

Listing 33: Branch Condition Generator BranchConditionGen.sv

```
1  `timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 07/28/2026 07:21:37 PM
7  // Design Name:
8  // Module Name: BranchConditionGen
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //      Determines the branch condition instructions
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module BranchConditionGen(
24     input    logic [31:0] RS1, RS2,
25     output   logic      BR_EQ,      // equal branch
26             logic      BR_LT,      // less than branch
27             logic      BR_LTU     // less then branch (unsigned)
28 );
29
30     // ~~~~ combinational logic ~~~~
31     assign BR_EQ    = (RS1 == RS2);
32     assign BR_LT    = ($signed(RS1) < $signed(RS2));
33     assign BR_LTU   = (RS1 < RS2);
34
35 endmodule
```

Listing 34: TestSim for Branch Condition Gen BranchConditionGen_tb.sv

```
1  `timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 08/04/2026 10:49:22 AM
7  // Design Name:
8  // Module Name: BranchConditionGen_tb
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module BranchConditionGen_tb();
24
25     // ~~~~ init local vars ~~~~
26     logic[31:0] a, b;
27     logic      eq, lts, ltu;
28
29     logic[31:0] testcase;      // testing var
30
31     // ~~~~ instances ~~~~
32     BranchConditionGen UUT
33     (
34         .RS1(a),
35         .RS2(b),
36         .BR_EQ(eq),
37         .BR_LT(lts),
38         .BR_LTU(ltu)
39     );
40
41     // ~~~~ testing ~~~~
42     initial
43     begin
44
45         // ~~ TC1 : equal are not less (simple) ~~
46         #5;
47         a = 32'h1234_5678;
48         b = 32'h1234_5678;
```

```

49     testcase = 1;
50     #5;
51     $display("TC%d:", testcase);
52     assert(eq === 1'b1) else $error("Equality failed");
53     assert(lts === 1'b0) else $error("LessThan Signed failed");
54     assert(ltu === 1'b0) else $error("LessThan Unsigned failed");
55
56
57     // ~~ TC2 : signed vs unsigned (simple) ~~
58     #5;
59     a = 32'hA50F_96C3; // is less than b if signed
60     b = 32'h5AF0_693C;
61     testcase++;
62     #5;
63     $display("TC%d:", testcase);
64     assert(eq === 1'b0) else $error("Equality failed");
65     assert(lts === 1'b1) else $error("LessThan Signed failed");
66     assert(ltu === 1'b0) else $error("LessThan Unsigned failed");
67
68
69     // ~~ TC3 : equal are not less (ceiling) ~~
70     #5;
71     a = 32'hFFFF_FFFF;
72     b = 32'hFFFF_FFFF;
73     testcase++;
74     #5;
75     $display("TC%d:", testcase);
76     assert(eq === 1'b1) else $error("Equality failed");
77     assert(lts === 1'b0) else $error("LessThan Signed failed");
78     assert(ltu === 1'b0) else $error("LessThan Unsigned failed");
79
80
81     // ~~ TC4 : equal are not less (floor) ~~
82     #5;
83     a = 32'h0000_0000;
84     b = 32'h0000_0000;
85     testcase++;
86     #5;
87     $display("TC%d:", testcase);
88     assert(eq === 1'b1) else $error("Equality failed");
89     assert(lts === 1'b0) else $error("LessThan Signed failed");
90     assert(ltu === 1'b0) else $error("LessThan Unsigned failed");
91
92
93     // ~~ TC5 : signed vs unsigned (simple reversed) ~~
94     #5;
95     a = 32'h5AF0_693C;
96     b = 32'hA50F_96C3;
97     testcase++;
98     #5;
99     $display("TC%d:", testcase);
100    assert(eq === 1'b0) else $error("Equality failed");
101    assert(lts === 1'b0) else $error("LessThan Signed failed");
102    assert(ltu === 1'b1) else $error("LessThan Unsigned failed");
103
104
105    // ~~ TC6 : signed vs unsigned (floor and ceiling 1) ~~
106    #5;
107    a = 32'h0000_0000;
108    b = 32'hFFFF_FFFF;
109    testcase++;
110    #5;
111    $display("TC%d:", testcase);
112    assert(eq === 1'b0) else $error("Equality failed");
113    assert(lts === 1'b0) else $error("LessThan Signed failed");
114    assert(ltu === 1'b1) else $error("LessThan Unsigned failed");
115
116
117    // ~~ TC7 : signed vs unsigned (floor and ceiling 2) ~~
118    #5;
119    a = 32'hFFFF_FFFF;
120    b = 32'h0000_0000;
121    testcase++;
122    #5;
123    $display("TC%d:", testcase);
124    assert(eq === 1'b0) else $error("Equality failed");
125    assert(lts === 1'b1) else $error("LessThan Signed failed");
126    assert(ltu === 1'b0) else $error("LessThan Unsigned failed");
127    end
128
129 endmodule

```

0.3.9 Control Unit Decoder

Listing 35: Control Unit Decoder CUnit_Decoder.sv

```
1  'timescale 1ns / 1ps
2  //////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 08/14/2026 02:38:59 PM
7  // Design Name:
8  // Module Name: CUnit_Decoder
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
21
22
23 module CUnit_Decoder(
24
25     // ~~~~ obtain inputs during/after EXEC ~~~~
26     input logic [6:0]   OPCODE,
27     input logic [2:0]   FUNC3,
28     input logic         FUNC7,
29     input logic BR_EQ, BR_LT, BR_LTU,
30
31     // ~~~~ MUX sel outs ~~~~
32     output logic [3:0]  ALU_FUNC,           // ALU operation sel
33     output logic        ALU_SRC_A,         // ALU srcA sel
34     output logic [1:0]  ALU_SRC_B,         // ALU srcB sel
35     output logic [1:0]  PC_SOURCE,         // input sel for program counter
36     output logic [1:0]  RF_WR_SEL         // register file write (data) source sel
37 );
38
39
40 always_comb
41 begin
42     case(OPCODE)
43
44         // ~~ loads (OP=3) ~~
45         7'b0000011:
46             begin
47                 ALU_FUNC    = 4'b0000;
48                 ALU_SRC_A   = 1'b0;
49                 ALU_SRC_B   = 2'b01;           // Select : immed (I-type)
50                 PC_SOURCE   = 2'b00;
51                 RF_WR_SEL   = 2'b10;           // Select : rd = DOUT2 = ram[rs1 +
52             end
53
54         // ~~ immed value (OP=19) ~~
55         7'b0010011:
56             begin
57                 ALU_FUNC = ((FUNC3 == 3'b101) && (FUNC7 == 1'b1)) ? ({1'b0, FUNC3} + 4'b1000) : {1'b0, FUNC3};
58                 ALU_SRC_A = 1'b0;
59                 ALU_SRC_B = 2'b01;           // Select immed I
60                 PC_SOURCE = 2'b00;
61                 RF_WR_SEL = 2'b11;           // Select rd = result
62             end
63
64         // ~~ add upper imm to pc (auipc) (OP=23) ~~
65         7'b0010111:
66             begin
67                 ALU_FUNC    = 4'b0000;
68                 ALU_SRC_A   = 1'b1;           // Select : imm (U-type)
69                 ALU_SRC_B   = 2'b11;           // Select : PC
70                 PC_SOURCE   = 2'b00;
71                 RF_WR_SEL   = 2'b11;           // Select : result
72             end
73
74         // ~~ store (OP=35) ~~
75         7'b0100011:
76             begin
77                 ALU_FUNC    = 4'b0000;
78                 ALU_SRC_A   = 1'b0;
79                 ALU_SRC_B   = 2'b10;           // Select : immed (S-type)
80                 PC_SOURCE   = 2'b00;
81                 RF_WR_SEL   = 2'b00;
82             end
83
84         // ~~ Registers as value (OP=51) ~~
85         7'b0110011:
86             begin
87
```

```

88         if((FUNC3 == 3'b000) || (FUNC3 == 3'b101))
89         begin
90             ALU_FUNC = (FUNC7 == 1'b1) ? ({1'b0,FUNC3}+4'b1000) : {1'b0,FUNC3};
91         end
92
93         else begin ALU_FUNC = {1'b0,FUNC3}; end
94
95         ALU_SRC_A = 1'b0;
96         ALU_SRC_B = 2'b00;
97         PC_SOURCE = 2'b00;
98         RF_WR_SEL = 2'b11; // Select rd = result
99     end
100
101     // ~~ lui (OP=55) ~~
102     7'b0110111:
103     begin
104         ALU_FUNC = 4'b1001; // Select : copy srcA
105         ALU_SRC_A = 1'b1; // Select : imm (U-type)
106         ALU_SRC_B = 2'b00;
107         PC_SOURCE = 2'b00;
108         RF_WR_SEL = 2'b11; // Select : result
109     end
110
111     // ~~ Branch (OP=99) ~~
112     7'b1100011:
113     begin
114         ALU_FUNC = 4'b0000;
115         ALU_SRC_A = 1'b0;
116         ALU_SRC_B = 2'b00;
117
118         case(FUNC3)
119             3'b000: begin PC_SOURCE = (BR_EQ) ? 2'b10 : 2'b00; end // BEQ : branch or PC
120             3'b001: begin PC_SOURCE = (~BR_EQ) ? 2'b10 : 2'b00; end // BNE : branch or PC
121             3'b100: begin PC_SOURCE = (BR_LT) ? 2'b10 : 2'b00; end // BLT : branch or PC
122             3'b101: begin PC_SOURCE = (~BR_LT) ? 2'b10 : 2'b00; end // BGE : branch or PC
123             3'b110: begin PC_SOURCE = (BR_LTU) ? 2'b10 : 2'b00; end // BLTU : branch or PC
124             3'b111: begin PC_SOURCE = (~BR_LTU)? 2'b10 : 2'b00; end // BGEU : branch or PC
125             default: begin PC_SOURCE = 2'b00; end
126         endcase
127
128         RF_WR_SEL = 2'b00;
129     end
130
131     // ~~ Jump and link register (OP=103) ~~
132     7'b1100111:
133     begin
134         ALU_FUNC = 4'b0000;
135         ALU_SRC_A = 1'b0;
136         ALU_SRC_B = 2'b00;
137         PC_SOURCE = 2'b01; // Select JALR
138         RF_WR_SEL = 2'b00;
139     end
140
141     // ~~ Jump and link (OP=111) ~~
142     7'b1101111:
143     begin
144         ALU_FUNC = 4'b0000;
145         ALU_SRC_A = 1'b0;
146         ALU_SRC_B = 2'b00;
147         PC_SOURCE = 2'b11; // Select JAL
148         RF_WR_SEL = 2'b00;
149     end
150
151     default:
152     begin
153         ALU_FUNC = 4'b0000; // Select : A + B
154         ALU_SRC_A = 1'b0; // A = rs1
155         ALU_SRC_B = 2'b00; // B = rs2
156         PC_SOURCE = 2'b00; // PC + 4
157         RF_WR_SEL = 2'b00; // PC + 4
158     end
159
160     endcase
161 end
162
163 endmodule

```

Listing 36: TestSim for Control Unit Decoder CUnit_Decoder_tb.sv

```

1  `timescale 1ns / 1ps
2  //////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 08/18/2026 09:18:24 AM
7  // Design Name:
8  // Module Name: CUnit_Decoder_tb
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:

```

```

13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
21
22
23 module CUnit_Decoder_tb();
24
25     // ~~~~ local vars ~~~~
26     logic [6:0] opcode;
27     logic [2:0] func3;
28     logic      func7;
29     logic      br_eq, br_lt, br_ltu;
30
31     logic [3:0] alu_func;
32     logic      src_a;
33     logic [1:0] src_b;
34     logic [1:0] pc_src;
35     logic [1:0] rf_wr_sel;
36
37     // ~~ testing ~~
38     logic[31:0] testcase;
39     logic [2:0] func3_arr [0:9];
40     logic [3:0] alu_arr [0:9];
41     int        opcodeCNT;
42
43
44     // ~~~~ DUT instance ~~~~
45     CUnit_Decoder UUT
46     (
47         // ~~ inputs ~~
48         .OPCODE(opcode),           // 7'b
49         .FUNC3(func3),             // 3'b
50         .FUNC7(func7),             // 1'b
51         .BR_EQ(br_eq),             // 1'b
52         .BR_LT(br_lt),             // 1'b
53         .BR_LTU(br_ltu),           // 1'b
54
55         // ~~ outputs ~~
56         .ALU_FUNC(alu_func),        // 4'b
57         .ALU_SRC_A(src_a),          // 1'b
58         .ALU_SRC_B(src_b),          // 2'b
59         .PC_SOURCE(pc_src),         // 2'b
60         .RF_WR_SEL(rf_wr_sel)       // 2'b
61     );
62
63
64     // ~~~~ testing ~~~~
65     initial
66     begin
67
68         // ~~ TC1 : unknown opcode is defaulted ~~
69         #5;
70         opcode = 7'b000_0000;
71         func3  = 3'bxxx;
72         func7  = 1'bx;
73         br_eq  = 1'bx;
74         br_lt  = 1'bx;
75         br_ltu = 1'bx;
76         testcase = 1;
77
78         #5;
79         $display("TC%d:", testcase);
80         assert(alu_func == 4'b0000) else $error("ALU_FUNC failed");
81         assert(src_a == 1'b0)      else $error("ALU_SRC_A failed");
82         assert(src_b == 2'b00)     else $error("ALU_SRC_B failed");
83         assert(pc_src == 2'b00)    else $error("PC_SOURCE failed");
84         assert(rf_wr_sel == 2'b00) else $error("RF_WR_SEL failed");
85
86
87
88
89         // ~~ TC2 : loading ~~
90         opcode = 7'b000_0011;
91         opcodeCNT = 5;
92         func3_arr =
93         '{
94             default : 3'b000,
95             0: 3'b000,           // lb
96             1: 3'b001,           // lh
97             2: 3'b010,           // lw
98             3: 3'b100,           // lbu
99             4: 3'b101            // lhu
100         };
101
102         testcase++;
103         for(int i = 0; i < opcodeCNT; i++)

```

```

104     begin
105
106         #5;
107         func3    = func3_arr[i];
108         func7    = 1'bx;
109         br_eq    = 1'bx;
110         br_lt    = 1'bx;
111         br_ltu   = 1'bx;
112
113         #5;
114         $display("TC%d: %d", testcase, i);
115         assert(alu_func === 4'b0000)    else $error("ALU_FUNC failed");
116         assert(src_a === 1'b0)          else $error("ALU_SRC_A failed");
117         assert(src_b === 2'b01)         else $error("ALU_SRC_B failed");
118         assert(pc_src === 2'b00)        else $error("PC_SOURCE failed");
119         assert(rf_wr_sel === 2'b10)     else $error("RF_WR_SEL failed");
120     end
121
122
123     // ~~ TC3 : immed ops ~~
124     opcode = 7'b001_0011;
125     opcodeCNT = 9;
126     func3_arr =
127     '{
128         default : 3'b000,
129         0: 3'b000,          // addi
130         1: 3'b001,          // slli
131         2: 3'b010,          // slti
132         3: 3'b011,          // sltiu
133         4: 3'b100,          // xori
134         5: 3'b101,          // srli
135         6: 3'b101,          // srai
136         7: 3'b110,          // ori
137         8: 3'b111           // andi
138     };
139
140     alu_arr =
141     '{
142         default : 4'b0000,
143         0: 4'b0000,          // addi
144         1: 4'b0001,          // slli
145         2: 4'b0010,          // slti
146         3: 4'b0011,          // sltiu
147         4: 4'b0100,          // xori
148         5: 4'b0101,          // srli
149         6: 4'b1101,          // srai
150         7: 4'b0110,          // ori
151         8: 4'b0111           // andi
152     };
153
154     testcase++;
155     for(int i = 0; i < opcodeCNT; i++)
156     begin
157
158         #5;
159         func3    = func3_arr[i];
160         func7    = (i==6) ? 1'b1 : 1'b0;    //srai needs func7
161         br_eq    = 1'bx;
162         br_lt    = 1'bx;
163         br_ltu   = 1'bx;
164
165         #5;
166         $display("TC%d: %d", testcase, i);
167         assert(alu_func === alu_arr[i])    else $error("ALU_FUNC failed");
168         assert(src_a === 1'b0)          else $error("ALU_SRC_A failed");
169         assert(src_b === 2'b01)         else $error("ALU_SRC_B failed");
170         assert(pc_src === 2'b00)        else $error("PC_SOURCE failed");
171         assert(rf_wr_sel === 2'b11)     else $error("RF_WR_SEL failed");
172     end
173
174
175
176
177     // ~~ TC4 : auipc ~~
178     #5;
179     opcode = 7'b001_0111;
180     func3    = 3'bxxx;
181     func7    = 1'bx;
182     br_eq    = 1'bx;
183     br_lt    = 1'bx;
184     br_ltu   = 1'bx;
185     testcase++;
186
187     #5;
188     $display("TC%d:", testcase);
189     assert(alu_func === 4'b0000)    else $error("ALU_FUNC failed");
190     assert(src_a === 1'b1)          else $error("ALU_SRC_A failed");
191     assert(src_b === 2'b11)         else $error("ALU_SRC_B failed");
192     assert(pc_src === 2'b00)        else $error("PC_SOURCE failed");
193     assert(rf_wr_sel === 2'b11)     else $error("RF_WR_SEL failed");
194

```

```

195
196
197 // ~~ TC5 : store ~~
198 opcode = 7'b010_0011;
199 opcodeCNT = 3;
200 func3_arr =
201 '{
202     default : 3'b000,
203     0: 3'b000, // sb
204     1: 3'b001, // sh
205     2: 3'b010 // sw
206 };
207
208 testcase++;
209 for(int i = 0; i < opcodeCNT; i++)
210 begin
211
212     #5;
213     func3 = func3_arr[i];
214     func7 = 1'bx;
215     br_eq = 1'bx;
216     br_lt = 1'bx;
217     br_ltu = 1'bx;
218
219     #5;
220     $display("TC%d: %d", testcase, i);
221     assert(alu_func === 4'b0000) else $error("ALU_FUNC failed");
222     assert(src_a === 1'b0) else $error("ALU_SRC_A failed");
223     assert(src_b === 2'b10) else $error("ALU_SRC_B failed");
224     assert(pc_src === 2'b00) else $error("PC_SOURCE failed");
225     assert(rf_wr_sel === 2'b00) else $error("RF_WR_SEL failed");
226 end
227
228
229
230
231 // ~~ TC6 : dual register ops ~~
232 opcode = 7'b011_0011;
233 opcodeCNT = 10;
234 func3_arr =
235 '{
236     default : 3'b000,
237     0: 3'b000, // add
238     1: 3'b000, // sub "
239     2: 3'b001, // sll
240     3: 3'b010, // slt
241     4: 3'b011, // sltu
242     5: 3'b100, // xor
243     6: 3'b101, // srl
244     7: 3'b101, // sra "
245     8: 3'b110, // or
246     9: 3'b111 // and
247 };
248
249 alu_arr =
250 '{
251     default : 4'b0000,
252     0: 4'b0000, // add
253     1: 4'b1000, // sub
254     2: 4'b0001, // sll
255     3: 4'b0010, // slt
256     4: 4'b0011, // sltu
257     5: 4'b0100, // xor
258     6: 4'b0101, // srl
259     7: 4'b1101, // sra
260     8: 4'b0110, // or
261     9: 4'b0111 // and
262 };
263
264 testcase++;
265 for(int i = 0; i < opcodeCNT; i++)
266 begin
267
268     #5;
269     func3 = func3_arr[i];
270     func7 = (i==1 || i==7) ? 1'b1 : 1'b0; //sub and sra need func7
271     br_eq = 1'bx;
272     br_lt = 1'bx;
273     br_ltu = 1'bx;
274
275     #5;
276     $display("TC%d: %d", testcase, i);
277     assert(alu_func === alu_arr[i]) else $error("ALU_FUNC failed");
278     assert(src_a === 1'b0) else $error("ALU_SRC_A failed");
279     assert(src_b === 2'b00) else $error("ALU_SRC_B failed");
280     assert(pc_src === 2'b00) else $error("PC_SOURCE failed");
281     assert(rf_wr_sel === 2'b11) else $error("RF_WR_SEL failed");
282 end
283
284
285

```

```

286 // ~~ TC7 : lui ~~
287 #5;
288 opcode = 7'b011_0111;
289 func3 = 3'bxxx;
290 func7 = 1'bx;
291 br_eq = 1'bx;
292 br_lt = 1'bx;
293 br_ltu = 1'bx;
294 testcase++;
295
296 #5;
297 $display("TC%d:", testcase);
298 assert(alu_func == 4'b1001) else $error("ALU_FUNC failed");
299 assert(src_a == 1'b1) else $error("ALU_SRC_A failed");
300 assert(src_b == 2'b00) else $error("ALU_SRC_B failed");
301 assert(pc_src == 2'b00) else $error("PC_SOURCE failed");
302 assert(rf_wr_sel == 2'b11) else $error("RF_WR_SEL failed");
303
304
305
306 // ~~ TC8 : branch ~~
307 opcode = 7'b110_0011;
308 testcase++;
309 opcodeCNT = 6;
310 func3_arr =
311 '{
312     default : 3'b000,
313     0: 3'b000, // beq
314     1: 3'b001, // bne
315     2: 3'b100, // blt
316     3: 3'b101, // bge
317     4: 3'b110, // bltu
318     5: 3'b111 // bgeu
319 };
320
321 func7 = 1'bx;
322
323 // ~ beq (0) ~
324 #5;
325 func3 = func3_arr[0];
326 br_eq = 1'b0;
327 br_lt = 1'bx;
328 br_ltu = 1'bx;
329
330 #5;
331 $display("TC%d: BEQ", testcase);
332 assert(alu_func == 4'b0000) else $error("ALU_FUNC failed");
333 assert(src_a == 1'b0) else $error("ALU_SRC_A failed");
334 assert(src_b == 2'b00) else $error("ALU_SRC_B failed");
335 assert(pc_src == 2'b00) else $error("PC_SOURCE failed");
336 assert(rf_wr_sel == 2'b00) else $error("RF_WR_SEL failed");
337
338 #5;
339 br_eq = 1'b1;
340
341 #5;
342 $display("TC%d: BEQ", testcase);
343 assert(alu_func == 4'b0000) else $error("ALU_FUNC failed");
344 assert(src_a == 1'b0) else $error("ALU_SRC_A failed");
345 assert(src_b == 2'b00) else $error("ALU_SRC_B failed");
346 assert(pc_src == 2'b10) else $error("PC_SOURCE failed");
347 assert(rf_wr_sel == 2'b00) else $error("RF_WR_SEL failed");
348
349
350 // ~ bne (1) ~
351 #5;
352 func3 = func3_arr[1];
353 br_eq = 1'b0;
354 br_lt = 1'bx;
355 br_ltu = 1'bx;
356
357 #5;
358 $display("TC%d: BNE", testcase);
359 assert(alu_func == 4'b0000) else $error("ALU_FUNC failed");
360 assert(src_a == 1'b0) else $error("ALU_SRC_A failed");
361 assert(src_b == 2'b00) else $error("ALU_SRC_B failed");
362 assert(pc_src == 2'b10) else $error("PC_SOURCE failed");
363 assert(rf_wr_sel == 2'b00) else $error("RF_WR_SEL failed");
364
365 #5;
366 br_eq = 1'b1;
367
368 #5;
369 $display("TC%d: BNE", testcase);
370 assert(alu_func == 4'b0000) else $error("ALU_FUNC failed");
371 assert(src_a == 1'b0) else $error("ALU_SRC_A failed");
372 assert(src_b == 2'b00) else $error("ALU_SRC_B failed");
373 assert(pc_src == 2'b00) else $error("PC_SOURCE failed");
374 assert(rf_wr_sel == 2'b00) else $error("RF_WR_SEL failed");
375
376

```

```

377 // ~ blt (2) ~
378 #5;
379 func3 = func3_arr[2];
380 br_eq = 1'bx;
381 br_lt = 1'b0;
382 br_ltu = 1'bx;
383
384 #5;
385 $display("TC%d: BLT", testcase);
386 assert(alu_func == 4'b0000) else $error("ALU_FUNC failed");
387 assert(src_a == 1'b0) else $error("ALU_SRC_A failed");
388 assert(src_b == 2'b00) else $error("ALU_SRC_B failed");
389 assert(pc_src == 2'b00) else $error("PC_SOURCE failed");
390 assert(rf_wr_sel == 2'b00) else $error("RF_WR_SEL failed");
391
392 #5;
393 br_lt = 1'b1;
394
395 #5;
396 $display("TC%d: BLT", testcase);
397 assert(alu_func == 4'b0000) else $error("ALU_FUNC failed");
398 assert(src_a == 1'b0) else $error("ALU_SRC_A failed");
399 assert(src_b == 2'b00) else $error("ALU_SRC_B failed");
400 assert(pc_src == 2'b10) else $error("PC_SOURCE failed");
401 assert(rf_wr_sel == 2'b00) else $error("RF_WR_SEL failed");
402
403 // ~ bge (3) ~
404 #5;
405 func3 = func3_arr[3];
406 br_eq = 1'bx;
407 br_lt = 1'b0;
408 br_ltu = 1'bx;
409
410 #5;
411 $display("TC%d: BGE", testcase);
412 assert(alu_func == 4'b0000) else $error("ALU_FUNC failed");
413 assert(src_a == 1'b0) else $error("ALU_SRC_A failed");
414 assert(src_b == 2'b00) else $error("ALU_SRC_B failed");
415 assert(pc_src == 2'b10) else $error("PC_SOURCE failed");
416 assert(rf_wr_sel == 2'b00) else $error("RF_WR_SEL failed");
417
418 #5;
419 br_lt = 1'b1;
420
421 #5;
422 $display("TC%d: BGE", testcase);
423 assert(alu_func == 4'b0000) else $error("ALU_FUNC failed");
424 assert(src_a == 1'b0) else $error("ALU_SRC_A failed");
425 assert(src_b == 2'b00) else $error("ALU_SRC_B failed");
426 assert(pc_src == 2'b00) else $error("PC_SOURCE failed");
427 assert(rf_wr_sel == 2'b00) else $error("RF_WR_SEL failed");
428
429 // ~ bltu (4) ~
430 #5;
431 func3 = func3_arr[4];
432 br_eq = 1'bx;
433 br_lt = 1'bx;
434 br_ltu = 1'b0;
435
436 #5;
437 $display("TC%d: BLTU", testcase);
438 assert(alu_func == 4'b0000) else $error("ALU_FUNC failed");
439 assert(src_a == 1'b0) else $error("ALU_SRC_A failed");
440 assert(src_b == 2'b00) else $error("ALU_SRC_B failed");
441 assert(pc_src == 2'b00) else $error("PC_SOURCE failed");
442 assert(rf_wr_sel == 2'b00) else $error("RF_WR_SEL failed");
443
444 #5;
445 br_ltu = 1'b1;
446
447 #5;
448 $display("TC%d: BLTU", testcase);
449 assert(alu_func == 4'b0000) else $error("ALU_FUNC failed");
450 assert(src_a == 1'b0) else $error("ALU_SRC_A failed");
451 assert(src_b == 2'b00) else $error("ALU_SRC_B failed");
452 assert(pc_src == 2'b10) else $error("PC_SOURCE failed");
453 assert(rf_wr_sel == 2'b00) else $error("RF_WR_SEL failed");
454
455 // ~ bgeu (5) ~
456 #5;
457 func3 = func3_arr[5];
458 br_eq = 1'bx;
459 br_lt = 1'bx;
460 br_ltu = 1'b0;
461
462 #5;

```

```

468     $display("TC%d: BGEU", testcase);
469     assert(alu_func == 4'b0000)      else $error("ALU_FUNC failed");
470     assert(src_a == 1'b0)            else $error("ALU_SRC_A failed");
471     assert(src_b == 2'b00)          else $error("ALU_SRC_B failed");
472     assert(pc_src == 2'b10)         else $error("PC_SOURCE failed");
473     assert(rf_wr_sel == 2'b00)      else $error("RF_WR_SEL failed");
474
475     #5;
476     br_ltu    = 1'b1;
477
478     #5;
479     $display("TC%d: BGEU", testcase);
480     assert(alu_func == 4'b0000)      else $error("ALU_FUNC failed");
481     assert(src_a == 1'b0)            else $error("ALU_SRC_A failed");
482     assert(src_b == 2'b00)          else $error("ALU_SRC_B failed");
483     assert(pc_src == 2'b00)         else $error("PC_SOURCE failed");
484     assert(rf_wr_sel == 2'b00)      else $error("RF_WR_SEL failed");
485
486
487     // ~~ TC9 : jalr ~~
488     #5;
489     opcode    = 7'b110_0111;
490     func3     = 3'bxxx;
491     func7     = 1'bx;
492     br_eq     = 1'bx;
493     br_lt     = 1'bx;
494     br_ltu    = 1'bx;
495     testcase++;
496
497     #5;
498     $display("TC%d:", testcase);
499     assert(alu_func == 4'b0000)      else $error("ALU_FUNC failed");
500     assert(src_a == 1'b0)            else $error("ALU_SRC_A failed");
501     assert(src_b == 2'b00)          else $error("ALU_SRC_B failed");
502     assert(pc_src == 2'b01)         else $error("PC_SOURCE failed");
503     assert(rf_wr_sel == 2'b00)      else $error("RF_WR_SEL failed");
504
505
506     // ~~ TC10 : jal ~~
507     #5;
508     opcode    = 7'b110_1111;
509     func3     = 3'bxxx;
510     func7     = 1'bx;
511     br_eq     = 1'bx;
512     br_lt     = 1'bx;
513     br_ltu    = 1'bx;
514     testcase++;
515
516     #5;
517     $display("TC%d:", testcase);
518     assert(alu_func == 4'b0000)      else $error("ALU_FUNC failed");
519     assert(src_a == 1'b0)            else $error("ALU_SRC_A failed");
520     assert(src_b == 2'b00)          else $error("ALU_SRC_B failed");
521     assert(pc_src == 2'b11)         else $error("PC_SOURCE failed");
522     assert(rf_wr_sel == 2'b00)      else $error("RF_WR_SEL failed");
523
524     end
525 endmodule

```

0.3.10 Control Unit Finite State Machine

Listing 37: Control Unit FSM CUnit_FSM.sv

```
1 `timescale 1ns / 1ps
2 //////////////////////////////////////////////////
3 // Company:
4 // Engineer:
5 //
6 // Create Date: 08/15/2026 11:47:24 AM
7 // Design Name:
8 // Module Name: CUnit_FSM
9 // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //////////////////////////////////////////////////
21
22
23 module CUnit_FSM(
24     input logic      CLK,
25     input logic      RST,           // high-level input : reset
26     //input logic     INTRR,        // high-level input : interrupt
27     input logic [6:0] OPCODE,      // shared with decoder + regfile
28
29     output logic      PC_WRITE,     // update PC
30     output logic      REG_WRITE,    // update dest reg value
31     output logic      MEM_RDEN1,    // fetch value at addr1
32     output logic      MEM_RDEN2,    // fetch value at addr2 (occurs after update if enacted)
33     output logic      MEM_WE2,      // update value at addr2
34     output logic      RESET        // program counter reset
35 );
36
37
38 // ~~~~ local vars ~~~~
39 typedef enum logic [1:0] {INIT, FETCH, EXEC, WRITEBACK} e_state;
40 e_state curr_state, next_state;
41
42 typedef enum logic {LOAD, LOAD_BAR} e_lsignal;
43 e_lsignal curr_signal, next_signal;
44
45 logic start;
46
47 // ~~~~ initial conditions ~~~~
48 initial
49 begin
50     start      = 1'b1;
51 end
52
53 // ~~~~ state engine ~~~~
54 always_ff @(posedge CLK or posedge RST) begin
55     if(RST == 1'b1 || start == 1'b1) begin
56         curr_state      <= INIT;
57         curr_signal      <= LOAD;
58         start           <= 1'b0;
59     end
60     else begin
61         curr_state <= next_state;
62         curr_signal <= next_signal;
63     end
64 end
65
66 // ~~~~ state definitions ~~~~
67 always_comb begin
68     case(curr_state)
69
70         // ~ INIT ~
71         //      - Defaults all signals
72         //      - Entry point for reset
73         INIT: begin
74             RESET      = 1'b1;           // reset PC
75             PC_WRITE    = 1'b0;
76             REG_WRITE   = 1'b0;
77             MEM_WE2     = 1'b0;
78             MEM_RDEN1   = 1'b0;           // Dont read current instr
79             MEM_RDEN2   = 1'b0;
80
81             next_state  = FETCH;
82             next_signal = LOAD;
83         end
84     end
85 end
86
87
```

```

88      // ~ FETCH ~
89      //      - Grabs instruction
90      //      - Set memory unit to grab instruction
91      FETCH: begin
92          RESET      = 1'b0;
93          PC_WRITE   = 1'b0;
94          REG_WRITE  = 1'b0;
95          MEM_WE2    = 1'b0;
96
97          // ~~ Repeat FETCH in extra cycle ~~
98          if(curr_signal == LOAD_BAR) begin
99              MEM_RDEN1 = 1'b0;
100             MEM_RDEN2 = 1'b1;
101             next_signal = LOAD_BAR;
102         end
103         // ~~ else, standard FETCH cycle ~~
104         else begin
105             MEM_RDEN1 = 1'b1;
106             MEM_RDEN2 = 1'b0;
107             next_signal = LOAD;
108         end
109
110         next_state = EXEC;
111     end
112
113     // ~ EXEC ~
114     //      - Set all control sigs to read memory
115     //      - may direct to FETCH for lengthy instr cycles
116     //      - OPCODE dictates what is toggled here
117     EXEC: begin
118         RESET      = 1'b0;
119         PC_WRITE   = 1'b0;
120         REG_WRITE  = 1'b0;
121         MEM_WE2    = 1'b0;
122         MEM_RDEN1 = 1'b0;
123         MEM_RDEN2 = 1'b0;
124
125         if(OPCODE == 7'b000_0011) begin          // load needs an extra cycle
126             if(curr_signal == LOAD) begin
127                 next_signal = LOAD_BAR;
128                 next_state = FETCH;
129             end
130
131             else begin
132                 next_signal = LOAD;
133                 next_state = WRITEBACK;
134             end
135         end
136
137         else begin
138             next_signal = LOAD;
139             next_state = WRITEBACK;
140         end
141     end
142
143
144
145
146     // ~ WRITEBACK ~
147     //      - Save contents to reg or memory
148     //      - update pc
149     WRITEBACK: begin
150         RESET      = 1'b0;
151         PC_WRITE   = 1'b1;
152
153         if(OPCODE == 7'b0100011) begin          // store (35)
154             REG_WRITE = 1'b0;
155             MEM_WE2   = 1'b1;
156         end
157
158         else if(OPCODE == 7'b1100011) begin // branch (99)
159             REG_WRITE = 1'b0;
160             MEM_WE2   = 1'b0;
161         end
162
163         else begin
164             REG_WRITE = 1'b1;
165             MEM_WE2   = 1'b0;
166         end
167
168         MEM_RDEN1 = 1'b0;
169         MEM_RDEN2 = 1'b0;
170
171         next_state = FETCH;
172         next_signal = LOAD;
173
174     end
175 endcase
176
177 end
178

```

Listing 38: TestSim for Control Unit FSM CUnit_FSM_tb.sv

```

1  'timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 08/18/2026 12:48:54 PM
7  // Design Name:
8  // Module Name: CUnit_FSM_tb
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module CUnit_FSM_tb();
24
25     // ~~~~ local vars ~~~~
26
27     // ~~ input ~~
28     logic      clk;
29     logic      rst;
30     //logic     interr;
31     logic [6:0] opcode;
32
33     // ~~ output ~~
34     logic      pc_write;
35     logic      reg_write;
36     logic      mem_rden1;
37     logic      mem_rden2;
38     logic      mem_we2;
39     logic      pc_reset;
40
41     // ~~ testing ~~
42     logic[31:0] testcase;
43     logic[6:0] opc_arr [0:8];
44     int opcCNT = 9;
45
46
47     // ~~~~ UUT instance ~~~~
48     CUnit_FSM UUT
49     (
50         .CLK(clk),                // 1'b
51         .RST(rst),                // 1'b
52         //.INTRR(),                // 1'b
53         .OPCODE(opcode),          // 7'b
54
55         .PC_WRITE(pc_write),      // 1'b
56         .REG_WRITE(reg_write),    // 1'b
57         .MEM_RDEN1(mem_rden1),    // 1'b
58         .MEM_RDEN2(mem_rden2),    // 1'b
59         .MEM_WE2(mem_we2),        // 1'b
60         .RESET(pc_reset)          // 1'b
61     );
62
63
64     // ~~ action : CLK (10ns period) ~~
65     always
66     begin
67         #5;
68         clk <= !clk;
69     end
70
71
72     // ~~~~ action : data ~~~~
73     initial begin
74
75         // ~~ init ~~
76         clk = 1'b1;
77         rst = 1'b0;
78         opcode = 7'bxxx_xxxx;
79         opc_arr =
80         '{
81             default : 7'b000_0000,
82             0:       7'b000_0011,          // load
83             1:       7'b001_0011,          // imm ops
84             2:       7'b001_0111,          // auipc
85             3:       7'b010_0011,          // store
86             4:       7'b011_0011,          // dual reg ops
87             5:       7'b011_0111,          // lui

```

```

88         6:          7'b110_0011,          // branch
89         7:          7'b110_0111,          // jalr
90         8:          7'b110_1111          // jal
91     };
92
93     // ~~ TC1 : load(3) ~~
94     testcase = 1;
95     $display("TC%d:", testcase);
96
97     // ~ INIT ~
98     $display("INIT");
99     #5;
100    assert(pc_reset === 1'b1)    else $error("PC_RESET failed");
101    assert(pc_write  === 1'b0)    else $error("PC_WRITE failed");
102    assert(reg_write === 1'b0)    else $error("REG_WRITE failed");
103    assert(mem_we2   === 1'b0)    else $error("MEM_WE2 failed");
104    assert(mem_rden1 === 1'b0)    else $error("MEM_RDEN1 failed");
105    assert(mem_rden2 === 1'b0)    else $error("MEM_RDEN2 failed");
106    #5;
107
108    // ~ FETCH 1 ~
109    $display("FETCH 1");
110    #5;
111    assert(pc_reset === 1'b0)    else $error("PC_RESET failed");
112    assert(pc_write  === 1'b0)    else $error("PC_WRITE failed");
113    assert(reg_write === 1'b0)    else $error("REG_WRITE failed");
114    assert(mem_we2   === 1'b0)    else $error("MEM_WE2 failed");
115    assert(mem_rden1 === 1'b1)    else $error("MEM_RDEN1 failed");
116    assert(mem_rden2 === 1'b0)    else $error("MEM_RDEN2 failed");
117    #5;
118
119    // ~ EXEC 1 ~
120    $display("EXEC 1");
121    opcode = opcd_arr[0];          // load opcode
122    #5;
123    assert(pc_reset === 1'b0)    else $error("PC_RESET failed");
124    assert(pc_write  === 1'b0)    else $error("PC_WRITE failed");
125    assert(reg_write === 1'b0)    else $error("REG_WRITE failed");
126    assert(mem_we2   === 1'b0)    else $error("MEM_WE2 failed");
127    assert(mem_rden1 === 1'b0)    else $error("MEM_RDEN1 failed");
128    assert(mem_rden2 === 1'b0)    else $error("MEM_RDEN2 failed");
129    #5;
130
131    // ~ FETCH 2 ~
132    $display("FETCH 2");
133    #5;
134    assert(pc_reset === 1'b0)    else $error("PC_RESET failed");
135    assert(pc_write  === 1'b0)    else $error("PC_WRITE failed");
136    assert(reg_write === 1'b0)    else $error("REG_WRITE failed");
137    assert(mem_we2   === 1'b0)    else $error("MEM_WE2 failed");
138    assert(mem_rden1 === 1'b0)    else $error("MEM_RDEN1 failed");
139    assert(mem_rden2 === 1'b1)    else $error("MEM_RDEN2 failed");
140    #5;
141
142    // ~ EXEC 2 ~
143    $display("EXEC 2");
144    #5;
145    assert(pc_reset === 1'b0)    else $error("PC_RESET failed");
146    assert(pc_write  === 1'b0)    else $error("PC_WRITE failed");
147    assert(reg_write === 1'b0)    else $error("REG_WRITE failed");
148    assert(mem_we2   === 1'b0)    else $error("MEM_WE2 failed");
149    assert(mem_rden1 === 1'b0)    else $error("MEM_RDEN1 failed");
150    assert(mem_rden2 === 1'b0)    else $error("MEM_RDEN2 failed");
151    #5;
152
153    // ~ WRITEBACK ~
154    $display("WRITEBACK");
155    #5;
156    assert(pc_reset === 1'b0)    else $error("PC_RESET failed");
157    assert(pc_write  === 1'b1)    else $error("PC_WRITE failed");
158    assert(reg_write === 1'b1)    else $error("REG_WRITE failed");
159    assert(mem_we2   === 1'b0)    else $error("MEM_WE2 failed");
160    assert(mem_rden1 === 1'b0)    else $error("MEM_RDEN1 failed");
161    assert(mem_rden2 === 1'b0)    else $error("MEM_RDEN2 failed");
162    #5;
163
164
165    // ~~ TC2 : imm ops (19) ~~
166    testcase++;
167    $display("TC%d:", testcase);
168
169    // ~ FETCH ~
170    $display("FETCH");
171    #5;
172    assert(pc_reset === 1'b0)    else $error("PC_RESET failed");
173    assert(pc_write  === 1'b0)    else $error("PC_WRITE failed");
174    assert(reg_write === 1'b0)    else $error("REG_WRITE failed");
175    assert(mem_we2   === 1'b0)    else $error("MEM_WE2 failed");
176    assert(mem_rden1 === 1'b1)    else $error("MEM_RDEN1 failed");
177    assert(mem_rden2 === 1'b0)    else $error("MEM_RDEN2 failed");
178    #5;

```

```

179
180 // ~ EXEC ~
181 $display("EXEC");
182 opcode = opc_arr[1]; // load opcode
183 #5;
184 assert(pc_reset === 1'b0) else $error("PC_RESET failed");
185 assert(pc_write === 1'b0) else $error("PC_WRITE failed");
186 assert(reg_write === 1'b0) else $error("REG_WRITE failed");
187 assert(mem_we2 === 1'b0) else $error("MEM_WE2 failed");
188 assert(mem_rden1 === 1'b0) else $error("MEM_RDEN1 failed");
189 assert(mem_rden2 === 1'b0) else $error("MEM_RDEN2 failed");
190 #5;
191
192 // ~ WRITEBACK ~
193 $display("WRITEBACK");
194 #5;
195 assert(pc_reset === 1'b0) else $error("PC_RESET failed");
196 assert(pc_write === 1'b1) else $error("PC_WRITE failed");
197 assert(reg_write === 1'b1) else $error("REG_WRITE failed");
198 assert(mem_we2 === 1'b0) else $error("MEM_WE2 failed");
199 assert(mem_rden1 === 1'b0) else $error("MEM_RDEN1 failed");
200 assert(mem_rden2 === 1'b0) else $error("MEM_RDEN2 failed");
201 #5;
202
203
204 // ~~ TC3 : auipc (23) ~~
205 testcase++;
206 $display("TC%d:", testcase);
207
208 // ~ FETCH ~
209 $display("FETCH");
210 #5;
211 assert(pc_reset === 1'b0) else $error("PC_RESET failed");
212 assert(pc_write === 1'b0) else $error("PC_WRITE failed");
213 assert(reg_write === 1'b0) else $error("REG_WRITE failed");
214 assert(mem_we2 === 1'b0) else $error("MEM_WE2 failed");
215 assert(mem_rden1 === 1'b1) else $error("MEM_RDEN1 failed");
216 assert(mem_rden2 === 1'b0) else $error("MEM_RDEN2 failed");
217 #5;
218
219 // ~ EXEC ~
220 $display("EXEC");
221 opcode = opc_arr[2]; // load opcode
222 #5;
223 assert(pc_reset === 1'b0) else $error("PC_RESET failed");
224 assert(pc_write === 1'b0) else $error("PC_WRITE failed");
225 assert(reg_write === 1'b0) else $error("REG_WRITE failed");
226 assert(mem_we2 === 1'b0) else $error("MEM_WE2 failed");
227 assert(mem_rden1 === 1'b0) else $error("MEM_RDEN1 failed");
228 assert(mem_rden2 === 1'b0) else $error("MEM_RDEN2 failed");
229 #5;
230
231 // ~ WRITEBACK ~
232 $display("WRITEBACK");
233 #5;
234 assert(pc_reset === 1'b0) else $error("PC_RESET failed");
235 assert(pc_write === 1'b1) else $error("PC_WRITE failed");
236 assert(reg_write === 1'b1) else $error("REG_WRITE failed");
237 assert(mem_we2 === 1'b0) else $error("MEM_WE2 failed");
238 assert(mem_rden1 === 1'b0) else $error("MEM_RDEN1 failed");
239 assert(mem_rden2 === 1'b0) else $error("MEM_RDEN2 failed");
240 #5;
241
242
243 // ~~ TC4 : store (35) ~~
244 testcase++;
245 $display("TC%d:", testcase);
246
247 // ~ FETCH ~
248 $display("FETCH");
249 #5;
250 assert(pc_reset === 1'b0) else $error("PC_RESET failed");
251 assert(pc_write === 1'b0) else $error("PC_WRITE failed");
252 assert(reg_write === 1'b0) else $error("REG_WRITE failed");
253 assert(mem_we2 === 1'b0) else $error("MEM_WE2 failed");
254 assert(mem_rden1 === 1'b1) else $error("MEM_RDEN1 failed");
255 assert(mem_rden2 === 1'b0) else $error("MEM_RDEN2 failed");
256 #5;
257
258 // ~ EXEC ~
259 $display("EXEC");
260 opcode = opc_arr[3]; // load opcode
261 #5;
262 assert(pc_reset === 1'b0) else $error("PC_RESET failed");
263 assert(pc_write === 1'b0) else $error("PC_WRITE failed");
264 assert(reg_write === 1'b0) else $error("REG_WRITE failed");
265 assert(mem_we2 === 1'b0) else $error("MEM_WE2 failed");
266 assert(mem_rden1 === 1'b0) else $error("MEM_RDEN1 failed");
267 assert(mem_rden2 === 1'b0) else $error("MEM_RDEN2 failed");
268 #5;
269

```

```

270 // ~ WRITEBACK ~
271 $display("WRITEBACK");
272 #5;
273 assert(pc_reset === 1'b0) else $error("PC_RESET failed");
274 assert(pc_write === 1'b1) else $error("PC_WRITE failed");
275 assert(reg_write === 1'b0) else $error("REG_WRITE failed");
276 assert(mem_we2 === 1'b1) else $error("MEM_WE2 failed");
277 assert(mem_rden1 === 1'b0) else $error("MEM_RDEN1 failed");
278 assert(mem_rden2 === 1'b0) else $error("MEM_RDEN2 failed");
279 #5;
280
281
282 // ~~ TC5 : dual reg ops (51) ~~
283 testcase++;
284 $display("TC%d:", testcase);
285
286 // ~ FETCH ~
287 $display("FETCH");
288 #5;
289 assert(pc_reset === 1'b0) else $error("PC_RESET failed");
290 assert(pc_write === 1'b0) else $error("PC_WRITE failed");
291 assert(reg_write === 1'b0) else $error("REG_WRITE failed");
292 assert(mem_we2 === 1'b0) else $error("MEM_WE2 failed");
293 assert(mem_rden1 === 1'b1) else $error("MEM_RDEN1 failed");
294 assert(mem_rden2 === 1'b0) else $error("MEM_RDEN2 failed");
295 #5;
296
297 // ~ EXEC ~
298 $display("EXEC");
299 opcode = opc_arr[4]; // load opcode
300 #5;
301 assert(pc_reset === 1'b0) else $error("PC_RESET failed");
302 assert(pc_write === 1'b0) else $error("PC_WRITE failed");
303 assert(reg_write === 1'b0) else $error("REG_WRITE failed");
304 assert(mem_we2 === 1'b0) else $error("MEM_WE2 failed");
305 assert(mem_rden1 === 1'b0) else $error("MEM_RDEN1 failed");
306 assert(mem_rden2 === 1'b0) else $error("MEM_RDEN2 failed");
307 #5;
308
309 // ~ WRITEBACK ~
310 $display("WRITEBACK");
311 #5;
312 assert(pc_reset === 1'b0) else $error("PC_RESET failed");
313 assert(pc_write === 1'b1) else $error("PC_WRITE failed");
314 assert(reg_write === 1'b1) else $error("REG_WRITE failed");
315 assert(mem_we2 === 1'b0) else $error("MEM_WE2 failed");
316 assert(mem_rden1 === 1'b0) else $error("MEM_RDEN1 failed");
317 assert(mem_rden2 === 1'b0) else $error("MEM_RDEN2 failed");
318 #5;
319
320
321 // ~~ TC6 : lui (55) ~~
322 testcase++;
323 $display("TC%d:", testcase);
324
325 // ~ FETCH ~
326 $display("FETCH");
327 #5;
328 assert(pc_reset === 1'b0) else $error("PC_RESET failed");
329 assert(pc_write === 1'b0) else $error("PC_WRITE failed");
330 assert(reg_write === 1'b0) else $error("REG_WRITE failed");
331 assert(mem_we2 === 1'b0) else $error("MEM_WE2 failed");
332 assert(mem_rden1 === 1'b1) else $error("MEM_RDEN1 failed");
333 assert(mem_rden2 === 1'b0) else $error("MEM_RDEN2 failed");
334 #5;
335
336 // ~ EXEC ~
337 $display("EXEC");
338 opcode = opc_arr[5]; // load opcode
339 #5;
340 assert(pc_reset === 1'b0) else $error("PC_RESET failed");
341 assert(pc_write === 1'b0) else $error("PC_WRITE failed");
342 assert(reg_write === 1'b0) else $error("REG_WRITE failed");
343 assert(mem_we2 === 1'b0) else $error("MEM_WE2 failed");
344 assert(mem_rden1 === 1'b0) else $error("MEM_RDEN1 failed");
345 assert(mem_rden2 === 1'b0) else $error("MEM_RDEN2 failed");
346 #5;
347
348 // ~ WRITEBACK ~
349 $display("WRITEBACK");
350 #5;
351 assert(pc_reset === 1'b0) else $error("PC_RESET failed");
352 assert(pc_write === 1'b1) else $error("PC_WRITE failed");
353 assert(reg_write === 1'b1) else $error("REG_WRITE failed");
354 assert(mem_we2 === 1'b0) else $error("MEM_WE2 failed");
355 assert(mem_rden1 === 1'b0) else $error("MEM_RDEN1 failed");
356 assert(mem_rden2 === 1'b0) else $error("MEM_RDEN2 failed");
357 #5;
358
359
360 // ~~ TC7 : branch (99) ~~

```

```

361     testcase++;
362     $display("TC%d:", testcase);
363
364     // ~ FETCH ~
365     $display("FETCH");
366     #5;
367     assert(pc_reset == 1'b0)    else $error("PC_RESET failed");
368     assert(pc_write == 1'b0)    else $error("PC_WRITE failed");
369     assert(reg_write == 1'b0)   else $error("REG_WRITE failed");
370     assert(mem_we2 == 1'b0)     else $error("MEM_WE2 failed");
371     assert(mem_rden1 == 1'b1)   else $error("MEM_RDEN1 failed");
372     assert(mem_rden2 == 1'b0)   else $error("MEM_RDEN2 failed");
373     #5;
374
375     // ~ EXEC ~
376     $display("EXEC");
377     opcode = opc_arr[6];        // load opcode
378     #5;
379     assert(pc_reset == 1'b0)    else $error("PC_RESET failed");
380     assert(pc_write == 1'b0)    else $error("PC_WRITE failed");
381     assert(reg_write == 1'b0)   else $error("REG_WRITE failed");
382     assert(mem_we2 == 1'b0)     else $error("MEM_WE2 failed");
383     assert(mem_rden1 == 1'b0)   else $error("MEM_RDEN1 failed");
384     assert(mem_rden2 == 1'b0)   else $error("MEM_RDEN2 failed");
385     #5;
386
387     // ~ WRITEBACK ~
388     $display("WRITEBACK");
389     #5;
390     assert(pc_reset == 1'b0)    else $error("PC_RESET failed");
391     assert(pc_write == 1'b1)    else $error("PC_WRITE failed");
392     assert(reg_write == 1'b0)   else $error("REG_WRITE failed");
393     assert(mem_we2 == 1'b0)     else $error("MEM_WE2 failed");
394     assert(mem_rden1 == 1'b0)   else $error("MEM_RDEN1 failed");
395     assert(mem_rden2 == 1'b0)   else $error("MEM_RDEN2 failed");
396     #5;
397
398
399
400     // ~~ TC8 : jalr (103) ~~
401     testcase++;
402     $display("TC%d:", testcase);
403
404     // ~ FETCH ~
405     $display("FETCH");
406     #5;
407     assert(pc_reset == 1'b0)    else $error("PC_RESET failed");
408     assert(pc_write == 1'b0)    else $error("PC_WRITE failed");
409     assert(reg_write == 1'b0)   else $error("REG_WRITE failed");
410     assert(mem_we2 == 1'b0)     else $error("MEM_WE2 failed");
411     assert(mem_rden1 == 1'b1)   else $error("MEM_RDEN1 failed");
412     assert(mem_rden2 == 1'b0)   else $error("MEM_RDEN2 failed");
413     #5;
414
415     // ~ EXEC ~
416     $display("EXEC");
417     opcode = opc_arr[7];        // load opcode
418     #5;
419     assert(pc_reset == 1'b0)    else $error("PC_RESET failed");
420     assert(pc_write == 1'b0)    else $error("PC_WRITE failed");
421     assert(reg_write == 1'b0)   else $error("REG_WRITE failed");
422     assert(mem_we2 == 1'b0)     else $error("MEM_WE2 failed");
423     assert(mem_rden1 == 1'b0)   else $error("MEM_RDEN1 failed");
424     assert(mem_rden2 == 1'b0)   else $error("MEM_RDEN2 failed");
425     #5;
426
427     // ~ WRITEBACK ~
428     $display("WRITEBACK");
429     #5;
430     assert(pc_reset == 1'b0)    else $error("PC_RESET failed");
431     assert(pc_write == 1'b1)    else $error("PC_WRITE failed");
432     assert(reg_write == 1'b1)   else $error("REG_WRITE failed");
433     assert(mem_we2 == 1'b0)     else $error("MEM_WE2 failed");
434     assert(mem_rden1 == 1'b0)   else $error("MEM_RDEN1 failed");
435     assert(mem_rden2 == 1'b0)   else $error("MEM_RDEN2 failed");
436     #5;
437
438
439
440     // ~~ TC9 : jal (111) ~~
441     testcase++;
442     $display("TC%d:", testcase);
443
444     // ~ FETCH ~
445     $display("FETCH");
446     #5;
447     assert(pc_reset == 1'b0)    else $error("PC_RESET failed");
448     assert(pc_write == 1'b0)    else $error("PC_WRITE failed");
449     assert(reg_write == 1'b0)   else $error("REG_WRITE failed");
450     assert(mem_we2 == 1'b0)     else $error("MEM_WE2 failed");
451     assert(mem_rden1 == 1'b1)   else $error("MEM_RDEN1 failed");

```

```

452     assert(mem_rden2 === 1'b0)    else $error("MEM_RDEN2 failed");
453     #5;
454
455     // ~ EXEC ~
456     $display("EXEC");
457     opcode = opc_arr[8];          // load opcode
458     #5;
459     assert(pc_reset === 1'b0)    else $error("PC_RESET failed");
460     assert(pc_write === 1'b0)    else $error("PC_WRITE failed");
461     assert(reg_write === 1'b0)   else $error("REG_WRITE failed");
462     assert(mem_we2 === 1'b0)     else $error("MEM_WE2 failed");
463     assert(mem_rden1 === 1'b0)   else $error("MEM_RDEN1 failed");
464     assert(mem_rden2 === 1'b0)   else $error("MEM_RDEN2 failed");
465     #5;
466
467     // ~ WRITEBACK ~
468     $display("WRITEBACK");
469     #5;
470     assert(pc_reset === 1'b0)    else $error("PC_RESET failed");
471     assert(pc_write === 1'b1)    else $error("PC_WRITE failed");
472     assert(reg_write === 1'b1)   else $error("REG_WRITE failed");
473     assert(mem_we2 === 1'b0)     else $error("MEM_WE2 failed");
474     assert(mem_rden1 === 1'b0)   else $error("MEM_RDEN1 failed");
475     assert(mem_rden2 === 1'b0)   else $error("MEM_RDEN2 failed");
476     #5;
477
478
479     // ~~ TC10 : reset check ~~
480     testcase++;
481     $display("TC%d:", testcase);
482
483     // ~ FETCH ~
484     $display("FETCH");
485     rst = 1'b1;
486     #5;
487     assert(pc_reset === 1'b1)    else $error("PC_RESET failed");
488     #5;
489     rst = 1'b0;
490     #5;
491     // we've reached fetch after reset
492     assert(pc_reset === 1'b0)    else $error("PC_RESET failed");
493     assert(pc_write === 1'b0)    else $error("PC_WRITE failed");
494     assert(reg_write === 1'b0)   else $error("REG_WRITE failed");
495     assert(mem_we2 === 1'b0)     else $error("MEM_WE2 failed");
496     assert(mem_rden1 === 1'b1)   else $error("MEM_RDEN1 failed");
497     assert(mem_rden2 === 1'b0)   else $error("MEM_RDEN2 failed");
498     #5;
499
500     // ~ EXEC ~
501     $display("EXEC");
502     rst = 1'b1;
503     #5;
504     assert(pc_reset === 1'b1)    else $error("PC_RESET failed");
505     #5;
506     rst = 1'b0;
507     #15;
508     // we've reached exec after reset
509     assert(pc_reset === 1'b0)    else $error("PC_RESET failed");
510     assert(pc_write === 1'b0)    else $error("PC_WRITE failed");
511     assert(reg_write === 1'b0)   else $error("REG_WRITE failed");
512     assert(mem_we2 === 1'b0)     else $error("MEM_WE2 failed");
513     assert(mem_rden1 === 1'b0)   else $error("MEM_RDEN1 failed");
514     assert(mem_rden2 === 1'b0)   else $error("MEM_RDEN2 failed");
515     #5;
516
517     // ~ WRITEBACK ~
518     $display("WRITEBACK");
519     rst = 1'b1;
520     #5;
521     assert(pc_reset === 1'b1)    else $error("PC_RESET failed");
522     #5;
523     rst = 1'b0;
524     #25;
525     assert(pc_reset === 1'b0)    else $error("PC_RESET failed");
526     assert(pc_write === 1'b1)    else $error("PC_WRITE failed");
527     assert(reg_write === 1'b1)   else $error("REG_WRITE failed");
528     assert(mem_we2 === 1'b0)     else $error("MEM_WE2 failed");
529     assert(mem_rden1 === 1'b0)   else $error("MEM_RDEN1 failed");
530     assert(mem_rden2 === 1'b0)   else $error("MEM_RDEN2 failed");
531     #5;
532
533     // ~~ TC11 : awkward reset timing ~~
534     testcase++;
535     $display("TC%d:", testcase);
536
537     // 1 clk(pos) to register reset
538     #3;
539     rst = 1'b1;
540     #1;
541     rst = 1'b0;
542     #1;

```

```
543     assert(pc_reset == 1'b1)    else $error("PC_RESET failed");
544     #6;
545
546
547
548     end
549
550
551 endmodule
```

0.3.11 MicroController

Listing 39: Otter MCU Otter_MCU.sv

```
1  'timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 08/19/2026 08:57:52 AM
7  // Design Name:
8  // Module Name: Otter_MCU
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module Otter_MCU(
24     input logic [31:0] IOBUS_IN,
25     input logic        RST,
26     input logic        INTRR,
27     input logic        CLK,
28
29     output logic [31:0] IOBUS_OUT,
30     output logic [31:0] IOBUS_ADDR,
31     output logic        IOBUS_WR
32 );
33
34
35 // ~~~~ inits / wires ~~~~
36
37 // ~~ pc + mux ~~
38 logic        reset;
39 logic        pc_write;
40 logic [1:0]   pc_source;
41 logic [31:0] pc;
42 logic [31:0] pc_p4;
43 logic [31:0] pcmux_out;
44
45 // ~~ memory ~~
46 logic        mem_rden1;
47 logic        mem_rden2;
48 logic        mem_we2;
49 logic [31:0] din2;
50 logic [31:0] ir, dout2;
51
52 // ~~ reg file + mux ~~
53 logic        reg_write;
54 logic [31:0] reg_w_data;
55 logic [31:0] rs1, rs2;
56 logic [1:0]  rf_wr_sel;
57
58 // ~~ immed gen ~~
59 logic [31:0] ut, it, st, jt, bt;
60
61 // ~~ branch addr gen ~~
62 logic [31:0] jalr, branch, jal;
63
64 // ~~ alu + mux ~~
65 logic        alu_src_a;
66 logic [1:0]   alu_src_b;
67 logic [3:0]   alu_func;
68 logic [31:0] src_a, src_b, result;
69
70 // ~~ csr ~~ ?????????????????????????????????????????????
71 logic [31:0] csr_reg;
72
73 // ~~ bcond + control decode and fsm ~~
74 logic        br_eq, br_lt, br_ltu;
75
76
77 // ~~~~ modules (l-2-r on diagram) ~~~~
78 PC_Reg PC(
79     .CLK(CLK),                // 1'b I
80     .PC_RST(reset),           // 1'b I
81     .PC_WE(pc_write),         // 1'b I
82     .PC_DIN(pcmux_out),       // 32'b I
83     .PC_COUNT(pc)             // 32'b 0
84 );
85
86 Mux_4N #(32) PC_MUX(
87     .D0(pc_p4),               // 32'b I
```

```

88     .D1(jalr),           // 32'b I
89     .D2(branch),        // 32'b I
90     .D3(jal),           // 32'b I
91     .S(pc_source),       // 2'b I
92     .Y(pcmux_out)        // 32'b 0
93 );
94
95 Plus4_1N #(32) P4(
96     .IN(pc),             // 32'b I
97     .OUT(pc_p4)          // 32'b 0
98 );
99
100 Memory MEMORY
101 (
102     .MEM_CLK(CLK),        // 1'b I
103     .MEM_RDEN1(mem_rden1), // 1'b I
104     .MEM_RDEN2(mem_rden2), // 1'b I
105     .MEM_WE2(mem_we2),    // 1'b I
106     .MEM_ADDR1(pc[15:2]), // 14'b I
107     .MEM_ADDR2(result),   // 32'b I
108     .MEM_DIN2(din2),      // 32'b I
109     .MEM_SIZE(ir[13:12]), // 2'b I
110     .MEM_SIGN(ir[14]),    // 1'b I
111     .IO_IN(IOBUS_IN),     // 32'b I
112
113     .IO_WR(IOBUS_WR),     // 1'b 0
114     .MEM_DOUT1(ir),       // 32'b 0
115     .MEM_DOUT2(dout2)     // 32'b 0
116 );
117
118 Mux_4N #(32) REG_MUX(
119     .D0(pc_p4),           // 32'b I
120     .D1(csr_reg),         // 32'b I
121     .D2(dout2),           // 32'b I
122     .D3(result),          // 32'b I
123     .S(rf_wr_sel),        // 2'b I
124     .Y(reg_w_data)        // 32'b 0
125 );
126
127 RegFile REG_FILE(
128     .EN(reg_write),       // 1'b I
129     .CLK(CLK),            // 1'b I
130     .ADR1(ir[19:15]),     // 5'b I
131     .ADR2(ir[24:20]),     // 5'b I
132     .W_ADR(ir[11:7]),     // 5'b I
133     .W_DATA(reg_w_data),  // 32'b I
134     .RS1(rs1),            // 32'b 0
135     .RS2(rs2)             // 32'b 0
136 );
137 assign IOBUS_OUT = rs2;
138
139 ImmedGen IMMED_GEN(
140     .INSTR(ir),           // 32'b I but only 25 are used
141     .U_TYPE(ut),          // 32'b 0
142     .I_TYPE(it),          // 32'b 0
143     .S_TYPE(st),          // 32'b 0
144     .J_TYPE(jt),          // 32'b 0
145     .B_TYPE(bt)           // 32'b 0
146 );
147
148 BranchAddrGen BRANCH_ADDR_GEN(
149     .RS1(rs1),            // 32'b I
150     .I_TYPE(it),          // 32'b I
151     .J_TYPE(jt),          // 32'b I
152     .B_TYPE(bt),          // 32'b I
153     .PC(pc),              // 32'b I
154     .JAL(jal),            // 32'b 0
155     .JALR(jalr),          // 32'b 0
156     .BRANCH(branch)       // 32'b 0
157 );
158
159 Mux_2N #(32) ALU_SRC_A_MUX(
160     .D0(rs1),             // 32'b I
161     .D1(ut),              // 32'b I
162     .S(alu_src_a),        // 2'b I
163     .Y(src_a)             // 32'b 0
164 );
165
166 Mux_4N #(32) ALU_SRC_B_MUX(
167     .D0(rs2),             // 32'b I
168     .D1(it),              // 32'b I
169     .D2(st),              // 32'b I
170     .D3(pc),              // 32'b I
171     .S(alu_src_b),        // 2'b I
172     .Y(src_b)             // 32'b 0
173 );
174
175 ALU ALU(
176     .ALU_FUN(alu_func),   // 4'b I
177     .SRC_A(src_a),        // 32'b I
178     .SRC_B(src_b),        // 32'b I

```

```

179     .ALU_RESULT(result)      // 32'b 0
180 );
181 assign IOBUS_ADDR = result;
182
183 BranchConditionGen BRANCH_COND_GEN(
184     .RS1(rs1),                // 32'b I
185     .RS2(rs2),                // 32'b I
186     .BR_EQ(br_eq),            // 1'b 0
187     .BR_LT(br_lt),            // 1'b 0
188     .BR_LTU(br_ltu)           // 1'b 0
189 );
190
191 CUnit_Decoder CU_DCDR(
192     .OPCODE(ir[6:0]),          // 7'b I
193     .FUNC3(ir[14:12]),         // 3'b I
194     .FUNC7(ir[30]),            // 1'b I
195     .BR_EQ(br_eq),            // 1'b I
196     .BR_LT(br_lt),            // 1'b I
197     .BR_LTU(br_ltu),          // 1'b I
198     .ALU_FUNC(alu_func),       // 4'b 0
199     .ALU_SRC_A(alu_src_a),     // 1'b 0
200     .ALU_SRC_B(alu_src_b),     // 2'b 0
201     .PC_SOURCE(pc_source),     // 2'b 0
202     .RF_WR_SEL(rf_wr_sel)     // 2'b 0
203 );
204
205 CUnit_FSM CU_FSM(
206     .CLK(CLK),                 // 1'b I
207     .RST(RST),                 // 1'b I
208     //.INTRR(),                 // 1'b I
209     .OPCODE(ir[6:0]),          // 7'b I
210     .PC_WRITE(pc_write),       // 1'b 0
211     .REG_WRITE(reg_write),     // 1'b 0
212     .MEM_RDEN1(mem_rden1),     // 1'b 0
213     .MEM_RDEN2(mem_rden2),     // 1'b 0
214     .MEM_WE2(mem_we2),         // 1'b 0
215     .RESET(reset)              // 1'b 0
216 );
217 endmodule

```

Listing 40: TestSim for MCU, CUnit_FSM_tb.sv, using RARS from demo.asm

```

1  'timescale 1ns / 1ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 08/19/2026 11:23:25 AM
7  // Design Name:
8  // Module Name: Otter_MCU_tb
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments: Ensure mem file is using sample code from hw6
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module Otter_MCU_tb();
24
25     // ~~~~ local vars ~~~~
26
27     // ~~ inputs ~~
28     logic      clk;
29     logic      rst;
30     logic      intrr;
31     logic [31:0]  din;
32
33     // ~~ outputs ~~
34     logic      write;
35     logic [31:0]  rs2;
36     logic [31:0]  alu_result;
37
38     // ~~ testing ~~
39     int testcase;
40     typedef enum {LUI, ADDI, SLLI, SLT, XOR_, BEQ} e_instr;
41     e_instr curr_instr;
42
43     // ~~~~ instances ~~~~
44     Otter_MCU UUT (
45         .IOBUS_IN(din),          // 32'b I
46         .RST(rst),               // 1'b I
47         .INTRR(intrr),           // 1'b I
48         .CLK(clk),               // 1'b I
49         .IOBUS_OUT(rs2),         // 32'b 0

```

```

50 .IOBUS_ADDR(alu_result),      // 32'b 0
51 .IOBUS_WR(write)             // 1'b 0      // 0 for non-MMIO
52 );
53
54 // ~~ action : CLK (10ns period) ~~
55 always
56 begin
57     #5;
58     clk <= !clk;
59 end
60
61 initial begin
62
63     // ~~ init ~~
64     testcase = 0;
65     clk =      1'b1;
66     rst =      1'b0;
67     intrr =    1'b0;
68     din =      32'h0000_0000;
69
70     // ~~ TC1 : 4x walkthrough ~~
71     testcase++;
72     #10;                // INIT complete
73     testall();
74
75     // ~~ TC2 : reset aand repeat TC1 ~~
76     testcase++;
77     rst =          1'b1;
78     #30;
79     rst =          1'b0;    // INIT already complete here
80     testall();
81 end
82
83 task testall();
84     for(int i = 0; i < 5; i++) begin
85         $display("%d", i);
86
87         // ~~ instr 1 : lui ~~
88         $display("lui"); curr_instr = LUI;
89         #10;                // FETCH complete
90         #10;                // EXEC complete
91         assert(alu_result == 32'hAA05_5000)    else $error("ALU_RESULT failed");
92         #10;                // WRITEBACK complete
93
94         // ~~ instr 2 : addi ~~
95         $display("addi"); curr_instr = ADDI;
96         #10;                // FETCH complete
97         #10;                // EXEC complete
98         assert(alu_result == 32'hAA05_5765)    else $error("ALU_RESULT failed");
99         assert(rs2 == 32'hAA05_5000)          else $error("RS2 failed");
100        #10;                // WRITEBACK complete
101
102        // ~~ instr 3 : slli ~~
103        $display("slli"); curr_instr = SLLI;
104        #10;                // FETCH complete
105        #10;                // EXEC complete
106        assert(alu_result == 32'h502A_BB28)    else $error("ALU_RESULT failed");
107        #10;                // WRITEBACK complete
108
109        // ~~ instr 4 : slt ~~
110        $display("slt"); curr_instr = SLT;
111        #10;                // FETCH complete
112        #10;                // EXEC complete
113        assert(alu_result == 32'h0000_0001)    else $error("ALU_RESULT failed");
114        assert(rs2 == 32'hAA05_5765)          else $error("RS2 failed");
115        #10;                // WRITEBACK complete
116
117        // ~~ instr 5 : xor ~~
118        $display("xor"); curr_instr = XOR_;
119        #10;                // FETCH complete
120        #10;                // EXEC complete
121        assert(alu_result == 32'hFA2F_EC4D)    else $error("ALU_RESULT failed");
122        assert(rs2 == 32'h502A_BB28)          else $error("RS2 failed");
123        #10;                // WRITEBACK complete
124
125        // ~~ instr 6 : beq ~~
126        $display("beq"); curr_instr = BEQ;
127        #10;                // FETCH complete
128        #10;                // EXEC complete
129        assert(rs2 == 32'h0000_0000)          else $error("RS2 failed");
130        #10;                // WRITEBACK complete
131    end
132 endtask
133
134
135 endmodule

```